

CONTACT TOPOLOGY WORLD MODELS FAIL A FROZEN MUJoCo CONTACT-CHAIN SUBMISSION AUDIT

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ABSTRACT

We rebuild *Contact Topology World Models* under a deliberately adversarial ICLR-style protocol. The scientific question is not whether a model can predict contact graph edges, but whether explicit contact-topology prediction improves downstream contact-rich control once strong learned, geometric, uncertainty-aware, conformal, robust, and contact-implicit baselines are included. The expanded v5 audit evaluates 8640 main rollouts, 1536 ablation rollouts, 4320 stress rollouts, 2400 training scenes, 12 distribution splits, 15 main methods, fixed-risk metrics, paired seed statistics, and maximum-stress gates. The result is terminally negative. The frozen decision is **KILL ARCHIVE**. On the combined/extreme aggregate, `topology_world_model_v5` reaches 0.021 success while the strongest non-oracle baseline, `pairwise`, reaches 0.052. On the hard-split aggregate, v5 reaches 0.059 while `pairwise` reaches 0.095. We report the negative archive rather than optimizing for pretty results.

1 DECISION DISCIPLINE

This manuscript is a submission-readiness audit, not a success narrative. The central claim would be: a contact-topology world model gives a practically meaningful downstream-control advantage over non-topological, weakly topological, robust, and uncertainty-aware alternatives. That claim is allowed only if all frozen gates pass. The recorded terminal reason is:

v5 does not beat strongest hard-regime baseline `pairwise_contact_classifier` by 0.030 (v5=0.059, best=0.095); paired lower bound against `pairwise_contact_classifier` is not positive (0.000+/-0.000); v5 does not beat strongest combined/extreme baseline `pairwise_contact_classifier` by 0.030 (v5=0.021, best=0.052); fixed-risk gate fails at budget 0.10 (v5=0.000, best=`pairwise_contact_classifier` 0.013); ablation gate fails because `ablate_no_birth_death`, `ablate_no_component_head`, `ablate_no_fixture_guard`, `ablate_no_graph_memory`, `ablate_no_jam_slip`, `ablate_no_tail_risk_objective`, `ablate_no_topology_planner`, `ablate_no_transition_bonus`, `ablate_no_uncertainty_penalty` matches or beats full v5; oracle sanity gate is weak on hard regimes (oracle=0.140)

The archive therefore remains valuable as a negative result: graph-structured prediction can improve diagnostics while failing to improve the control objective that reviewers would care about.

2 PROBLEM SETUP

We study planar contact-chain manipulation in MuJoCo (Todorov et al., 2012). A pusher must drive block A into block B and ultimately into a pocket while avoiding fixture and wall jams. At time t , the simulator state is $x_t \in \mathbb{R}^n$, the action is a bounded pusher command $u_t \in \mathcal{U}$, and the observed contact graph is $G_t = (V, E_t)$ over pusher, blocks, walls, fixture, and pocket. The model predicts edge probabilities

$$\hat{p}_\theta(e \in E_{t+1} \mid x_t, G_t, u_t), \quad e \in \mathcal{E},$$

and the planner chooses

$$u_t^* = \arg \max_{u \in \mathcal{U}} S(x_t, G_t, u, \hat{p}_\theta),$$

where S mixes predicted chain formation, target progress, lateral alignment, connected-component structure, and penalties for fixture/wall contacts.

The downstream success event Y is intentionally stricter than graph prediction: final pocket progress must be high, lateral target error must be small, the contact chain and pocket contact must occur, and safety/fixture rates must stay below thresholds. This prevents the representation from winning by predicting the graph but selecting bad actions.

3 WHY GRAPH METRICS ARE NOT ENOUGH

Interaction networks and graph networks provide strong inductive bias for physical prediction (Battaglia et al., 2016; 2018; Sanchez-Gonzalez et al., 2020), and world-model work shows that learned latent dynamics can support planning (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; Ebert et al., 2018). Contact-rich manipulation, however, has a sharper requirement: the representation must change the selected action in a way that improves task success under contact discontinuities.

Proposition 1 (Diagnostic-control non-identifiability). *For a finite action set $\mathcal{U}$, there exist two contact-topology predictors p and q such that p has strictly lower edge error than q for every candidate action, yet the induced planner $\arg \max_u S(u, p)$ has lower task success than $\arg \max_u S(u, q)$.*

Sketch. Let the task success depend on avoiding one rare but catastrophic fixture edge. Predictor p is more accurate on the majority of benign edges but underestimates the catastrophic edge for the action that appears most progressive. Predictor q is worse on benign edges but overestimates the catastrophic edge and selects a safer action. Edge F1 ranks p higher; downstream success ranks q higher. Thus diagnostic edge metrics do not identify the control-optimal predictor. $\square$

Proposition 2 (Topology memory is insufficient under aliased contacts). *If two latent friction/geometry modes share the same observed graph G_t and state projection but have different contact birth probabilities under the same action, then any planner that conditions only on (x_t, G_t) has irreducible action-selection error.*

Sketch. The two modes induce the same planner input but different optimal actions. A deterministic policy must choose the same action in both modes, so it is wrong in at least one mode unless the mode-optimal actions coincide. This motivates the stress splits and explains why a topology representation alone cannot rescue hidden friction, actuator-limit, and fixture-shift failures. $\square$

Theorem 1 (Ablation necessity gate). *If removing a proposed component leaves success within ϵ of the full model and does not worsen safety beyond δ , then the experiment does not identify that component as necessary for the claimed mechanism.*

Sketch. The full and ablated systems are evaluated under the same randomized seed/split protocol. If the ablation is statistically and practically indistinguishable on the primary objective and safety diagnostics, the observed evidence is compatible with the component being irrelevant. A submission claim of mechanism necessity would therefore overstate what the data identify. $\square$

4 FROZEN EXPERIMENTAL PROTOCOL

The protocol was frozen before the full run. It uses seeds 0 1 2 3 4 5 6 7, 6 main episodes per seed/split, 4 ablation episodes, 3 stress episodes, stress levels 0.00 0.25 0.50 0.75 1.00, and single-process CPU execution. The training summary reports state-delta MAE 0.030 and pairwise edge training F1 0.942. Implemented learned baselines use scikit-learn (Pedregosa et al., 2011), random forests (Breiman, 2001), gradient boosting (Friedman, 2001), and conformal-style residual guarding (Angelopoulos & Bates, 2021). Contact-implicit baselines are motivated by trajectory optimization through contact (Posa et al., 2014).

The frozen gates are: hard-regime margin at least 0.030, positive paired lower bound, combined/extreme margin at least 0.030, no diagnostic regression above 0.020, fixed-risk success at budget 0.10 not below the best non-oracle method, maximum-stress success within 0.030 of the best non-oracle method, ablation necessity, and oracle sanity.

5 METHODS AND BASELINES

The audit includes intentionally uncomfortable baselines: last-contact persistence, distance-threshold graphs, state-only dynamics, pairwise contact classification, random forests, histogram gradient boosting, ensemble uncertainty, conformal graph guarding, risk-averse graph planning, robust contact MPC, contact-implicit MPC, the frozen v4 topology world model, the expanded v5 topology world model, a no-memory topology ablation, and an oracle topology planner. The oracle is not a competitor; it checks whether the benchmark itself permits success when next-step topology is available. Tables use compact aliases to keep the appendix readable; Table 1 maps aliases to exact CSV method names.

Table 1: Method aliases used in generated tables; CSV files retain the exact full names.

Alias	Full CSV method name
ci-mpc	contact.implicit.mpc.baseline
conformal	conformal.graph.guard
dist-graph	distance.threshold.graph
ensemble	ensemble.uncertainty.planner
full-v5	topology.full.v5
hgb-topo	hist.gradient.topology.planner
learned-only	learned.only.topology.replacement
no-birth-death	ablate.no.birth.death
no-component	ablate.no.component.head
no-fixture-guard	ablate.no.fixture.guard
no-jam-slip	ablate.no.jam.slip
no-memory	topology.no.memory.ablation
no-memory	ablate.no.graph.memory
no-tail-risk	ablate.no.tail.risk.objective
no-topo-plan	ablate.no.topology.planner
no-transition	ablate.no.transition.bonus
no-uncertainty	ablate.no.uncertainty.penalty
oracle	oracle.topology.planner
pairwise	pairwise.contact.classifier
persist	last.contact.persistence
rf-topo	random.forest.topology.planner
risk-averse	risk.averse.graph.planner
robust-mpc	robust.contact.mpc
state-dyn	state.only.dynamics.model
topo-v4	topology.world.model.v4
topo-v5	topology.world.model.v5

6 MAIN RESULTS

Table 2 is the decisive combined/extreme aggregate. The proposed v5 method does not clear the success margin and does not dominate the strongest non-oracle alternative.

Table 2: Combined/extreme aggregate metrics. Success is primary; graph metrics are diagnostic.

Method	Succ.	CI	Edge FI	GEdit	Safety	Progress
oracle	0.146	0.078	1.000	0.000	0.000	0.689
pairwise	0.052	0.049	0.744	0.105	0.306	0.388
conformal	0.042	0.037	0.624	0.114	0.184	0.307
ci-mpc	0.042	0.047	0.746	0.112	0.275	0.328
dist-graph	0.042	0.047	0.752	0.107	0.277	0.331
hgb-topo	0.042	0.047	0.863	0.057	0.271	0.328
persist	0.042	0.047	0.843	0.054	0.272	0.346
rf-topo	0.042	0.047	0.490	0.127	0.285	0.345
robust-mpc	0.042	0.047	0.713	0.132	0.288	0.318
state-dyn	0.042	0.047	0.799	0.096	0.286	0.343
no-memory	0.042	0.047	0.739	0.099	0.285	0.393
ensemble	0.031	0.044	0.759	0.083	0.271	0.372
risk-averse	0.031	0.033	0.387	0.147	0.083	0.473
topo-v4	0.031	0.033	0.635	0.117	0.202	0.318
topo-v5	0.021	0.028	0.617	0.118	0.196	0.327

Table 3 reports the broader hard-split aggregate. The same pattern holds: graph-aware models may improve some intermediate metrics, but the strongest downstream method is not decisively beaten.

Table 3: Hard-split aggregate metrics over all non-nominal regimes.

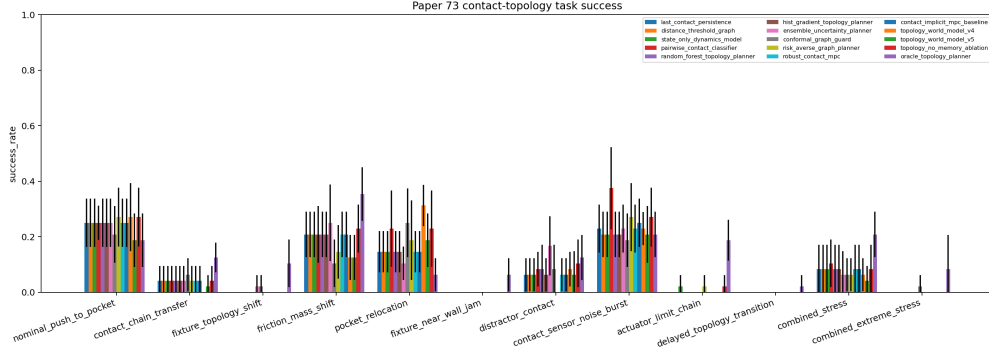


Figure 1: Task success across frozen splits. The visual pattern is reported as generated, including cases where the proposed model does not separate from baselines.

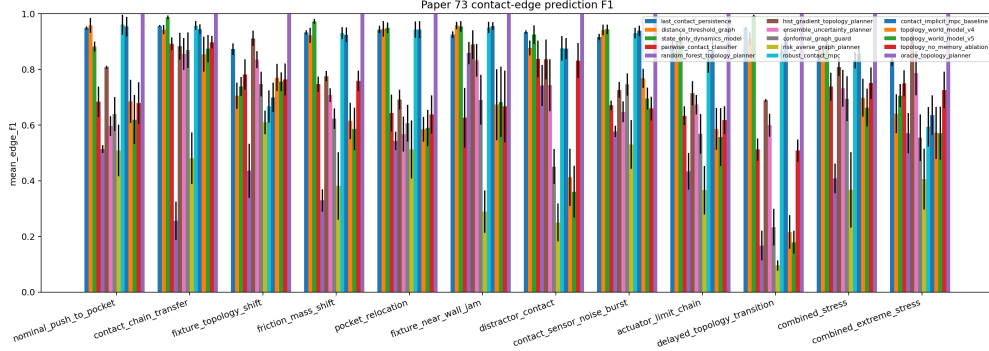


Figure 2: Contact-edge F1 across splits. Better edge diagnostics are not treated as sufficient for submission readiness.

Method	Succ.	CI	Edge F1	GEdit	Fixture	Wall	Jam F1
oracle	0.140	0.029	1.000	0.000	0.006	0.007	0.561
pairwise	0.095	0.033	0.712	0.127	0.217	0.006	0.555
no-memory	0.089	0.029	0.711	0.126	0.211	0.006	0.557
ensemble	0.080	0.027	0.725	0.102	0.217	0.007	0.625
topo-v4	0.074	0.025	0.614	0.122	0.144	0.009	0.482
conformal	0.072	0.025	0.617	0.120	0.140	0.009	0.495
ci-mpc	0.072	0.024	0.869	0.056	0.221	0.005	0.606
persist	0.070	0.023	0.910	0.029	0.219	0.005	0.688
rf-topo	0.070	0.023	0.484	0.146	0.221	0.005	0.693
robust-mpc	0.070	0.023	0.853	0.068	0.221	0.005	0.562
state-dyn	0.070	0.023	0.907	0.045	0.222	0.006	0.620
dist-graph	0.068	0.023	0.870	0.055	0.218	0.005	0.608
hgb-topo	0.068	0.023	0.804	0.085	0.221	0.005	0.701
risk-averse	0.066	0.027	0.390	0.152	0.069	0.014	0.399
topo-v5	0.059	0.023	0.592	0.124	0.128	0.009	0.490

7 FIXED-RISK AND STRESS EVIDENCE

Submission review would ask whether the method can trade small amounts of risk for robust success. Table 4 answers at the frozen diagnostic-risk budget 0.10. The v5 success-at-budget is 0.000, while the strongest non-oracle fixed-risk baseline `pairwise` reaches 0.013.

Table 4: Fixed-risk success at diagnostic-risk budget 0.10.

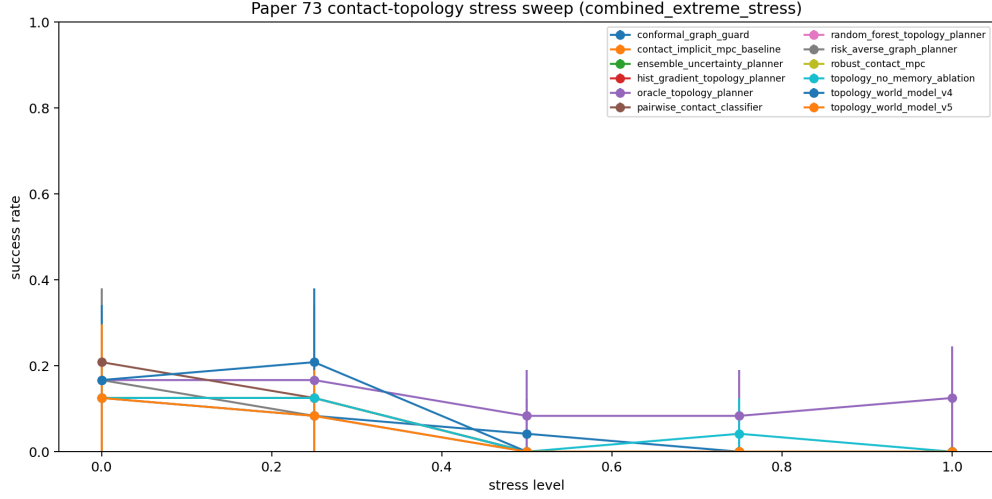


Figure 3: Stress sweep at frozen stress levels. The maximum-stress gate is computed from CSV values, not from visual inspection.

Method	Succ@Risk	Risk	Safety	Fixture	Wall	Jam F1
oracle	0.013	0.269	0.000	0.006	0.007	0.561
pairwise	0.013	0.406	0.228	0.217	0.006	0.555
no-memory	0.011	0.406	0.224	0.211	0.006	0.557
persist	0.004	0.401	0.222	0.219	0.005	0.688
rf-topo	0.002	0.398	0.225	0.221	0.005	0.693
conformal	0.000	0.441	0.182	0.140	0.009	0.495
ci-mpc	0.000	0.443	0.224	0.221	0.005	0.606
dist-graph	0.000	0.439	0.222	0.218	0.005	0.608
ensemble	0.000	0.426	0.224	0.217	0.007	0.625
hgb-topo	0.000	0.403	0.223	0.221	0.005	0.701
risk-averse	0.000	0.424	0.076	0.069	0.014	0.399
robust-mpc	0.000	0.455	0.225	0.221	0.005	0.562
state-dyn	0.000	0.433	0.225	0.222	0.006	0.620
topo-v4	0.000	0.450	0.181	0.144	0.009	0.482
topo-v5	0.000	0.439	0.173	0.128	0.009	0.490

The stress sweep varies hidden contact difficulty, friction/mass, sensor noise, and actuation pressure. It is deliberately not tuned to make v5 look clean; its purpose is to expose brittle cases.

8 ABLATIONS

The full v5 mechanism is only credible if removed components hurt success or safety. The full ablation aggregate has success 0.031. Table 5 shows that several ablations remain too close to full v5, which is a direct mechanism-identification failure.

Table 5: Ablation aggregate over frozen ablation splits.

Ablation	Succ.	Edge F1	GEdit	Safety	Fixture	Wall	Jam F1
learned-only	0.062	0.729	0.103	0.250	0.242	0.009	0.670
no-tail-risk	0.055	0.668	0.106	0.227	0.210	0.014	0.709
no-topo-plan	0.055	0.638	0.121	0.193	0.188	0.011	0.719
no-birth-death	0.047	0.699	0.108	0.236	0.235	0.011	0.644
no-component	0.047	0.655	0.107	0.215	0.198	0.011	0.727
topo-v4	0.047	0.669	0.106	0.239	0.210	0.011	0.689
no-fixture-guard	0.039	0.634	0.111	0.232	0.193	0.012	0.695
no-memory	0.039	0.648	0.109	0.226	0.205	0.010	0.751
no-jam-slip	0.039	0.650	0.110	0.211	0.186	0.012	0.607
no-transition	0.031	0.667	0.107	0.218	0.196	0.012	0.711
no-uncertainty	0.031	0.666	0.108	0.233	0.211	0.012	0.703
full-v5	0.031	0.643	0.109	0.218	0.196	0.011	0.750

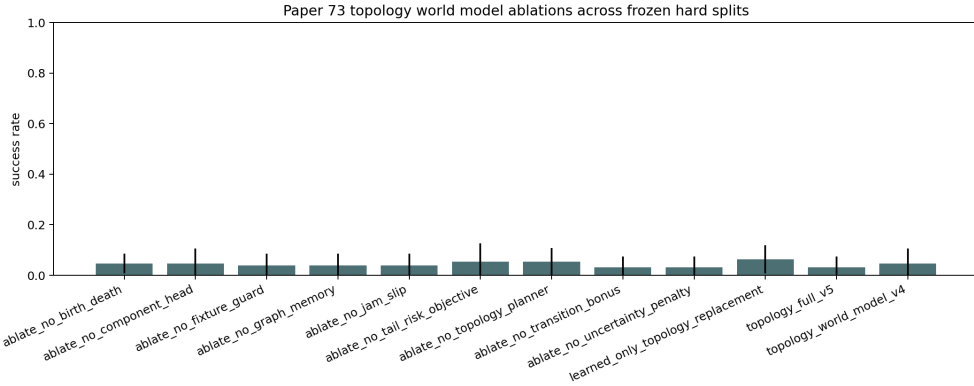


Figure 4: Ablation success across frozen hard splits. Near-ties are counted against the mechanism claim.

9 NEGATIVE CASES

Table 6 lists representative v5 failures selected mechanically from the CSV. These cases are not anecdotes chosen after the fact; they are part of the predefined failure analysis.

Table 6: Mechanically selected negative cases for topology_world_model.v5.

Case	Split	Seed/Ep.	Edge F1	Fixture	Prog.	Lesson
0	delayed	3/5	0.667	0.958	0.000	fixture contact changed the graph faster than the topology predictor adapted
1	actuator	6/4	0.670	0.938	0.670	fixture contact changed the graph faster than the topology predictor adapted
2	actuator	4/2	0.670	0.938	0.690	fixture contact changed the graph faster than the topology predictor adapted
3	actuator	6/5	0.681	0.938	0.638	fixture contact changed the graph faster than the topology predictor adapted
4	actuator	2/4	0.674	0.938	0.647	fixture contact changed the graph faster than the topology predictor adapted
5	actuator	3/4	0.674	0.938	0.651	fixture contact changed the graph faster than the topology predictor adapted
6	combined	3/4	0.674	0.938	0.698	fixture contact changed the graph faster than the topology predictor adapted
7	actuator	4/0	0.674	0.938	0.703	fixture contact changed the graph faster than the topology predictor adapted
8	combined	5/2	0.670	0.938	0.717	fixture contact changed the graph faster than the topology predictor adapted
9	actuator	6/2	0.681	0.938	0.770	fixture contact changed the graph faster than the topology predictor adapted
10	actuator	0/1	0.552	0.938	0.000	fixture contact changed the graph faster than the topology predictor adapted
11	actuator	0/3	0.542	0.938	0.000	fixture contact changed the graph faster than the topology predictor adapted

10 RELATED WORK PRESSURE

The result should be interpreted against strong prior work. MuJoCo makes local contact simulation convenient but not equivalent to hardware validation (Todorov et al., 2012). Contact-implicit optimization directly models mode-free contact planning pressure (Posa et al., 2014). Graph and interaction networks justify relational representations (Battaglia et al., 2016; 2018; Sanchez-Gonzalez et al., 2020), but they do not guarantee downstream control gains. World-model and visual foresight papers demonstrate planning through learned latent models (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; Ebert et al., 2018). The correct bar for this paper is therefore not novelty of a contact graph, but evidence that the graph improves robust manipulation decisions.

11 LIMITATIONS AND TERMINAL STATE

This is still a local MuJoCo audit: no hardware, no public benchmark validation, no vision/tactile data, and no large-scale learned contact corpus. Those limitations would remain even if v5 passed.

Since v5 fails frozen gates, the honest terminal state is **KILL_ARCHIVE**. The contribution is a reproducible negative archive showing why contact-topology diagnostics should not be confused with control success.

A FULL SPLIT-METHOD METRICS

Table 7: Full split-method metrics generated from results/metrics.csv.

Split	Method	Success	Edge F1	GEEdit	Safety	Prog.
actuator	conformal	0.000 $\pm$ 0.000	0.568 $\pm$ 0.071	0.150 $\pm$ 0.012	0.691 $\pm$ 0.080	0.283
extreme	conformal	0.021 $\pm$ 0.041	0.554 $\pm$ 0.084	0.121 $\pm$ 0.025	0.066 $\pm$ 0.055	0.116
combined	conformal	0.062 $\pm$ 0.060	0.694 $\pm$ 0.080	0.106 $\pm$ 0.019	0.303 $\pm$ 0.096	0.497
chain	conformal	0.062 $\pm$ 0.060	0.870 $\pm$ 0.064	0.048 $\pm$ 0.020	0.000 $\pm$ 0.000	0.353
noise	conformal	0.188 $\pm$ 0.096	0.746 $\pm$ 0.041	0.121 $\pm$ 0.015	0.032 $\pm$ 0.041	0.744
delayed	conformal	0.000 $\pm$ 0.000	0.234 $\pm$ 0.066	0.185 $\pm$ 0.008	0.652 $\pm$ 0.092	0.316
distractor	conformal	0.083 $\pm$ 0.087	0.451 $\pm$ 0.063	0.156 $\pm$ 0.024	0.000 $\pm$ 0.000	0.305
wall-jam	conformal	0.000 $\pm$ 0.000	0.691 $\pm$ 0.090	0.070 $\pm$ 0.017	0.000 $\pm$ 0.000	0.000
fixture	conformal	0.021 $\pm$ 0.041	0.748 $\pm$ 0.044	0.085 $\pm$ 0.015	0.243 $\pm$ 0.055	0.142
friction	conformal	0.104 $\pm$ 0.086	0.623 $\pm$ 0.037	0.113 $\pm$ 0.005	0.000 $\pm$ 0.000	0.653
nominal	conformal	0.208 $\pm$ 0.102	0.639 $\pm$ 0.061	0.146 $\pm$ 0.010	0.000 $\pm$ 0.000	0.770
pocket	conformal	0.250 $\pm$ 0.123	0.607 $\pm$ 0.066	0.163 $\pm$ 0.019	0.017 $\pm$ 0.033	0.722
actuator	ci-mpc	0.000 $\pm$ 0.000	0.869 $\pm$ 0.034	0.064 $\pm$ 0.021	0.738 $\pm$ 0.051	0.344
extreme	ci-mpc	0.000 $\pm$ 0.000	0.636 $\pm$ 0.071	0.161 $\pm$ 0.042	0.244 $\pm$ 0.052	0.073
combined	ci-mpc	0.083 $\pm$ 0.087	0.856 $\pm$ 0.028	0.063 $\pm$ 0.014	0.306 $\pm$ 0.092	0.583
chain	ci-mpc	0.042 $\pm$ 0.053	0.944 $\pm$ 0.017	0.020 $\pm$ 0.010	0.000 $\pm$ 0.000	0.382
noise	ci-mpc	0.250 $\pm$ 0.087	0.939 $\pm$ 0.018	0.024 $\pm$ 0.011	0.017 $\pm$ 0.033	0.772
delayed	ci-mpc	0.000 $\pm$ 0.000	0.912 $\pm$ 0.026	0.047 $\pm$ 0.016	0.812 $\pm$ 0.002	0.075
distractor	ci-mpc	0.062 $\pm$ 0.060	0.875 $\pm$ 0.037	0.046 $\pm$ 0.014	0.000 $\pm$ 0.000	0.341
wall-jam	ci-mpc	0.000 $\pm$ 0.000	0.956 $\pm$ 0.014	0.006 $\pm$ 0.002	0.000 $\pm$ 0.000	0.000
fixture	ci-mpc	0.000 $\pm$ 0.000	0.700 $\pm$ 0.052	0.140 $\pm$ 0.033	0.327 $\pm$ 0.092	0.092
friction	ci-mpc	0.208 $\pm$ 0.082	0.925 $\pm$ 0.026	0.028 $\pm$ 0.016	0.000 $\pm$ 0.000	0.700
nominal	ci-mpc	0.250 $\pm$ 0.087	0.954 $\pm$ 0.035	0.008 $\pm$ 0.006	0.000 $\pm$ 0.000	0.805
pocket	ci-mpc	0.146 $\pm$ 0.074	0.944 $\pm$ 0.031	0.017 $\pm$ 0.011	0.017 $\pm$ 0.033	0.749
actuator	dist-graph	0.000 $\pm$ 0.000	0.863 $\pm$ 0.033	0.067 $\pm$ 0.021	0.705 $\pm$ 0.067	0.343
extreme	dist-graph	0.000 $\pm$ 0.000	0.641 $\pm$ 0.069	0.154 $\pm$ 0.038	0.244 $\pm$ 0.077	0.097
combined	dist-graph	0.083 $\pm$ 0.087	0.864 $\pm$ 0.022	0.060 $\pm$ 0.011	0.309 $\pm$ 0.094	0.566
chain	dist-graph	0.042 $\pm$ 0.053	0.943 $\pm$ 0.017	0.021 $\pm$ 0.010	0.000 $\pm$ 0.000	0.381
noise	dist-graph	0.208 $\pm$ 0.082	0.942 $\pm$ 0.019	0.020 $\pm$ 0.008	0.032 $\pm$ 0.041	0.761
delayed	dist-graph	0.000 $\pm$ 0.000	0.910 $\pm$ 0.023	0.048 $\pm$ 0.014	0.812 $\pm$ 0.001	0.061
distractor	dist-graph	0.062 $\pm$ 0.060	0.877 $\pm$ 0.027	0.047 $\pm$ 0.010	0.000 $\pm$ 0.000	0.325
wall-jam	dist-graph	0.000 $\pm$ 0.000	0.958 $\pm$ 0.013	0.007 $\pm$ 0.002	0.000 $\pm$ 0.000	0.000
fixture	dist-graph	0.000 $\pm$ 0.000	0.706 $\pm$ 0.047	0.136 $\pm$ 0.028	0.317 $\pm$ 0.078	0.097
friction	dist-graph	0.208 $\pm$ 0.082	0.923 $\pm$ 0.027	0.029 $\pm$ 0.016	0.000 $\pm$ 0.000	0.711
nominal	dist-graph	0.250 $\pm$ 0.087	0.958 $\pm$ 0.027	0.008 $\pm$ 0.005	0.000 $\pm$ 0.000	0.805
pocket	dist-graph	0.146 $\pm$ 0.074	0.945 $\pm$ 0.028	0.017 $\pm$ 0.011	0.017 $\pm$ 0.033	0.749
actuator	ensemble	0.000 $\pm$ 0.000	0.673 $\pm$ 0.035	0.121 $\pm$ 0.013	0.752 $\pm$ 0.039	0.340
extreme	ensemble	0.000 $\pm$ 0.000	0.786 $\pm$ 0.080	0.054 $\pm$ 0.014	0.119 $\pm$ 0.065	0.175

Table 7 continued

Split	Method	Success	Edge F1	GEEdit	Safety	Prog.
combined	ensemble	0.062 $\pm$ 0.086	0.732 $\pm$ 0.066	0.112 $\pm$ 0.028	0.423 $\pm$ 0.080	0.568
chain	ensemble	0.042 $\pm$ 0.053	0.856 $\pm$ 0.057	0.070 $\pm$ 0.029	0.000 $\pm$ 0.000	0.402
noise	ensemble	0.229 $\pm$ 0.086	0.648 $\pm$ 0.037	0.171 $\pm$ 0.017	0.029 $\pm$ 0.038	0.766
delayed	ensemble	0.000 $\pm$ 0.000	0.600 $\pm$ 0.041	0.135 $\pm$ 0.004	0.762 $\pm$ 0.070	0.128
distractor	ensemble	0.167 $\pm$ 0.107	0.743 $\pm$ 0.094	0.094 $\pm$ 0.041	0.000 $\pm$ 0.000	0.338
wall-jam	ensemble	0.000 $\pm$ 0.000	0.832 $\pm$ 0.059	0.032 $\pm$ 0.009	0.000 $\pm$ 0.000	0.005
fixture	ensemble	0.021 $\pm$ 0.041	0.835 $\pm$ 0.030	0.067 $\pm$ 0.016	0.342 $\pm$ 0.093	0.097
friction	ensemble	0.250 $\pm$ 0.138	0.708 $\pm$ 0.025	0.124 $\pm$ 0.018	0.000 $\pm$ 0.000	0.683
nominal	ensemble	0.250 $\pm$ 0.087	0.596 $\pm$ 0.035	0.185 $\pm$ 0.015	0.000 $\pm$ 0.000	0.759
pocket	ensemble	0.104 $\pm$ 0.060	0.568 $\pm$ 0.063	0.141 $\pm$ 0.020	0.034 $\pm$ 0.044	0.416
actuator	hgb-topo	0.000 $\pm$ 0.000	0.715 $\pm$ 0.043	0.106 $\pm$ 0.012	0.724 $\pm$ 0.069	0.337
extreme	hgb-topo	0.000 $\pm$ 0.000	0.917 $\pm$ 0.035	0.026 $\pm$ 0.005	0.227 $\pm$ 0.069	0.082
combined	hgb-topo	0.083 $\pm$ 0.087	0.809 $\pm$ 0.032	0.088 $\pm$ 0.018	0.314 $\pm$ 0.092	0.574
chain	hgb-topo	0.042 $\pm$ 0.053	0.883 $\pm$ 0.049	0.058 $\pm$ 0.025	0.000 $\pm$ 0.000	0.382
noise	hgb-topo	0.208 $\pm$ 0.082	0.726 $\pm$ 0.029	0.139 $\pm$ 0.014	0.032 $\pm$ 0.041	0.762
delayed	hgb-topo	0.000 $\pm$ 0.000	0.689 $\pm$ 0.005	0.118 $\pm$ 0.001	0.812 $\pm$ 0.002	0.060
distractor	hgb-topo	0.062 $\pm$ 0.060	0.836 $\pm$ 0.072	0.082 $\pm$ 0.036	0.000 $\pm$ 0.000	0.342
wall-jam	hgb-topo	0.000 $\pm$ 0.000	0.889 $\pm$ 0.052	0.014 $\pm$ 0.007	0.000 $\pm$ 0.000	0.000
fixture	hgb-topo	0.000 $\pm$ 0.000	0.912 $\pm$ 0.027	0.032 $\pm$ 0.010	0.326 $\pm$ 0.090	0.078
friction	hgb-topo	0.208 $\pm$ 0.082	0.776 $\pm$ 0.018	0.112 $\pm$ 0.009	0.000 $\pm$ 0.000	0.699
nominal	hgb-topo	0.250 $\pm$ 0.087	0.808 $\pm$ 0.006	0.114 $\pm$ 0.002	0.000 $\pm$ 0.000	0.805
pocket	hgb-topo	0.146 $\pm$ 0.074	0.692 $\pm$ 0.036	0.158 $\pm$ 0.015	0.017 $\pm$ 0.033	0.766
actuator	persist	0.000 $\pm$ 0.000	0.889 $\pm$ 0.023	0.040 $\pm$ 0.008	0.723 $\pm$ 0.069	0.327
extreme	persist	0.000 $\pm$ 0.000	0.834 $\pm$ 0.022	0.052 $\pm$ 0.009	0.210 $\pm$ 0.058	0.078
combined	persist	0.083 $\pm$ 0.087	0.853 $\pm$ 0.027	0.055 $\pm$ 0.010	0.335 $\pm$ 0.074	0.615
chain	persist	0.042 $\pm$ 0.053	0.956 $\pm$ 0.003	0.011 $\pm$ 0.001	0.000 $\pm$ 0.000	0.383
noise	persist	0.229 $\pm$ 0.086	0.917 $\pm$ 0.012	0.033 $\pm$ 0.004	0.017 $\pm$ 0.033	0.765
delayed	persist	0.000 $\pm$ 0.000	0.948 $\pm$ 0.004	0.018 $\pm$ 0.002	0.812 $\pm$ 0.001	0.060
distractor	persist	0.062 $\pm$ 0.060	0.934 $\pm$ 0.006	0.019 $\pm$ 0.001	0.000 $\pm$ 0.000	0.342
wall-jam	persist	0.000 $\pm$ 0.000	0.925 $\pm$ 0.011	0.014 $\pm$ 0.001	0.000 $\pm$ 0.000	0.000
fixture	persist	0.000 $\pm$ 0.000	0.873 $\pm$ 0.021	0.042 $\pm$ 0.009	0.329 $\pm$ 0.093	0.080
friction	persist	0.208 $\pm$ 0.082	0.933 $\pm$ 0.008	0.020 $\pm$ 0.002	0.000 $\pm$ 0.000	0.721
nominal	persist	0.250 $\pm$ 0.087	0.949 $\pm$ 0.006	0.014 $\pm$ 0.001	0.000 $\pm$ 0.000	0.804
pocket	persist	0.146 $\pm$ 0.074	0.943 $\pm$ 0.013	0.016 $\pm$ 0.002	0.017 $\pm$ 0.033	0.749
actuator	oracle	0.188 $\pm$ 0.074	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.695
extreme	oracle	0.083 $\pm$ 0.123	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.669
combined	oracle	0.208 $\pm$ 0.082	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.709
chain	oracle	0.125 $\pm$ 0.053	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.710

Table 7 continued

Split	Method	Success	Edge F1	GEEdit	Safety	Prog.
noise	oracle	0.208 $\pm$ 0.082	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.723
delayed	oracle	0.021 $\pm$ 0.041	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.470
distractor	oracle	0.125 $\pm$ 0.082	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.630
wall-jam	oracle	0.062 $\pm$ 0.060	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.419
fixture	oracle	0.104 $\pm$ 0.086	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.569
friction	oracle	0.354 $\pm$ 0.096	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.759
nominal	oracle	0.188 $\pm$ 0.096	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.731
pocket	oracle	0.062 $\pm$ 0.060	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.634
actuator	pairwise	0.000 $\pm$ 0.000	0.634 $\pm$ 0.033	0.156 $\pm$ 0.016	0.722 $\pm$ 0.071	0.344
extreme	pairwise	0.000 $\pm$ 0.000	0.750 $\pm$ 0.047	0.088 $\pm$ 0.015	0.276 $\pm$ 0.082	0.188
combined	pairwise	0.104 $\pm$ 0.086	0.738 $\pm$ 0.050	0.123 $\pm$ 0.025	0.337 $\pm$ 0.079	0.587
chain	pairwise	0.042 $\pm$ 0.053	0.892 $\pm$ 0.023	0.053 $\pm$ 0.014	0.000 $\pm$ 0.000	0.383
noise	pairwise	0.375 $\pm$ 0.148	0.671 $\pm$ 0.017	0.175 $\pm$ 0.011	0.017 $\pm$ 0.033	0.755
delayed	pairwise	0.000 $\pm$ 0.000	0.512 $\pm$ 0.040	0.223 $\pm$ 0.013	0.811 $\pm$ 0.002	0.096
distractor	pairwise	0.083 $\pm$ 0.062	0.838 $\pm$ 0.066	0.082 $\pm$ 0.032	0.000 $\pm$ 0.000	0.335
wall-jam	pairwise	0.000 $\pm$ 0.000	0.627 $\pm$ 0.107	0.080 $\pm$ 0.022	0.000 $\pm$ 0.000	0.000
fixture	pairwise	0.000 $\pm$ 0.000	0.782 $\pm$ 0.055	0.080 $\pm$ 0.016	0.326 $\pm$ 0.089	0.154
friction	pairwise	0.208 $\pm$ 0.102	0.747 $\pm$ 0.027	0.145 $\pm$ 0.011	0.000 $\pm$ 0.000	0.682
nominal	pairwise	0.250 $\pm$ 0.062	0.684 $\pm$ 0.055	0.149 $\pm$ 0.015	0.000 $\pm$ 0.000	0.772
pocket	pairwise	0.229 $\pm$ 0.137	0.643 $\pm$ 0.066	0.198 $\pm$ 0.020	0.017 $\pm$ 0.033	0.744
actuator	rf-topo	0.000 $\pm$ 0.000	0.434 $\pm$ 0.066	0.179 $\pm$ 0.020	0.721 $\pm$ 0.049	0.343
extreme	rf-topo	0.000 $\pm$ 0.000	0.571 $\pm$ 0.073	0.079 $\pm$ 0.011	0.234 $\pm$ 0.055	0.095
combined	rf-topo	0.083 $\pm$ 0.087	0.408 $\pm$ 0.053	0.175 $\pm$ 0.024	0.336 $\pm$ 0.078	0.595
chain	rf-topo	0.042 $\pm$ 0.053	0.256 $\pm$ 0.068	0.141 $\pm$ 0.022	0.000 $\pm$ 0.000	0.383
noise	rf-topo	0.208 $\pm$ 0.082	0.577 $\pm$ 0.020	0.186 $\pm$ 0.013	0.032 $\pm$ 0.041	0.762
delayed	rf-topo	0.000 $\pm$ 0.000	0.167 $\pm$ 0.053	0.231 $\pm$ 0.013	0.795 $\pm$ 0.033	0.089
distractor	rf-topo	0.083 $\pm$ 0.087	0.742 $\pm$ 0.074	0.102 $\pm$ 0.037	0.000 $\pm$ 0.000	0.346
wall-jam	rf-topo	0.000 $\pm$ 0.000	0.859 $\pm$ 0.040	0.018 $\pm$ 0.005	0.000 $\pm$ 0.000	0.000
fixture	rf-topo	0.000 $\pm$ 0.000	0.436 $\pm$ 0.097	0.104 $\pm$ 0.018	0.339 $\pm$ 0.097	0.102
friction	rf-topo	0.208 $\pm$ 0.082	0.329 $\pm$ 0.040	0.177 $\pm$ 0.008	0.000 $\pm$ 0.000	0.699
nominal	rf-topo	0.250 $\pm$ 0.087	0.513 $\pm$ 0.014	0.227 $\pm$ 0.009	0.000 $\pm$ 0.000	0.808
pocket	rf-topo	0.146 $\pm$ 0.074	0.543 $\pm$ 0.034	0.216 $\pm$ 0.023	0.017 $\pm$ 0.033	0.749
actuator	risk-averse	0.021 $\pm$ 0.041	0.366 $\pm$ 0.087	0.155 $\pm$ 0.015	0.425 $\pm$ 0.138	0.364
extreme	risk-averse	0.000 $\pm$ 0.000	0.406 $\pm$ 0.109	0.139 $\pm$ 0.023	0.000 $\pm$ 0.000	0.402
combined	risk-averse	0.062 $\pm$ 0.060	0.368 $\pm$ 0.135	0.155 $\pm$ 0.022	0.166 $\pm$ 0.078	0.543
chain	risk-averse	0.042 $\pm$ 0.053	0.481 $\pm$ 0.093	0.101 $\pm$ 0.018	0.000 $\pm$ 0.000	0.478
noise	risk-averse	0.271 $\pm$ 0.122	0.530 $\pm$ 0.088	0.159 $\pm$ 0.009	0.000 $\pm$ 0.000	0.694
delayed	risk-averse	0.000 $\pm$ 0.000	0.096 $\pm$ 0.018	0.182 $\pm$ 0.012	0.023 $\pm$ 0.034	0.471

Table 7 continued

Split	Method	Success	Edge F1	GEEdit	Safety	Prog.
distractor	risk-averse	0.000 $\pm$ 0.000	0.250 $\pm$ 0.068	0.173 $\pm$ 0.016	0.000 $\pm$ 0.000	0.256
wall-jam	risk-averse	0.000 $\pm$ 0.000	0.289 $\pm$ 0.076	0.182 $\pm$ 0.028	0.016 $\pm$ 0.031	0.304
fixture	risk-averse	0.000 $\pm$ 0.000	0.610 $\pm$ 0.043	0.117 $\pm$ 0.015	0.188 $\pm$ 0.081	0.269
friction	risk-averse	0.146 $\pm$ 0.096	0.381 $\pm$ 0.121	0.148 $\pm$ 0.013	0.000 $\pm$ 0.000	0.641
nominal	risk-averse	0.271 $\pm$ 0.106	0.509 $\pm$ 0.092	0.166 $\pm$ 0.018	0.000 $\pm$ 0.000	0.706
pocket	risk-averse	0.188 $\pm$ 0.144	0.513 $\pm$ 0.104	0.162 $\pm$ 0.016	0.017 $\pm$ 0.034	0.713
actuator	robust-mpc	0.000 $\pm$ 0.000	0.834 $\pm$ 0.046	0.088 $\pm$ 0.029	0.706 $\pm$ 0.067	0.341
extreme	robust-mpc	0.000 $\pm$ 0.000	0.593 $\pm$ 0.071	0.182 $\pm$ 0.041	0.231 $\pm$ 0.066	0.066
combined	robust-mpc	0.083 $\pm$ 0.087	0.833 $\pm$ 0.029	0.081 $\pm$ 0.015	0.345 $\pm$ 0.087	0.570
chain	robust-mpc	0.042 $\pm$ 0.053	0.957 $\pm$ 0.016	0.014 $\pm$ 0.008	0.000 $\pm$ 0.000	0.381
noise	robust-mpc	0.229 $\pm$ 0.086	0.930 $\pm$ 0.019	0.033 $\pm$ 0.011	0.032 $\pm$ 0.041	0.767
delayed	robust-mpc	0.000 $\pm$ 0.000	0.864 $\pm$ 0.025	0.085 $\pm$ 0.018	0.812 $\pm$ 0.002	0.074
distractor	robust-mpc	0.062 $\pm$ 0.060	0.876 $\pm$ 0.045	0.046 $\pm$ 0.017	0.000 $\pm$ 0.000	0.342
wall-jam	robust-mpc	0.000 $\pm$ 0.000	0.950 $\pm$ 0.020	0.007 $\pm$ 0.003	0.000 $\pm$ 0.000	0.000
fixture	robust-mpc	0.000 $\pm$ 0.000	0.667 $\pm$ 0.058	0.170 $\pm$ 0.042	0.328 $\pm$ 0.091	0.081
friction	robust-mpc	0.208 $\pm$ 0.082	0.930 $\pm$ 0.022	0.029 $\pm$ 0.014	0.000 $\pm$ 0.000	0.716
nominal	robust-mpc	0.250 $\pm$ 0.087	0.961 $\pm$ 0.036	0.008 $\pm$ 0.008	0.000 $\pm$ 0.000	0.806
pocket	robust-mpc	0.146 $\pm$ 0.074	0.943 $\pm$ 0.029	0.017 $\pm$ 0.008	0.017 $\pm$ 0.033	0.749
actuator	state-dyn	0.021 $\pm$ 0.041	0.917 $\pm$ 0.024	0.044 $\pm$ 0.014	0.704 $\pm$ 0.043	0.357
extreme	state-dyn	0.000 $\pm$ 0.000	0.706 $\pm$ 0.041	0.131 $\pm$ 0.027	0.253 $\pm$ 0.054	0.081
combined	state-dyn	0.083 $\pm$ 0.087	0.891 $\pm$ 0.018	0.060 $\pm$ 0.010	0.319 $\pm$ 0.090	0.605
chain	state-dyn	0.042 $\pm$ 0.053	0.987 $\pm$ 0.007	0.007 $\pm$ 0.005	0.000 $\pm$ 0.000	0.395
noise	state-dyn	0.208 $\pm$ 0.082	0.944 $\pm$ 0.017	0.039 $\pm$ 0.014	0.044 $\pm$ 0.059	0.766
delayed	state-dyn	0.000 $\pm$ 0.000	0.993 $\pm$ 0.003	0.003 $\pm$ 0.001	0.812 $\pm$ 0.002	0.060
distractor	state-dyn	0.062 $\pm$ 0.060	0.925 $\pm$ 0.034	0.029 $\pm$ 0.013	0.000 $\pm$ 0.000	0.342
wall-jam	state-dyn	0.000 $\pm$ 0.000	0.955 $\pm$ 0.019	0.007 $\pm$ 0.003	0.000 $\pm$ 0.000	0.000
fixture	state-dyn	0.000 $\pm$ 0.000	0.738 $\pm$ 0.034	0.121 $\pm$ 0.021	0.326 $\pm$ 0.094	0.080
friction	state-dyn	0.208 $\pm$ 0.082	0.973 $\pm$ 0.009	0.014 $\pm$ 0.006	0.000 $\pm$ 0.000	0.702
nominal	state-dyn	0.250 $\pm$ 0.087	0.882 $\pm$ 0.017	0.090 $\pm$ 0.012	0.000 $\pm$ 0.000	0.804
pocket	state-dyn	0.146 $\pm$ 0.074	0.950 $\pm$ 0.019	0.036 $\pm$ 0.017	0.017 $\pm$ 0.033	0.749
actuator	no-memory	0.021 $\pm$ 0.041	0.618 $\pm$ 0.050	0.164 $\pm$ 0.026	0.705 $\pm$ 0.067	0.344
extreme	no-memory	0.000 $\pm$ 0.000	0.726 $\pm$ 0.065	0.083 $\pm$ 0.018	0.235 $\pm$ 0.078	0.222
combined	no-memory	0.083 $\pm$ 0.087	0.751 $\pm$ 0.057	0.115 $\pm$ 0.023	0.335 $\pm$ 0.100	0.564
chain	no-memory	0.042 $\pm$ 0.053	0.898 $\pm$ 0.022	0.049 $\pm$ 0.013	0.000 $\pm$ 0.000	0.365
noise	no-memory	0.271 $\pm$ 0.106	0.659 $\pm$ 0.043	0.172 $\pm$ 0.015	0.032 $\pm$ 0.041	0.740
delayed	no-memory	0.000 $\pm$ 0.000	0.508 $\pm$ 0.040	0.222 $\pm$ 0.014	0.811 $\pm$ 0.003	0.100
distractor	no-memory	0.104 $\pm$ 0.086	0.832 $\pm$ 0.063	0.082 $\pm$ 0.029	0.000 $\pm$ 0.000	0.320
wall-jam	no-memory	0.000 $\pm$ 0.000	0.667 $\pm$ 0.129	0.070 $\pm$ 0.024	0.000 $\pm$ 0.000	0.000

Table 7 continued

Split	Method	Success	Edge F1	GEdit	Safety	Prog.
fixture	no-memory	0.000 $\pm$ 0.000	0.764 $\pm$ 0.057	0.086 $\pm$ 0.019	0.329 $\pm$ 0.093	0.115
friction	no-memory	0.229 $\pm$ 0.086	0.759 $\pm$ 0.037	0.145 $\pm$ 0.016	0.000 $\pm$ 0.000	0.700
nominal	no-memory	0.271 $\pm$ 0.106	0.679 $\pm$ 0.075	0.148 $\pm$ 0.017	0.000 $\pm$ 0.000	0.770
pocket	no-memory	0.229 $\pm$ 0.137	0.638 $\pm$ 0.069	0.200 $\pm$ 0.021	0.017 $\pm$ 0.033	0.734
actuator	topo-v4	0.000 $\pm$ 0.000	0.586 $\pm$ 0.075	0.153 $\pm$ 0.023	0.706 $\pm$ 0.067	0.294
extreme	topo-v4	0.000 $\pm$ 0.000	0.572 $\pm$ 0.093	0.119 $\pm$ 0.025	0.082 $\pm$ 0.046	0.083
combined	topo-v4	0.062 $\pm$ 0.060	0.699 $\pm$ 0.067	0.114 $\pm$ 0.021	0.322 $\pm$ 0.091	0.554
chain	topo-v4	0.000 $\pm$ 0.000	0.855 $\pm$ 0.063	0.052 $\pm$ 0.021	0.000 $\pm$ 0.000	0.358
noise	topo-v4	0.229 $\pm$ 0.060	0.768 $\pm$ 0.034	0.118 $\pm$ 0.013	0.032 $\pm$ 0.041	0.756
delayed	topo-v4	0.000 $\pm$ 0.000	0.216 $\pm$ 0.060	0.186 $\pm$ 0.009	0.597 $\pm$ 0.095	0.314
distractor	topo-v4	0.083 $\pm$ 0.062	0.412 $\pm$ 0.103	0.154 $\pm$ 0.031	0.000 $\pm$ 0.000	0.314
wall-jam	topo-v4	0.000 $\pm$ 0.000	0.673 $\pm$ 0.128	0.073 $\pm$ 0.027	0.000 $\pm$ 0.000	0.000
fixture	topo-v4	0.000 $\pm$ 0.000	0.771 $\pm$ 0.049	0.077 $\pm$ 0.015	0.230 $\pm$ 0.066	0.121
friction	topo-v4	0.125 $\pm$ 0.082	0.615 $\pm$ 0.066	0.117 $\pm$ 0.012	0.000 $\pm$ 0.000	0.638
nominal	topo-v4	0.271 $\pm$ 0.122	0.685 $\pm$ 0.078	0.141 $\pm$ 0.011	0.000 $\pm$ 0.000	0.790
pocket	topo-v4	0.312 $\pm$ 0.074	0.585 $\pm$ 0.044	0.178 $\pm$ 0.023	0.017 $\pm$ 0.033	0.725
actuator	topo-v5	0.000 $\pm$ 0.000	0.557 $\pm$ 0.104	0.159 $\pm$ 0.029	0.691 $\pm$ 0.080	0.307
extreme	topo-v5	0.000 $\pm$ 0.000	0.571 $\pm$ 0.095	0.115 $\pm$ 0.026	0.060 $\pm$ 0.040	0.100
combined	topo-v5	0.042 $\pm$ 0.053	0.663 $\pm$ 0.068	0.121 $\pm$ 0.018	0.331 $\pm$ 0.107	0.554
chain	topo-v5	0.021 $\pm$ 0.041	0.875 $\pm$ 0.050	0.046 $\pm$ 0.016	0.000 $\pm$ 0.000	0.341
noise	topo-v5	0.208 $\pm$ 0.102	0.696 $\pm$ 0.039	0.138 $\pm$ 0.011	0.017 $\pm$ 0.033	0.738
delayed	topo-v5	0.000 $\pm$ 0.000	0.179 $\pm$ 0.041	0.185 $\pm$ 0.009	0.576 $\pm$ 0.078	0.334
distractor	topo-v5	0.062 $\pm$ 0.086	0.361 $\pm$ 0.094	0.167 $\pm$ 0.031	0.000 $\pm$ 0.000	0.311
wall-jam	topo-v5	0.000 $\pm$ 0.000	0.683 $\pm$ 0.126	0.071 $\pm$ 0.024	0.000 $\pm$ 0.000	0.000
fixture	topo-v5	0.000 $\pm$ 0.000	0.756 $\pm$ 0.034	0.077 $\pm$ 0.011	0.206 $\pm$ 0.081	0.097
friction	topo-v5	0.125 $\pm$ 0.082	0.586 $\pm$ 0.076	0.122 $\pm$ 0.012	0.000 $\pm$ 0.000	0.609
nominal	topo-v5	0.188 $\pm$ 0.096	0.620 $\pm$ 0.088	0.150 $\pm$ 0.013	0.000 $\pm$ 0.000	0.755
pocket	topo-v5	0.188 $\pm$ 0.096	0.590 $\pm$ 0.066	0.157 $\pm$ 0.015	0.017 $\pm$ 0.033	0.692

B AGGREGATE PAIRWISE COMPARISONS

Table 8: Aggregate topology_world_model.v5 minus comparison method by group.

Group	Comparison	Succ Diff	Edge Diff	GEdit Red.	Safety Red.
all_splits	conformal	-0.014	-0.024	-0.004	0.009
all_splits	ci-mpc	-0.017	-0.281	-0.074	0.047
all_splits	dist-graph	-0.014	-0.283	-0.075	0.045
all_splits	ensemble	-0.024	-0.120	-0.017	0.047
all_splits	hgb-topo	-0.014	-0.210	-0.038	0.046
all_splits	persist	-0.016	-0.318	-0.098	0.045
all_splits	oracle	-0.075	-0.405	-0.126	-0.158
all_splits	pairwise	-0.038	-0.115	0.003	0.051
all_splits	rf-topo	-0.016	0.108	0.027	0.048
all_splits	risk-averse	-0.014	0.195	0.027	-0.089
all_splits	robust-mpc	-0.016	-0.267	-0.062	0.048
all_splits	state-dyn	-0.016	-0.310	-0.077	0.048
all_splits	no-memory	-0.035	-0.114	0.002	0.047
all_splits	topo-v4	-0.021	-0.025	-0.002	0.007
combined_and_extreme	conformal	-0.021	-0.007	-0.004	-0.011
combined_and_extreme	ci-mpc	-0.021	-0.129	-0.006	0.080
combined_and_extreme	dist-graph	-0.021	-0.135	-0.010	0.081
combined_and_extreme	ensemble	-0.010	-0.142	-0.035	0.076
combined_and_extreme	hgb-topo	-0.021	-0.246	-0.061	0.075
combined_and_extreme	persist	-0.021	-0.227	-0.064	0.077
combined_and_extreme	oracle	-0.125	-0.383	-0.118	-0.196
combined_and_extreme	pairwise	-0.031	-0.127	-0.012	0.111
combined_and_extreme	rf-topo	-0.021	0.127	0.009	0.089
combined_and_extreme	risk-averse	-0.010	0.230	0.029	-0.113
combined_and_extreme	robust-mpc	-0.021	-0.096	0.014	0.093
combined_and_extreme	state-dyn	-0.021	-0.182	-0.022	0.091
combined_and_extreme	no-memory	-0.021	-0.122	-0.019	0.089
combined_and_extreme	topo-v4	-0.010	-0.019	-0.001	0.007
hard_splits	conformal	-0.013	-0.025	-0.004	0.010
hard_splits	ci-mpc	-0.013	-0.276	-0.067	0.051
hard_splits	dist-graph	-0.009	-0.278	-0.068	0.049
hard_splits	ensemble	-0.021	-0.133	-0.022	0.051
hard_splits	hgb-topo	-0.009	-0.212	-0.039	0.050
hard_splits	persist	-0.011	-0.317	-0.094	0.050
hard_splits	oracle	-0.081	-0.408	-0.124	-0.173
hard_splits	pairwise	-0.036	-0.120	0.004	0.055
hard_splits	rf-topo	-0.011	0.108	0.023	0.052
hard_splits	risk-averse	-0.008	0.202	0.028	-0.097

Table 8 continued

Group	Comparison	Succ Diff	Edge Diff	GEdit Red.	Safety Red.
hard_splits	robust-mpc	-0.011	-0.260	-0.055	0.052
hard_splits	state-dyn	-0.011	-0.315	-0.079	0.052
hard_splits	no-memory	-0.030	-0.119	0.003	0.051
hard_splits	topo-v4	-0.015	-0.022	-0.002	0.008

C PAIRED SEED COMPARISONS

Table 9: Per-split paired seed comparisons against topology_world_model.v5.

Split	Comparison	Succ Diff	Edge Diff	GEEdit Red.	Safety Red.	Better Seeds
actuator	conformal	0.000 $\pm$ 0.000	-0.012	-0.009	-0.000	0/8
actuator	ci-mpc	0.000 $\pm$ 0.000	-0.313	-0.095	0.046	0/8
actuator	dist-graph	0.000 $\pm$ 0.000	-0.306	-0.092	0.013	0/8
actuator	ensemble	0.000 $\pm$ 0.000	-0.117	-0.038	0.060	0/8
actuator	hgb-topo	0.000 $\pm$ 0.000	-0.158	-0.053	0.032	0/8
actuator	persist	0.000 $\pm$ 0.000	-0.332	-0.119	0.031	0/8
actuator	oracle	-0.188 $\pm$ 0.074	-0.443	-0.159	-0.691	0/8
actuator	pairwise	0.000 $\pm$ 0.000	-0.077	-0.003	0.031	0/8
actuator	rf-topo	0.000 $\pm$ 0.000	0.123	0.020	0.030	0/8
actuator	risk-averse	-0.021 $\pm$ 0.041	0.191	-0.004	-0.267	0/8
actuator	robust-mpc	0.000 $\pm$ 0.000	-0.277	-0.071	0.014	0/8
actuator	state-dyn	-0.021 $\pm$ 0.041	-0.360	-0.116	0.012	0/8
actuator	no-memory	-0.021 $\pm$ 0.041	-0.062	0.004	0.014	0/8
actuator	topo-v4	0.000 $\pm$ 0.000	-0.029	-0.006	0.014	0/8
extreme	conformal	-0.021 $\pm$ 0.041	0.017	0.006	0.006	0/8
extreme	ci-mpc	0.000 $\pm$ 0.000	-0.065	0.046	0.184	0/8
extreme	dist-graph	0.000 $\pm$ 0.000	-0.070	0.039	0.184	0/8
extreme	ensemble	0.000 $\pm$ 0.000	-0.215	-0.061	0.059	0/8
extreme	hgb-topo	0.000 $\pm$ 0.000	-0.346	-0.089	0.167	0/8
extreme	persist	0.000 $\pm$ 0.000	-0.263	-0.062	0.150	0/8
extreme	oracle	-0.083 $\pm$ 0.123	-0.429	-0.115	-0.060	0/8
extreme	pairwise	0.000 $\pm$ 0.000	-0.179	-0.027	0.216	0/8
extreme	rf-topo	0.000 $\pm$ 0.000	-0.000	-0.036	0.174	0/8
extreme	risk-averse	0.000 $\pm$ 0.000	0.165	0.024	-0.060	0/8
extreme	robust-mpc	0.000 $\pm$ 0.000	-0.022	0.067	0.171	0/8
extreme	state-dyn	0.000 $\pm$ 0.000	-0.135	0.017	0.194	0/8
extreme	no-memory	0.000 $\pm$ 0.000	-0.155	-0.032	0.175	0/8
extreme	topo-v4	0.000 $\pm$ 0.000	-0.001	0.004	0.022	0/8
combined	conformal	-0.021 $\pm$ 0.041	-0.032	-0.015	-0.028	0/8
combined	ci-mpc	-0.042 $\pm$ 0.053	-0.194	-0.058	-0.025	0/8
combined	dist-graph	-0.042 $\pm$ 0.053	-0.201	-0.060	-0.022	0/8
combined	ensemble	-0.021 $\pm$ 0.041	-0.069	-0.009	0.092	0/8
combined	hgb-topo	-0.042 $\pm$ 0.053	-0.146	-0.033	-0.017	0/8
combined	persist	-0.042 $\pm$ 0.053	-0.190	-0.066	0.004	0/8
combined	oracle	-0.167 $\pm$ 0.062	-0.337	-0.121	-0.331	0/8
combined	pairwise	-0.062 $\pm$ 0.060	-0.076	0.002	0.006	0/8
combined	rf-topo	-0.042 $\pm$ 0.053	0.254	0.054	0.005	0/8
combined	risk-averse	-0.021 $\pm$ 0.041	0.295	0.034	-0.165	0/8

Table 9 continued

Split	Comparison	Succ Diff	Edge Diff	GEEdit Red.	Safety Red.	Better Seeds
combined	robust-mpc	-0.042 $\pm$ 0.053	-0.170	-0.039	0.014	0/8
combined	state-dyn	-0.042 $\pm$ 0.053	-0.229	-0.060	-0.012	0/8
combined	no-memory	-0.042 $\pm$ 0.053	-0.088	-0.006	0.003	0/8
combined	topo-v4	-0.021 $\pm$ 0.041	-0.036	-0.006	-0.009	0/8
chain	conformal	-0.042 $\pm$ 0.082	0.005	0.003	0.000	1/8
chain	ci-mpc	-0.021 $\pm$ 0.074	-0.070	-0.025	0.000	1/8
chain	dist-graph	-0.021 $\pm$ 0.074	-0.069	-0.025	0.000	1/8
chain	ensemble	-0.021 $\pm$ 0.074	0.019	0.024	0.000	1/8
chain	hgb-topo	-0.021 $\pm$ 0.074	-0.009	0.012	0.000	1/8
chain	persist	-0.021 $\pm$ 0.074	-0.082	-0.035	0.000	1/8
chain	oracle	-0.104 $\pm$ 0.086	-0.125	-0.046	0.000	1/8
chain	pairwise	-0.021 $\pm$ 0.074	-0.018	0.007	0.000	1/8
chain	rf-topo	-0.021 $\pm$ 0.074	0.619	0.096	0.000	1/8
chain	risk-averse	-0.021 $\pm$ 0.074	0.394	0.055	0.000	1/8
chain	robust-mpc	-0.021 $\pm$ 0.074	-0.083	-0.032	0.000	1/8
chain	state-dyn	-0.021 $\pm$ 0.074	-0.112	-0.039	0.000	1/8
chain	no-memory	-0.021 $\pm$ 0.074	-0.023	0.003	0.000	1/8
chain	topo-v4	0.021 $\pm$ 0.041	0.020	0.007	0.000	1/8
noise	conformal	0.021 $\pm$ 0.074	-0.050	-0.017	0.015	2/8
noise	ci-mpc	-0.042 $\pm$ 0.053	-0.243	-0.114	0.000	0/8
noise	dist-graph	-0.000 $\pm$ 0.062	-0.246	-0.118	0.015	1/8
noise	ensemble	-0.021 $\pm$ 0.074	0.048	0.033	0.012	1/8
noise	hgb-topo	-0.000 $\pm$ 0.062	-0.031	0.001	0.015	1/8
noise	persist	-0.021 $\pm$ 0.041	-0.221	-0.105	0.000	0/8
noise	oracle	-0.000 $\pm$ 0.138	-0.304	-0.138	-0.017	2/8
noise	pairwise	-0.167 $\pm$ 0.123	0.025	0.037	0.000	1/8
noise	rf-topo	-0.000 $\pm$ 0.062	0.118	0.048	0.015	1/8
noise	risk-averse	-0.062 $\pm$ 0.086	0.165	0.021	-0.017	1/8
noise	robust-mpc	-0.021 $\pm$ 0.074	-0.235	-0.105	0.015	1/8
noise	state-dyn	-0.000 $\pm$ 0.062	-0.248	-0.099	0.027	1/8
noise	no-memory	-0.062 $\pm$ 0.106	0.036	0.034	0.015	1/8
noise	topo-v4	-0.021 $\pm$ 0.074	-0.073	-0.020	0.015	1/8
delayed	conformal	0.000 $\pm$ 0.000	-0.055	-0.000	0.076	0/8
delayed	ci-mpc	0.000 $\pm$ 0.000	-0.733	-0.138	0.237	0/8
delayed	dist-graph	0.000 $\pm$ 0.000	-0.731	-0.137	0.237	0/8
delayed	ensemble	0.000 $\pm$ 0.000	-0.421	-0.050	0.186	0/8
delayed	hgb-topo	0.000 $\pm$ 0.000	-0.510	-0.068	0.237	0/8
delayed	persist	0.000 $\pm$ 0.000	-0.769	-0.168	0.237	0/8

Table 9 continued

Split	Comparison	Succ Diff	Edge Diff	GEEdit Red.	Safety Red.	Better Seeds
delayed	oracle	-0.021 ± 0.041	-0.821	-0.185	-0.576	0/8
delayed	pairwise	0.000 ± 0.000	-0.333	0.038	0.236	0/8
delayed	rf-topo	0.000 ± 0.000	0.012	0.046	0.219	0/8
delayed	risk-averse	0.000 ± 0.000	0.083	-0.004	-0.552	0/8
delayed	robust-mpc	0.000 ± 0.000	-0.685	-0.101	0.236	0/8
delayed	state-dyn	0.000 ± 0.000	-0.814	-0.182	0.237	0/8
delayed	no-memory	0.000 ± 0.000	-0.329	0.037	0.236	0/8
delayed	topo-v4	0.000 ± 0.000	-0.037	0.001	0.021	0/8
distractor	conformal	-0.021 ± 0.096	-0.090	-0.012	0.000	1/8
distractor	ci-mpc	-0.000 ± 0.062	-0.514	-0.121	0.000	1/8
distractor	dist-graph	-0.000 ± 0.062	-0.516	-0.121	0.000	1/8
distractor	ensemble	-0.104 ± 0.106	-0.383	-0.073	0.000	0/8
distractor	hgb-topo	-0.000 ± 0.062	-0.475	-0.085	0.000	1/8
distractor	persist	-0.000 ± 0.062	-0.574	-0.148	0.000	1/8
distractor	oracle	-0.062 ± 0.106	-0.639	-0.167	0.000	1/8
distractor	pairwise	-0.021 ± 0.074	-0.477	-0.086	0.000	1/8
distractor	rf-topo	-0.021 ± 0.074	-0.382	-0.066	0.000	1/8
distractor	risk-averse	0.062 ± 0.086	0.110	0.006	0.000	2/8
distractor	robust-mpc	-0.000 ± 0.062	-0.515	-0.121	0.000	1/8
distractor	state-dyn	-0.000 ± 0.062	-0.564	-0.138	0.000	1/8
distractor	no-memory	-0.042 ± 0.135	-0.471	-0.086	0.000	1/8
distractor	topo-v4	-0.021 ± 0.074	-0.052	-0.014	0.000	1/8
wall-jam	conformal	0.000 ± 0.000	-0.008	-0.001	0.000	0/8
wall-jam	ci-mpc	0.000 ± 0.000	-0.273	-0.064	0.000	0/8
wall-jam	dist-graph	0.000 ± 0.000	-0.275	-0.064	0.000	0/8
wall-jam	ensemble	0.000 ± 0.000	-0.149	-0.039	0.000	0/8
wall-jam	hgb-topo	0.000 ± 0.000	-0.206	-0.057	0.000	0/8
wall-jam	persist	0.000 ± 0.000	-0.241	-0.057	0.000	0/8
wall-jam	oracle	-0.062 ± 0.060	-0.317	-0.071	0.000	0/8
wall-jam	pairwise	0.000 ± 0.000	0.057	0.009	0.000	0/8
wall-jam	rf-topo	0.000 ± 0.000	-0.176	-0.052	0.000	0/8
wall-jam	risk-averse	0.000 ± 0.000	0.394	0.111	0.016	0/8
wall-jam	robust-mpc	0.000 ± 0.000	-0.267	-0.064	0.000	0/8
wall-jam	state-dyn	0.000 ± 0.000	-0.272	-0.064	0.000	0/8
wall-jam	no-memory	0.000 ± 0.000	0.016	-0.000	0.000	0/8
wall-jam	topo-v4	0.000 ± 0.000	0.010	0.002	0.000	0/8
fixture	conformal	-0.021 ± 0.041	0.008	0.007	0.036	0/8
fixture	ci-mpc	0.000 ± 0.000	0.056	0.063	0.121	0/8

Table 9 continued

Split	Comparison	Succ Diff	Edge Diff	GEEdit Red.	Safety Red.	Better Seeds
fixture	dist-graph	0.000 ± 0.000	0.050	0.058	0.111	0/8
fixture	ensemble	-0.021 ± 0.041	-0.079	-0.011	0.136	0/8
fixture	hgb-topo	0.000 ± 0.000	-0.156	-0.045	0.119	0/8
fixture	persist	0.000 ± 0.000	-0.117	-0.036	0.123	0/8
fixture	oracle	-0.104 ± 0.086	-0.244	-0.077	-0.206	0/8
fixture	pairwise	0.000 ± 0.000	-0.026	0.002	0.119	0/8
fixture	rf-topo	0.000 ± 0.000	0.320	0.026	0.133	0/8
fixture	risk-averse	0.000 ± 0.000	0.146	0.039	-0.019	0/8
fixture	robust-mpc	0.000 ± 0.000	0.089	0.092	0.122	0/8
fixture	state-dyn	0.000 ± 0.000	0.018	0.043	0.120	0/8
fixture	no-memory	0.000 ± 0.000	-0.008	0.008	0.123	0/8
fixture	topo-v4	0.000 ± 0.000	-0.015	-0.001	0.024	0/8
friction	conformal	0.021 ± 0.041	-0.037	-0.009	0.000	1/8
friction	ci-mpc	-0.083 ± 0.062	-0.338	-0.094	0.000	0/8
friction	dist-graph	-0.083 ± 0.062	-0.336	-0.093	0.000	0/8
friction	ensemble	-0.125 ± 0.135	-0.122	0.002	0.000	1/8
friction	hgb-topo	-0.083 ± 0.062	-0.190	-0.010	0.000	0/8
friction	persist	-0.083 ± 0.062	-0.347	-0.102	0.000	0/8
friction	oracle	-0.229 ± 0.122	-0.414	-0.122	0.000	0/8
friction	pairwise	-0.083 ± 0.107	-0.161	0.023	0.000	1/8
friction	rf-topo	-0.083 ± 0.062	0.257	0.055	0.000	0/8
friction	risk-averse	-0.021 ± 0.096	0.205	0.026	0.000	1/8
friction	robust-mpc	-0.083 ± 0.062	-0.344	-0.093	0.000	0/8
friction	state-dyn	-0.083 ± 0.062	-0.387	-0.108	0.000	0/8
friction	no-memory	-0.104 ± 0.086	-0.173	0.023	0.000	0/8
friction	topo-v4	0.000 ± 0.062	-0.029	-0.005	0.000	1/8
nominal	conformal	-0.021 ± 0.041	-0.020	-0.004	0.000	0/8
nominal	ci-mpc	-0.062 ± 0.060	-0.334	-0.142	0.000	0/8
nominal	dist-graph	-0.062 ± 0.060	-0.339	-0.142	0.000	0/8
nominal	ensemble	-0.062 ± 0.086	0.024	0.035	0.000	0/8
nominal	hgb-topo	-0.062 ± 0.060	-0.188	-0.036	0.000	0/8
nominal	persist	-0.062 ± 0.060	-0.330	-0.136	0.000	0/8
nominal	oracle	0.000 ± 0.087	-0.380	-0.150	0.000	2/8
nominal	pairwise	-0.062 ± 0.086	-0.064	-0.001	0.000	1/8
nominal	rf-topo	-0.062 ± 0.060	0.106	0.077	0.000	0/8
nominal	risk-averse	-0.083 ± 0.123	0.111	0.016	0.000	1/8
nominal	robust-mpc	-0.062 ± 0.060	-0.341	-0.142	0.000	0/8
nominal	state-dyn	-0.062 ± 0.060	-0.262	-0.061	0.000	0/8

Table 9 continued

Split	Comparison	Succ Diff	Edge Diff	GEEdit Red.	Safety Red.	Better Seeds
nominal	no-memory	-0.083 ± 0.087	-0.059	-0.002	0.000	1/8
nominal	topo-v4	-0.083 ± 0.062	-0.066	-0.009	0.000	0/8
pocket	conformal	-0.062 ± 0.106	-0.017	0.005	0.000	1/8
pocket	ci-mpc	0.042 ± 0.082	-0.354	-0.140	0.000	3/8
pocket	dist-graph	0.042 ± 0.082	-0.356	-0.140	0.000	3/8
pocket	ensemble	0.083 ± 0.138	0.022	-0.017	0.017	4/8
pocket	hgb-topo	0.042 ± 0.082	-0.102	0.000	0.000	3/8
pocket	persist	0.042 ± 0.082	-0.354	-0.141	0.000	3/8
pocket	oracle	0.125 ± 0.120	-0.410	-0.157	-0.017	5/8
pocket	pairwise	-0.042 ± 0.120	-0.054	0.041	0.000	2/8
pocket	rf-topo	0.042 ± 0.082	0.047	0.059	0.000	3/8
pocket	risk-averse	-0.000 ± 0.123	0.077	0.004	0.000	3/8
pocket	robust-mpc	0.042 ± 0.082	-0.354	-0.141	0.000	3/8
pocket	state-dyn	0.042 ± 0.082	-0.360	-0.122	0.000	3/8
pocket	no-memory	-0.042 ± 0.120	-0.049	0.042	0.000	2/8
pocket	topo-v4	-0.125 ± 0.082	0.005	0.021	0.000	0/8

D FULL ABLATION METRICS

Table 10: Full ablation metrics by split and method.

Split	Method	Success	Edge F1	GEEdit	Safety	Jam F1
actuator	no-birth-death	0.062 ± 0.080	0.691 ± 0.048	0.142 ± 0.025	0.594 ± 0.128	0.821
extreme	no-birth-death	0.031 ± 0.061	0.643 ± 0.085	0.117 ± 0.020	0.141 ± 0.064	0.344
combined	no-birth-death	0.094 ± 0.090	0.766 ± 0.077	0.106 ± 0.034	0.210 ± 0.109	0.630
wall-jam	no-birth-death	0.000 ± 0.000	0.695 ± 0.123	0.069 ± 0.027	0.000 ± 0.000	0.781
actuator	no-component	0.062 ± 0.080	0.642 ± 0.082	0.137 ± 0.021	0.594 ± 0.127	0.781
extreme	no-component	0.000 ± 0.000	0.477 ± 0.027	0.137 ± 0.019	0.059 ± 0.066	0.562
combined	no-component	0.125 ± 0.093	0.750 ± 0.086	0.096 ± 0.030	0.208 ± 0.129	0.688
wall-jam	no-component	0.000 ± 0.000	0.751 ± 0.117	0.057 ± 0.025	0.000 ± 0.000	0.875
actuator	no-fixture-guard	0.062 ± 0.080	0.620 ± 0.079	0.139 ± 0.018	0.593 ± 0.127	0.812
extreme	no-fixture-guard	0.000 ± 0.000	0.527 ± 0.050	0.128 ± 0.021	0.105 ± 0.058	0.438
combined	no-fixture-guard	0.094 ± 0.129	0.701 ± 0.110	0.108 ± 0.039	0.229 ± 0.154	0.656
wall-jam	no-fixture-guard	0.000 ± 0.000	0.688 ± 0.108	0.070 ± 0.021	0.000 ± 0.000	0.875
actuator	no-memory	0.062 ± 0.080	0.601 ± 0.112	0.145 ± 0.025	0.569 ± 0.116	0.812
extreme	no-memory	0.000 ± 0.000	0.543 ± 0.071	0.126 ± 0.015	0.094 ± 0.076	0.625
combined	no-memory	0.094 ± 0.090	0.740 ± 0.084	0.101 ± 0.028	0.240 ± 0.132	0.661
wall-jam	no-memory	0.000 ± 0.000	0.708 ± 0.125	0.066 ± 0.027	0.000 ± 0.000	0.906
actuator	no-jam-slip	0.094 ± 0.090	0.648 ± 0.072	0.142 ± 0.023	0.594 ± 0.128	0.734
extreme	no-jam-slip	0.000 ± 0.000	0.488 ± 0.063	0.145 ± 0.025	0.069 ± 0.068	0.312
combined	no-jam-slip	0.062 ± 0.080	0.733 ± 0.094	0.093 ± 0.026	0.181 ± 0.119	0.598
wall-jam	no-jam-slip	0.000 ± 0.000	0.732 ± 0.137	0.062 ± 0.029	0.000 ± 0.000	0.781
actuator	no-tail-risk	0.062 ± 0.080	0.649 ± 0.057	0.140 ± 0.019	0.615 ± 0.116	0.812
extreme	no-tail-risk	0.000 ± 0.000	0.510 ± 0.078	0.134 ± 0.019	0.055 ± 0.041	0.531
combined	no-tail-risk	0.156 ± 0.129	0.780 ± 0.064	0.091 ± 0.025	0.240 ± 0.115	0.710
wall-jam	no-tail-risk	0.000 ± 0.000	0.734 ± 0.118	0.061 ± 0.026	0.000 ± 0.000	0.781
actuator	no-topo-plan	0.062 ± 0.080	0.634 ± 0.070	0.152 ± 0.023	0.574 ± 0.157	0.844
extreme	no-topo-plan	0.031 ± 0.061	0.655 ± 0.048	0.121 ± 0.021	0.083 ± 0.066	0.594
combined	no-topo-plan	0.125 ± 0.131	0.819 ± 0.043	0.087 ± 0.029	0.117 ± 0.068	0.656
wall-jam	no-topo-plan	0.000 ± 0.000	0.443 ± 0.106	0.125 ± 0.023	0.000 ± 0.000	0.781
actuator	no-transition	0.031 ± 0.061	0.638 ± 0.079	0.140 ± 0.022	0.594 ± 0.127	0.812
extreme	no-transition	0.000 ± 0.000	0.489 ± 0.042	0.142 ± 0.018	0.079 ± 0.065	0.500
combined	no-transition	0.094 ± 0.129	0.788 ± 0.070	0.087 ± 0.027	0.198 ± 0.100	0.688
wall-jam	no-transition	0.000 ± 0.000	0.752 ± 0.120	0.058 ± 0.026	0.000 ± 0.000	0.844
actuator	no-uncertainty	0.031 ± 0.061	0.660 ± 0.066	0.134 ± 0.021	0.594 ± 0.127	0.812
extreme	no-uncertainty	0.000 ± 0.000	0.564 ± 0.094	0.134 ± 0.021	0.117 ± 0.094	0.438
combined	no-uncertainty	0.094 ± 0.090	0.703 ± 0.087	0.103 ± 0.029	0.223 ± 0.130	0.656
wall-jam	no-uncertainty	0.000 ± 0.000	0.735 ± 0.121	0.060 ± 0.026	0.000 ± 0.000	0.906
actuator	learned-only	0.094 ± 0.090	0.679 ± 0.044	0.149 ± 0.020	0.615 ± 0.116	0.758
extreme	learned-only	0.031 ± 0.061	0.734 ± 0.070	0.102 ± 0.023	0.152 ± 0.097	0.375

Table 10 continued

Split	Method	Success	Edge F1	GEEdit	Safety	Jam F1
combined	learned-only	0.125 ± 0.131	0.754 ± 0.075	0.108 ± 0.034	0.232 ± 0.134	0.704
wall-jam	learned-only	0.000 ± 0.000	0.751 ± 0.111	0.055 ± 0.021	0.000 ± 0.000	0.844
actuator	full-v5	0.031 ± 0.061	0.637 ± 0.078	0.134 ± 0.023	0.618 ± 0.118	0.812
extreme	full-v5	0.000 ± 0.000	0.513 ± 0.068	0.136 ± 0.026	0.054 ± 0.040	0.594
combined	full-v5	0.094 ± 0.090	0.720 ± 0.108	0.099 ± 0.036	0.199 ± 0.140	0.689
wall-jam	full-v5	0.000 ± 0.000	0.704 ± 0.099	0.069 ± 0.020	0.000 ± 0.000	0.906
actuator	topo-v4	0.062 ± 0.080	0.642 ± 0.074	0.143 ± 0.021	0.594 ± 0.127	0.812
extreme	topo-v4	0.000 ± 0.000	0.559 ± 0.059	0.125 ± 0.020	0.133 ± 0.084	0.438
combined	topo-v4	0.125 ± 0.131	0.734 ± 0.085	0.098 ± 0.030	0.229 ± 0.154	0.661
wall-jam	topo-v4	0.000 ± 0.000	0.742 ± 0.096	0.058 ± 0.022	0.000 ± 0.000	0.844

E FULL STRESS SWEEP

Table 11: Stress sweep metrics by split, method, and stress level.

Split	Method	Level	Success	Edge F1	Safety	Jam F1
extreme	conformal	0.00	0.125 $\pm$ 0.172	0.682 $\pm$ 0.106	0.000 $\pm$ 0.000	0.044
extreme	conformal	0.25	0.083 $\pm$ 0.107	0.792 $\pm$ 0.097	0.031 $\pm$ 0.061	0.417
extreme	conformal	0.50	0.042 $\pm$ 0.082	0.820 $\pm$ 0.054	0.219 $\pm$ 0.150	0.792
extreme	conformal	0.75	0.000 $\pm$ 0.000	0.672 $\pm$ 0.105	0.318 $\pm$ 0.131	0.667
extreme	conformal	1.00	0.000 $\pm$ 0.000	0.455 $\pm$ 0.112	0.101 $\pm$ 0.091	0.792
combined	conformal	0.00	0.125 $\pm$ 0.120	0.631 $\pm$ 0.098	0.000 $\pm$ 0.000	0.299
combined	conformal	0.25	0.167 $\pm$ 0.175	0.795 $\pm$ 0.108	0.000 $\pm$ 0.000	0.333
combined	conformal	0.50	0.042 $\pm$ 0.082	0.770 $\pm$ 0.081	0.326 $\pm$ 0.126	0.792
combined	conformal	0.75	0.000 $\pm$ 0.000	0.713 $\pm$ 0.060	0.353 $\pm$ 0.126	0.750
combined	conformal	1.00	0.000 $\pm$ 0.000	0.427 $\pm$ 0.091	0.020 $\pm$ 0.039	0.625
fixture	conformal	0.00	0.167 $\pm$ 0.123	0.589 $\pm$ 0.126	0.000 $\pm$ 0.000	0.125
fixture	conformal	0.25	0.167 $\pm$ 0.175	0.832 $\pm$ 0.031	0.068 $\pm$ 0.087	0.461
fixture	conformal	0.50	0.083 $\pm$ 0.107	0.723 $\pm$ 0.092	0.494 $\pm$ 0.127	0.792
fixture	conformal	0.75	0.000 $\pm$ 0.000	0.748 $\pm$ 0.164	0.269 $\pm$ 0.115	0.917
fixture	conformal	1.00	0.000 $\pm$ 0.000	0.448 $\pm$ 0.113	0.068 $\pm$ 0.057	0.708
extreme	ci-mpc	0.00	0.125 $\pm$ 0.120	0.959 $\pm$ 0.019	0.000 $\pm$ 0.000	0.042
extreme	ci-mpc	0.25	0.125 $\pm$ 0.120	0.944 $\pm$ 0.024	0.000 $\pm$ 0.000	0.417
extreme	ci-mpc	0.50	0.000 $\pm$ 0.000	0.852 $\pm$ 0.043	0.254 $\pm$ 0.197	0.833
extreme	ci-mpc	0.75	0.000 $\pm$ 0.000	0.701 $\pm$ 0.055	0.440 $\pm$ 0.122	0.792
extreme	ci-mpc	1.00	0.000 $\pm$ 0.000	0.528 $\pm$ 0.087	0.205 $\pm$ 0.105	0.917
combined	ci-mpc	0.00	0.167 $\pm$ 0.214	0.958 $\pm$ 0.019	0.000 $\pm$ 0.000	0.292
combined	ci-mpc	0.25	0.208 $\pm$ 0.172	0.902 $\pm$ 0.029	0.000 $\pm$ 0.000	0.333
combined	ci-mpc	0.50	0.042 $\pm$ 0.082	0.859 $\pm$ 0.040	0.359 $\pm$ 0.165	0.792
combined	ci-mpc	0.75	0.000 $\pm$ 0.000	0.709 $\pm$ 0.043	0.387 $\pm$ 0.157	0.708
combined	ci-mpc	1.00	0.000 $\pm$ 0.000	0.511 $\pm$ 0.091	0.251 $\pm$ 0.119	0.958
fixture	ci-mpc	0.00	0.250 $\pm$ 0.163	0.971 $\pm$ 0.008	0.000 $\pm$ 0.000	0.125
fixture	ci-mpc	0.25	0.083 $\pm$ 0.107	0.951 $\pm$ 0.022	0.102 $\pm$ 0.097	0.500
fixture	ci-mpc	0.50	0.083 $\pm$ 0.107	0.857 $\pm$ 0.052	0.496 $\pm$ 0.127	0.875
fixture	ci-mpc	0.75	0.000 $\pm$ 0.000	0.634 $\pm$ 0.068	0.320 $\pm$ 0.143	0.875
fixture	ci-mpc	1.00	0.000 $\pm$ 0.000	0.529 $\pm$ 0.061	0.253 $\pm$ 0.094	0.792
extreme	ensemble	0.00	0.125 $\pm$ 0.172	0.580 $\pm$ 0.038	0.000 $\pm$ 0.000	0.042
extreme	ensemble	0.25	0.125 $\pm$ 0.120	0.714 $\pm$ 0.083	0.000 $\pm$ 0.000	0.417
extreme	ensemble	0.50	0.000 $\pm$ 0.000	0.829 $\pm$ 0.058	0.254 $\pm$ 0.197	0.875
extreme	ensemble	0.75	0.000 $\pm$ 0.000	0.827 $\pm$ 0.033	0.451 $\pm$ 0.112	0.750
extreme	ensemble	1.00	0.000 $\pm$ 0.000	0.760 $\pm$ 0.095	0.294 $\pm$ 0.138	0.875
combined	ensemble	0.00	0.167 $\pm$ 0.214	0.670 $\pm$ 0.083	0.000 $\pm$ 0.000	0.292
combined	ensemble	0.25	0.208 $\pm$ 0.172	0.690 $\pm$ 0.102	0.000 $\pm$ 0.000	0.333
combined	ensemble	0.50	0.042 $\pm$ 0.082	0.712 $\pm$ 0.084	0.326 $\pm$ 0.126	0.750

Table 11 continued

Split	Method	Level	Success	Edge F1	Safety	Jam F1
combined	ensemble	0.75	0.000 $\pm$ 0.000	0.780 $\pm$ 0.115	0.378 $\pm$ 0.111	1.000
combined	ensemble	1.00	0.000 $\pm$ 0.000	0.737 $\pm$ 0.081	0.211 $\pm$ 0.109	0.833
fixture	ensemble	0.00	0.208 $\pm$ 0.120	0.617 $\pm$ 0.061	0.028 $\pm$ 0.054	0.125
fixture	ensemble	0.25	0.083 $\pm$ 0.107	0.738 $\pm$ 0.084	0.068 $\pm$ 0.087	0.458
fixture	ensemble	0.50	0.083 $\pm$ 0.107	0.746 $\pm$ 0.048	0.497 $\pm$ 0.128	0.833
fixture	ensemble	0.75	0.000 $\pm$ 0.000	0.851 $\pm$ 0.082	0.310 $\pm$ 0.115	0.833
fixture	ensemble	1.00	0.000 $\pm$ 0.000	0.660 $\pm$ 0.112	0.220 $\pm$ 0.086	0.917
extreme	hgb-topo	0.00	0.125 $\pm$ 0.120	0.708 $\pm$ 0.054	0.000 $\pm$ 0.000	0.042
extreme	hgb-topo	0.25	0.125 $\pm$ 0.120	0.790 $\pm$ 0.077	0.000 $\pm$ 0.000	0.417
extreme	hgb-topo	0.50	0.000 $\pm$ 0.000	0.854 $\pm$ 0.054	0.253 $\pm$ 0.195	0.875
extreme	hgb-topo	0.75	0.000 $\pm$ 0.000	0.893 $\pm$ 0.033	0.438 $\pm$ 0.119	1.000
extreme	hgb-topo	1.00	0.000 $\pm$ 0.000	0.880 $\pm$ 0.017	0.253 $\pm$ 0.128	1.000
combined	hgb-topo	0.00	0.167 $\pm$ 0.214	0.825 $\pm$ 0.047	0.000 $\pm$ 0.000	0.292
combined	hgb-topo	0.25	0.208 $\pm$ 0.172	0.715 $\pm$ 0.101	0.000 $\pm$ 0.000	0.333
combined	hgb-topo	0.50	0.042 $\pm$ 0.082	0.770 $\pm$ 0.068	0.359 $\pm$ 0.165	0.833
combined	hgb-topo	0.75	0.000 $\pm$ 0.000	0.894 $\pm$ 0.032	0.380 $\pm$ 0.154	0.958
combined	hgb-topo	1.00	0.000 $\pm$ 0.000	0.866 $\pm$ 0.062	0.244 $\pm$ 0.120	1.000
fixture	hgb-topo	0.00	0.208 $\pm$ 0.120	0.792 $\pm$ 0.060	0.000 $\pm$ 0.000	0.167
fixture	hgb-topo	0.25	0.083 $\pm$ 0.107	0.768 $\pm$ 0.064	0.102 $\pm$ 0.097	0.500
fixture	hgb-topo	0.50	0.083 $\pm$ 0.107	0.827 $\pm$ 0.029	0.467 $\pm$ 0.115	0.875
fixture	hgb-topo	0.75	0.000 $\pm$ 0.000	0.916 $\pm$ 0.040	0.313 $\pm$ 0.145	1.000
fixture	hgb-topo	1.00	0.000 $\pm$ 0.000	0.914 $\pm$ 0.025	0.228 $\pm$ 0.072	1.000
extreme	oracle	0.00	0.167 $\pm$ 0.123	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.417
extreme	oracle	0.25	0.167 $\pm$ 0.175	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.375
extreme	oracle	0.50	0.083 $\pm$ 0.107	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.417
extreme	oracle	0.75	0.083 $\pm$ 0.107	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.792
extreme	oracle	1.00	0.125 $\pm$ 0.120	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.736
combined	oracle	0.00	0.292 $\pm$ 0.148	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.375
combined	oracle	0.25	0.333 $\pm$ 0.175	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.250
combined	oracle	0.50	0.208 $\pm$ 0.172	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.375
combined	oracle	0.75	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.708
combined	oracle	1.00	0.083 $\pm$ 0.107	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.917
fixture	oracle	0.00	0.208 $\pm$ 0.120	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.458
fixture	oracle	0.25	0.208 $\pm$ 0.172	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.417
fixture	oracle	0.50	0.125 $\pm$ 0.120	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.500
fixture	oracle	0.75	0.083 $\pm$ 0.107	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.583
fixture	oracle	1.00	0.042 $\pm$ 0.082	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.833
extreme	pairwise	0.00	0.208 $\pm$ 0.120	0.673 $\pm$ 0.113	0.000 $\pm$ 0.000	0.348

Table 11 continued

Split	Method	Level	Success	Edge F1	Safety	Jam F1
extreme	pairwise	0.25	0.125 $\pm$ 0.120	0.794 $\pm$ 0.095	0.000 $\pm$ 0.000	0.545
extreme	pairwise	0.50	0.000 $\pm$ 0.000	0.784 $\pm$ 0.048	0.254 $\pm$ 0.197	0.836
extreme	pairwise	0.75	0.000 $\pm$ 0.000	0.781 $\pm$ 0.080	0.439 $\pm$ 0.124	0.667
extreme	pairwise	1.00	0.000 $\pm$ 0.000	0.600 $\pm$ 0.106	0.270 $\pm$ 0.108	0.417
combined	pairwise	0.00	0.167 $\pm$ 0.214	0.789 $\pm$ 0.050	0.000 $\pm$ 0.000	0.380
combined	pairwise	0.25	0.292 $\pm$ 0.148	0.721 $\pm$ 0.108	0.000 $\pm$ 0.000	0.468
combined	pairwise	0.50	0.167 $\pm$ 0.123	0.733 $\pm$ 0.072	0.293 $\pm$ 0.149	0.853
combined	pairwise	0.75	0.000 $\pm$ 0.000	0.805 $\pm$ 0.049	0.343 $\pm$ 0.138	0.635
combined	pairwise	1.00	0.000 $\pm$ 0.000	0.609 $\pm$ 0.081	0.251 $\pm$ 0.094	0.542
fixture	pairwise	0.00	0.250 $\pm$ 0.163	0.677 $\pm$ 0.111	0.000 $\pm$ 0.000	0.253
fixture	pairwise	0.25	0.208 $\pm$ 0.120	0.782 $\pm$ 0.050	0.068 $\pm$ 0.087	0.679
fixture	pairwise	0.50	0.083 $\pm$ 0.107	0.768 $\pm$ 0.035	0.494 $\pm$ 0.127	0.789
fixture	pairwise	0.75	0.000 $\pm$ 0.000	0.819 $\pm$ 0.095	0.333 $\pm$ 0.142	0.875
fixture	pairwise	1.00	0.000 $\pm$ 0.000	0.684 $\pm$ 0.095	0.253 $\pm$ 0.076	0.583
extreme	rf-topo	0.00	0.125 $\pm$ 0.120	0.481 $\pm$ 0.040	0.000 $\pm$ 0.000	0.052
extreme	rf-topo	0.25	0.125 $\pm$ 0.120	0.436 $\pm$ 0.076	0.000 $\pm$ 0.000	0.433
extreme	rf-topo	0.50	0.000 $\pm$ 0.000	0.497 $\pm$ 0.112	0.254 $\pm$ 0.197	0.885
extreme	rf-topo	0.75	0.000 $\pm$ 0.000	0.560 $\pm$ 0.084	0.444 $\pm$ 0.117	0.875
extreme	rf-topo	1.00	0.000 $\pm$ 0.000	0.647 $\pm$ 0.118	0.309 $\pm$ 0.101	1.000
combined	rf-topo	0.00	0.208 $\pm$ 0.212	0.395 $\pm$ 0.095	0.000 $\pm$ 0.000	0.297
combined	rf-topo	0.25	0.208 $\pm$ 0.172	0.512 $\pm$ 0.088	0.000 $\pm$ 0.000	0.339
combined	rf-topo	0.50	0.042 $\pm$ 0.082	0.518 $\pm$ 0.094	0.326 $\pm$ 0.126	0.802
combined	rf-topo	0.75	0.000 $\pm$ 0.000	0.583 $\pm$ 0.126	0.339 $\pm$ 0.131	0.917
combined	rf-topo	1.00	0.000 $\pm$ 0.000	0.645 $\pm$ 0.137	0.259 $\pm$ 0.137	0.958
fixture	rf-topo	0.00	0.250 $\pm$ 0.163	0.444 $\pm$ 0.057	0.000 $\pm$ 0.000	0.138
fixture	rf-topo	0.25	0.083 $\pm$ 0.107	0.415 $\pm$ 0.127	0.068 $\pm$ 0.087	0.513
fixture	rf-topo	0.50	0.083 $\pm$ 0.107	0.425 $\pm$ 0.081	0.499 $\pm$ 0.131	0.903
fixture	rf-topo	0.75	0.000 $\pm$ 0.000	0.662 $\pm$ 0.109	0.335 $\pm$ 0.144	1.000
fixture	rf-topo	1.00	0.000 $\pm$ 0.000	0.655 $\pm$ 0.124	0.226 $\pm$ 0.082	1.000
extreme	risk-averse	0.00	0.167 $\pm$ 0.214	0.603 $\pm$ 0.094	0.000 $\pm$ 0.000	0.054
extreme	risk-averse	0.25	0.083 $\pm$ 0.107	0.608 $\pm$ 0.091	0.031 $\pm$ 0.061	0.375
extreme	risk-averse	0.50	0.000 $\pm$ 0.000	0.406 $\pm$ 0.111	0.123 $\pm$ 0.092	0.750
extreme	risk-averse	0.75	0.000 $\pm$ 0.000	0.584 $\pm$ 0.090	0.178 $\pm$ 0.097	0.625
extreme	risk-averse	1.00	0.000 $\pm$ 0.000	0.367 $\pm$ 0.107	0.008 $\pm$ 0.015	0.833
combined	risk-averse	0.00	0.125 $\pm$ 0.120	0.436 $\pm$ 0.158	0.000 $\pm$ 0.000	0.294
combined	risk-averse	0.25	0.167 $\pm$ 0.175	0.690 $\pm$ 0.115	0.000 $\pm$ 0.000	0.333
combined	risk-averse	0.50	0.000 $\pm$ 0.000	0.412 $\pm$ 0.112	0.218 $\pm$ 0.167	0.750
combined	risk-averse	0.75	0.000 $\pm$ 0.000	0.484 $\pm$ 0.158	0.156 $\pm$ 0.126	0.750

Table 11 continued

Split	Method	Level	Success	Edge F1	Safety	Jam F1
combined	risk-averse	1.00	0.000 $\pm$ 0.000	0.485 $\pm$ 0.065	0.035 $\pm$ 0.034	0.667
fixture	risk-averse	0.00	0.125 $\pm$ 0.120	0.512 $\pm$ 0.127	0.000 $\pm$ 0.000	0.250
fixture	risk-averse	0.25	0.083 $\pm$ 0.107	0.613 $\pm$ 0.140	0.034 $\pm$ 0.066	0.421
fixture	risk-averse	0.50	0.083 $\pm$ 0.107	0.370 $\pm$ 0.093	0.126 $\pm$ 0.094	0.729
fixture	risk-averse	0.75	0.000 $\pm$ 0.000	0.594 $\pm$ 0.108	0.232 $\pm$ 0.117	0.833
fixture	risk-averse	1.00	0.000 $\pm$ 0.000	0.351 $\pm$ 0.083	0.041 $\pm$ 0.046	0.833
extreme	robust-mpc	0.00	0.125 $\pm$ 0.120	0.974 $\pm$ 0.011	0.000 $\pm$ 0.000	0.042
extreme	robust-mpc	0.25	0.125 $\pm$ 0.120	0.957 $\pm$ 0.012	0.000 $\pm$ 0.000	0.417
extreme	robust-mpc	0.50	0.000 $\pm$ 0.000	0.818 $\pm$ 0.068	0.254 $\pm$ 0.197	0.792
extreme	robust-mpc	0.75	0.000 $\pm$ 0.000	0.694 $\pm$ 0.061	0.433 $\pm$ 0.123	0.750
extreme	robust-mpc	1.00	0.000 $\pm$ 0.000	0.517 $\pm$ 0.073	0.273 $\pm$ 0.132	0.917
combined	robust-mpc	0.00	0.167 $\pm$ 0.214	0.965 $\pm$ 0.018	0.000 $\pm$ 0.000	0.292
combined	robust-mpc	0.25	0.208 $\pm$ 0.172	0.916 $\pm$ 0.024	0.034 $\pm$ 0.066	0.333
combined	robust-mpc	0.50	0.042 $\pm$ 0.082	0.845 $\pm$ 0.044	0.359 $\pm$ 0.165	0.792
combined	robust-mpc	0.75	0.000 $\pm$ 0.000	0.687 $\pm$ 0.057	0.370 $\pm$ 0.145	0.667
combined	robust-mpc	1.00	0.000 $\pm$ 0.000	0.519 $\pm$ 0.099	0.260 $\pm$ 0.124	0.917
fixture	robust-mpc	0.00	0.208 $\pm$ 0.120	0.976 $\pm$ 0.008	0.000 $\pm$ 0.000	0.167
fixture	robust-mpc	0.25	0.083 $\pm$ 0.107	0.956 $\pm$ 0.026	0.068 $\pm$ 0.087	0.500
fixture	robust-mpc	0.50	0.083 $\pm$ 0.107	0.844 $\pm$ 0.062	0.496 $\pm$ 0.129	0.833
fixture	robust-mpc	0.75	0.000 $\pm$ 0.000	0.626 $\pm$ 0.062	0.331 $\pm$ 0.139	0.750
fixture	robust-mpc	1.00	0.000 $\pm$ 0.000	0.497 $\pm$ 0.069	0.251 $\pm$ 0.093	0.667
extreme	no-memory	0.00	0.125 $\pm$ 0.120	0.669 $\pm$ 0.128	0.000 $\pm$ 0.000	0.337
extreme	no-memory	0.25	0.125 $\pm$ 0.120	0.775 $\pm$ 0.099	0.000 $\pm$ 0.000	0.541
extreme	no-memory	0.50	0.000 $\pm$ 0.000	0.780 $\pm$ 0.063	0.254 $\pm$ 0.197	0.912
extreme	no-memory	0.75	0.042 $\pm$ 0.082	0.799 $\pm$ 0.060	0.420 $\pm$ 0.115	0.792
extreme	no-memory	1.00	0.000 $\pm$ 0.000	0.535 $\pm$ 0.108	0.246 $\pm$ 0.118	0.458
combined	no-memory	0.00	0.167 $\pm$ 0.214	0.800 $\pm$ 0.046	0.000 $\pm$ 0.000	0.385
combined	no-memory	0.25	0.333 $\pm$ 0.175	0.748 $\pm$ 0.080	0.000 $\pm$ 0.000	0.499
combined	no-memory	0.50	0.125 $\pm$ 0.120	0.721 $\pm$ 0.066	0.326 $\pm$ 0.126	0.848
combined	no-memory	0.75	0.000 $\pm$ 0.000	0.781 $\pm$ 0.069	0.400 $\pm$ 0.134	0.669
combined	no-memory	1.00	0.000 $\pm$ 0.000	0.526 $\pm$ 0.165	0.223 $\pm$ 0.094	0.417
fixture	no-memory	0.00	0.333 $\pm$ 0.123	0.699 $\pm$ 0.094	0.000 $\pm$ 0.000	0.236
fixture	no-memory	0.25	0.208 $\pm$ 0.172	0.759 $\pm$ 0.048	0.102 $\pm$ 0.097	0.641
fixture	no-memory	0.50	0.083 $\pm$ 0.107	0.768 $\pm$ 0.024	0.464 $\pm$ 0.104	0.872
fixture	no-memory	0.75	0.000 $\pm$ 0.000	0.848 $\pm$ 0.061	0.369 $\pm$ 0.170	0.833
fixture	no-memory	1.00	0.000 $\pm$ 0.000	0.644 $\pm$ 0.144	0.211 $\pm$ 0.074	0.542
extreme	topo-v4	0.00	0.167 $\pm$ 0.175	0.661 $\pm$ 0.109	0.000 $\pm$ 0.000	0.047
extreme	topo-v4	0.25	0.208 $\pm$ 0.172	0.772 $\pm$ 0.093	0.000 $\pm$ 0.000	0.417

Table 11 continued

Split	Method	Level	Success	Edge F1	Safety	Jam F1
extreme	topo-v4	0.50	0.000 $\pm$ 0.000	0.732 $\pm$ 0.075	0.250 $\pm$ 0.193	0.750
extreme	topo-v4	0.75	0.000 $\pm$ 0.000	0.730 $\pm$ 0.110	0.355 $\pm$ 0.189	0.792
extreme	topo-v4	1.00	0.000 $\pm$ 0.000	0.611 $\pm$ 0.099	0.170 $\pm$ 0.091	0.625
combined	topo-v4	0.00	0.125 $\pm$ 0.120	0.714 $\pm$ 0.130	0.000 $\pm$ 0.000	0.294
combined	topo-v4	0.25	0.125 $\pm$ 0.120	0.810 $\pm$ 0.087	0.000 $\pm$ 0.000	0.339
combined	topo-v4	0.50	0.042 $\pm$ 0.082	0.740 $\pm$ 0.063	0.351 $\pm$ 0.153	0.750
combined	topo-v4	0.75	0.000 $\pm$ 0.000	0.721 $\pm$ 0.089	0.405 $\pm$ 0.062	0.875
combined	topo-v4	1.00	0.000 $\pm$ 0.000	0.494 $\pm$ 0.095	0.082 $\pm$ 0.086	0.458
fixture	topo-v4	0.00	0.250 $\pm$ 0.163	0.604 $\pm$ 0.122	0.000 $\pm$ 0.000	0.169
fixture	topo-v4	0.25	0.125 $\pm$ 0.120	0.816 $\pm$ 0.063	0.068 $\pm$ 0.087	0.458
fixture	topo-v4	0.50	0.042 $\pm$ 0.082	0.748 $\pm$ 0.069	0.464 $\pm$ 0.115	0.792
fixture	topo-v4	0.75	0.000 $\pm$ 0.000	0.794 $\pm$ 0.105	0.332 $\pm$ 0.113	0.875
fixture	topo-v4	1.00	0.000 $\pm$ 0.000	0.479 $\pm$ 0.081	0.094 $\pm$ 0.057	0.458
extreme	topo-v5	0.00	0.125 $\pm$ 0.172	0.642 $\pm$ 0.104	0.000 $\pm$ 0.000	0.042
extreme	topo-v5	0.25	0.083 $\pm$ 0.107	0.802 $\pm$ 0.085	0.000 $\pm$ 0.000	0.417
extreme	topo-v5	0.50	0.000 $\pm$ 0.000	0.728 $\pm$ 0.063	0.254 $\pm$ 0.197	0.792
extreme	topo-v5	0.75	0.000 $\pm$ 0.000	0.674 $\pm$ 0.095	0.379 $\pm$ 0.157	0.708
extreme	topo-v5	1.00	0.000 $\pm$ 0.000	0.573 $\pm$ 0.121	0.154 $\pm$ 0.114	0.833
combined	topo-v5	0.00	0.167 $\pm$ 0.214	0.637 $\pm$ 0.102	0.000 $\pm$ 0.000	0.315
combined	topo-v5	0.25	0.125 $\pm$ 0.120	0.766 $\pm$ 0.118	0.000 $\pm$ 0.000	0.333
combined	topo-v5	0.50	0.000 $\pm$ 0.000	0.693 $\pm$ 0.114	0.327 $\pm$ 0.127	0.750
combined	topo-v5	0.75	0.000 $\pm$ 0.000	0.698 $\pm$ 0.114	0.282 $\pm$ 0.105	0.875
combined	topo-v5	1.00	0.000 $\pm$ 0.000	0.462 $\pm$ 0.103	0.050 $\pm$ 0.065	0.417
fixture	topo-v5	0.00	0.250 $\pm$ 0.163	0.566 $\pm$ 0.142	0.000 $\pm$ 0.000	0.125
fixture	topo-v5	0.25	0.125 $\pm$ 0.172	0.806 $\pm$ 0.047	0.102 $\pm$ 0.097	0.511
fixture	topo-v5	0.50	0.042 $\pm$ 0.082	0.732 $\pm$ 0.069	0.472 $\pm$ 0.112	0.792
fixture	topo-v5	0.75	0.000 $\pm$ 0.000	0.737 $\pm$ 0.178	0.310 $\pm$ 0.123	0.875
fixture	topo-v5	1.00	0.000 $\pm$ 0.000	0.417 $\pm$ 0.070	0.096 $\pm$ 0.076	0.458

F SEED-LEVEL EVIDENCE LEDGER

Table 12: Seed-level metrics for reproducibility and paired-statistic auditing.

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
actuator	conformal	0	6	0.000	0.645	0.158	0.764	0.114
actuator	conformal	1	6	0.000	0.531	0.158	0.764	0.140
actuator	conformal	2	6	0.000	0.641	0.145	0.524	0.351
actuator	conformal	3	6	0.000	0.710	0.120	0.663	0.459
actuator	conformal	4	6	0.000	0.470	0.136	0.531	0.542
actuator	conformal	5	6	0.000	0.393	0.176	0.670	0.313
actuator	conformal	6	6	0.000	0.590	0.146	0.809	0.343
actuator	conformal	7	6	0.000	0.569	0.163	0.806	0.000
extreme	conformal	0	6	0.167	0.472	0.151	0.000	0.175
extreme	conformal	1	6	0.000	0.511	0.145	0.177	0.222
extreme	conformal	2	6	0.000	0.505	0.133	0.080	0.000
extreme	conformal	3	6	0.000	0.423	0.151	0.191	0.244
extreme	conformal	4	6	0.000	0.441	0.148	0.000	0.000
extreme	conformal	5	6	0.000	0.650	0.097	0.069	0.109
extreme	conformal	6	6	0.000	0.719	0.063	0.000	0.179
extreme	conformal	7	6	0.000	0.713	0.081	0.007	0.000
combined	conformal	0	6	0.167	0.636	0.109	0.135	0.627
combined	conformal	1	6	0.000	0.771	0.074	0.236	0.415
combined	conformal	2	6	0.167	0.810	0.101	0.229	0.622
combined	conformal	3	6	0.000	0.475	0.155	0.351	0.408
combined	conformal	4	6	0.000	0.821	0.075	0.500	0.442
combined	conformal	5	6	0.167	0.725	0.089	0.226	0.506
combined	conformal	6	6	0.000	0.614	0.117	0.233	0.566
combined	conformal	7	6	0.000	0.703	0.130	0.514	0.392
chain	conformal	0	6	0.000	0.955	0.019	0.000	0.380
chain	conformal	1	6	0.000	0.841	0.046	0.000	0.123
chain	conformal	2	6	0.167	0.938	0.031	0.000	0.355
chain	conformal	3	6	0.000	0.844	0.060	0.000	0.462
chain	conformal	4	6	0.000	0.969	0.012	0.000	0.185
chain	conformal	5	6	0.167	0.771	0.090	0.000	0.587
chain	conformal	6	6	0.167	0.922	0.040	0.000	0.372
chain	conformal	7	6	0.000	0.717	0.087	0.000	0.362
noise	conformal	0	6	0.333	0.801	0.117	0.000	0.793
noise	conformal	1	6	0.000	0.683	0.148	0.000	0.741
noise	conformal	2	6	0.167	0.822	0.093	0.118	0.744
noise	conformal	3	6	0.000	0.744	0.112	0.000	0.637
noise	conformal	4	6	0.333	0.703	0.132	0.135	0.769
noise	conformal	5	6	0.167	0.715	0.131	0.000	0.780

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
noise	conformal	6	6	0.167	0.684	0.146	0.000	0.787
noise	conformal	7	6	0.333	0.814	0.090	0.000	0.697
delayed	conformal	0	6	0.000	0.243	0.179	0.406	0.380
delayed	conformal	1	6	0.000	0.146	0.185	0.809	0.353
delayed	conformal	2	6	0.000	0.258	0.207	0.542	0.233
delayed	conformal	3	6	0.000	0.368	0.175	0.677	0.257
delayed	conformal	4	6	0.000	0.190	0.180	0.677	0.400
delayed	conformal	5	6	0.000	0.154	0.179	0.674	0.384
delayed	conformal	6	6	0.000	0.373	0.199	0.809	0.229
delayed	conformal	7	6	0.000	0.137	0.178	0.622	0.296
distractor	conformal	0	6	0.000	0.588	0.108	0.000	0.202
distractor	conformal	1	6	0.000	0.384	0.180	0.000	0.398
distractor	conformal	2	6	0.167	0.406	0.205	0.000	0.534
distractor	conformal	3	6	0.000	0.516	0.127	0.000	0.238
distractor	conformal	4	6	0.333	0.453	0.167	0.000	0.503
distractor	conformal	5	6	0.000	0.416	0.150	0.000	0.048
distractor	conformal	6	6	0.000	0.534	0.124	0.000	0.081
distractor	conformal	7	6	0.167	0.309	0.185	0.000	0.433
wall-jam	conformal	0	6	0.000	0.749	0.054	0.000	0.000
wall-jam	conformal	1	6	0.000	0.755	0.053	0.000	0.000
wall-jam	conformal	2	6	0.000	0.829	0.042	0.000	0.000
wall-jam	conformal	3	6	0.000	0.466	0.112	0.000	0.000
wall-jam	conformal	4	6	0.000	0.673	0.072	0.000	0.000
wall-jam	conformal	5	6	0.000	0.524	0.104	0.000	0.000
wall-jam	conformal	6	6	0.000	0.745	0.061	0.000	0.000
wall-jam	conformal	7	6	0.000	0.786	0.060	0.000	0.000
fixture	conformal	0	6	0.000	0.733	0.091	0.243	0.213
fixture	conformal	1	6	0.000	0.643	0.118	0.309	0.246
fixture	conformal	2	6	0.000	0.689	0.104	0.295	0.098
fixture	conformal	3	6	0.000	0.779	0.086	0.274	0.272
fixture	conformal	4	6	0.167	0.754	0.080	0.267	0.138
fixture	conformal	5	6	0.000	0.735	0.082	0.271	0.000
fixture	conformal	6	6	0.000	0.848	0.051	0.056	0.171
fixture	conformal	7	6	0.000	0.799	0.065	0.226	0.000
friction	conformal	0	6	0.000	0.666	0.105	0.000	0.619
friction	conformal	1	6	0.000	0.539	0.107	0.000	0.564
friction	conformal	2	6	0.167	0.664	0.117	0.000	0.768
friction	conformal	3	6	0.000	0.637	0.110	0.000	0.490

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
friction	conformal	4	6	0.167	0.614	0.106	0.000	0.830
friction	conformal	5	6	0.333	0.653	0.122	0.000	0.652
friction	conformal	6	6	0.000	0.545	0.120	0.000	0.698
friction	conformal	7	6	0.167	0.666	0.116	0.000	0.600
nominal	conformal	0	6	0.167	0.712	0.137	0.000	0.801
nominal	conformal	1	6	0.167	0.527	0.142	0.000	0.707
nominal	conformal	2	6	0.333	0.716	0.136	0.000	0.805
nominal	conformal	3	6	0.333	0.546	0.159	0.000	0.634
nominal	conformal	4	6	0.333	0.554	0.170	0.000	0.799
nominal	conformal	5	6	0.333	0.736	0.129	0.000	0.809
nominal	conformal	6	6	0.000	0.618	0.154	0.000	0.806
nominal	conformal	7	6	0.000	0.706	0.142	0.000	0.803
pocket	conformal	0	6	0.167	0.564	0.133	0.135	0.434
pocket	conformal	1	6	0.167	0.710	0.140	0.000	0.804
pocket	conformal	2	6	0.000	0.599	0.163	0.000	0.806
pocket	conformal	3	6	0.167	0.412	0.210	0.000	0.750
pocket	conformal	4	6	0.167	0.565	0.189	0.000	0.791
pocket	conformal	5	6	0.500	0.660	0.138	0.000	0.680
pocket	conformal	6	6	0.500	0.667	0.175	0.000	0.827
pocket	conformal	7	6	0.333	0.676	0.154	0.000	0.682
actuator	ci-mpc	0	6	0.000	0.863	0.064	0.764	0.114
actuator	ci-mpc	1	6	0.000	0.804	0.106	0.764	0.177
actuator	ci-mpc	2	6	0.000	0.829	0.088	0.635	0.350
actuator	ci-mpc	3	6	0.000	0.943	0.023	0.660	0.469
actuator	ci-mpc	4	6	0.000	0.874	0.041	0.663	0.662
actuator	ci-mpc	5	6	0.000	0.824	0.099	0.806	0.433
actuator	ci-mpc	6	6	0.000	0.924	0.038	0.806	0.429
actuator	ci-mpc	7	6	0.000	0.895	0.052	0.806	0.118
extreme	ci-mpc	0	6	0.000	0.639	0.163	0.323	0.116
extreme	ci-mpc	1	6	0.000	0.759	0.105	0.250	0.000
extreme	ci-mpc	2	6	0.000	0.544	0.154	0.125	0.000
extreme	ci-mpc	3	6	0.000	0.709	0.118	0.260	0.277
extreme	ci-mpc	4	6	0.000	0.534	0.201	0.312	0.000
extreme	ci-mpc	5	6	0.000	0.637	0.171	0.267	0.110
extreme	ci-mpc	6	6	0.000	0.503	0.280	0.132	0.083
extreme	ci-mpc	7	6	0.000	0.766	0.092	0.281	0.000
combined	ci-mpc	0	6	0.333	0.909	0.031	0.260	0.676
combined	ci-mpc	1	6	0.167	0.844	0.063	0.243	0.519

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
combined	ci-mpc	2	6	0.000	0.901	0.044	0.229	0.625
combined	ci-mpc	3	6	0.000	0.834	0.077	0.365	0.633
combined	ci-mpc	4	6	0.000	0.786	0.089	0.469	0.489
combined	ci-mpc	5	6	0.167	0.860	0.072	0.247	0.435
combined	ci-mpc	6	6	0.000	0.877	0.046	0.122	0.776
combined	ci-mpc	7	6	0.000	0.839	0.081	0.517	0.510
chain	ci-mpc	0	6	0.000	0.930	0.030	0.000	0.427
chain	ci-mpc	1	6	0.000	0.976	0.005	0.000	0.137
chain	ci-mpc	2	6	0.000	0.964	0.009	0.000	0.352
chain	ci-mpc	3	6	0.000	0.966	0.008	0.000	0.463
chain	ci-mpc	4	6	0.000	0.956	0.010	0.000	0.185
chain	ci-mpc	5	6	0.000	0.928	0.031	0.000	0.608
chain	ci-mpc	6	6	0.167	0.903	0.043	0.000	0.380
chain	ci-mpc	7	6	0.167	0.931	0.026	0.000	0.502
noise	ci-mpc	0	6	0.333	0.961	0.012	0.000	0.795
noise	ci-mpc	1	6	0.000	0.965	0.010	0.000	0.792
noise	ci-mpc	2	6	0.333	0.891	0.049	0.000	0.782
noise	ci-mpc	3	6	0.333	0.937	0.026	0.000	0.680
noise	ci-mpc	4	6	0.333	0.937	0.026	0.135	0.772
noise	ci-mpc	5	6	0.167	0.959	0.013	0.000	0.784
noise	ci-mpc	6	6	0.167	0.909	0.043	0.000	0.794
noise	ci-mpc	7	6	0.333	0.950	0.012	0.000	0.778
delayed	ci-mpc	0	6	0.000	0.972	0.009	0.812	0.228
delayed	ci-mpc	1	6	0.000	0.913	0.044	0.812	0.000
delayed	ci-mpc	2	6	0.000	0.868	0.072	0.816	0.144
delayed	ci-mpc	3	6	0.000	0.960	0.016	0.809	0.000
delayed	ci-mpc	4	6	0.000	0.918	0.048	0.809	0.108
delayed	ci-mpc	5	6	0.000	0.900	0.062	0.812	0.000
delayed	ci-mpc	6	6	0.000	0.884	0.062	0.812	0.000
delayed	ci-mpc	7	6	0.000	0.881	0.064	0.812	0.119
distractor	ci-mpc	0	6	0.000	0.875	0.042	0.000	0.159
distractor	ci-mpc	1	6	0.000	0.929	0.029	0.000	0.438
distractor	ci-mpc	2	6	0.167	0.830	0.071	0.000	0.542
distractor	ci-mpc	3	6	0.000	0.946	0.015	0.000	0.437
distractor	ci-mpc	4	6	0.167	0.919	0.032	0.000	0.554
distractor	ci-mpc	5	6	0.000	0.790	0.074	0.000	0.137
distractor	ci-mpc	6	6	0.000	0.845	0.054	0.000	0.036
distractor	ci-mpc	7	6	0.167	0.866	0.054	0.000	0.426

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
wall-jam	ci-mpc	0	6	0.000	0.944	0.010	0.000	0.000
wall-jam	ci-mpc	1	6	0.000	0.959	0.006	0.000	0.000
wall-jam	ci-mpc	2	6	0.000	0.983	0.002	0.000	0.000
wall-jam	ci-mpc	3	6	0.000	0.943	0.009	0.000	0.000
wall-jam	ci-mpc	4	6	0.000	0.934	0.009	0.000	0.000
wall-jam	ci-mpc	5	6	0.000	0.951	0.006	0.000	0.000
wall-jam	ci-mpc	6	6	0.000	0.942	0.008	0.000	0.000
wall-jam	ci-mpc	7	6	0.000	0.990	0.001	0.000	0.000
fixture	ci-mpc	0	6	0.000	0.701	0.140	0.354	0.094
fixture	ci-mpc	1	6	0.000	0.745	0.112	0.566	0.100
fixture	ci-mpc	2	6	0.000	0.766	0.086	0.295	0.098
fixture	ci-mpc	3	6	0.000	0.805	0.094	0.208	0.288
fixture	ci-mpc	4	6	0.000	0.716	0.115	0.337	0.070
fixture	ci-mpc	5	6	0.000	0.582	0.213	0.253	0.000
fixture	ci-mpc	6	6	0.000	0.613	0.201	0.156	0.089
fixture	ci-mpc	7	6	0.000	0.668	0.164	0.448	0.000
friction	ci-mpc	0	6	0.000	0.853	0.076	0.000	0.655
friction	ci-mpc	1	6	0.167	0.918	0.032	0.000	0.780
friction	ci-mpc	2	6	0.167	0.938	0.018	0.000	0.798
friction	ci-mpc	3	6	0.167	0.958	0.009	0.000	0.531
friction	ci-mpc	4	6	0.167	0.911	0.033	0.000	0.677
friction	ci-mpc	5	6	0.333	0.957	0.011	0.000	0.681
friction	ci-mpc	6	6	0.333	0.902	0.038	0.000	0.821
friction	ci-mpc	7	6	0.333	0.960	0.010	0.000	0.655
nominal	ci-mpc	0	6	0.167	0.973	0.005	0.000	0.814
nominal	ci-mpc	1	6	0.333	0.829	0.029	0.000	0.733
nominal	ci-mpc	2	6	0.333	0.967	0.007	0.000	0.814
nominal	ci-mpc	3	6	0.333	0.973	0.007	0.000	0.816
nominal	ci-mpc	4	6	0.333	0.964	0.006	0.000	0.817
nominal	ci-mpc	5	6	0.333	0.972	0.005	0.000	0.818
nominal	ci-mpc	6	6	0.167	0.978	0.005	0.000	0.816
nominal	ci-mpc	7	6	0.000	0.978	0.005	0.000	0.811
pocket	ci-mpc	0	6	0.167	0.849	0.054	0.135	0.423
pocket	ci-mpc	1	6	0.167	0.965	0.010	0.000	0.816
pocket	ci-mpc	2	6	0.000	0.942	0.026	0.000	0.815
pocket	ci-mpc	3	6	0.167	0.978	0.005	0.000	0.817
pocket	ci-mpc	4	6	0.000	0.967	0.009	0.000	0.812
pocket	ci-mpc	5	6	0.333	0.911	0.020	0.000	0.678

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
pocket	ci-mpc	6	6	0.167	0.960	0.010	0.000	0.815
pocket	ci-mpc	7	6	0.167	0.977	0.006	0.000	0.818
actuator	dist-graph	0	6	0.000	0.847	0.071	0.753	0.114
actuator	dist-graph	1	6	0.000	0.816	0.099	0.764	0.208
actuator	dist-graph	2	6	0.000	0.824	0.092	0.521	0.351
actuator	dist-graph	3	6	0.000	0.943	0.022	0.660	0.468
actuator	dist-graph	4	6	0.000	0.864	0.043	0.660	0.649
actuator	dist-graph	5	6	0.000	0.811	0.106	0.670	0.448
actuator	dist-graph	6	6	0.000	0.917	0.042	0.806	0.433
actuator	dist-graph	7	6	0.000	0.881	0.060	0.806	0.070
extreme	dist-graph	0	6	0.000	0.633	0.166	0.323	0.120
extreme	dist-graph	1	6	0.000	0.718	0.119	0.316	0.000
extreme	dist-graph	2	6	0.000	0.476	0.147	0.083	0.000
extreme	dist-graph	3	6	0.000	0.701	0.122	0.146	0.277
extreme	dist-graph	4	6	0.000	0.644	0.158	0.399	0.130
extreme	dist-graph	5	6	0.000	0.660	0.163	0.267	0.109
extreme	dist-graph	6	6	0.000	0.519	0.271	0.132	0.140
extreme	dist-graph	7	6	0.000	0.775	0.088	0.288	0.000
combined	dist-graph	0	6	0.333	0.905	0.036	0.260	0.679
combined	dist-graph	1	6	0.167	0.841	0.064	0.243	0.393
combined	dist-graph	2	6	0.000	0.902	0.044	0.236	0.621
combined	dist-graph	3	6	0.000	0.865	0.059	0.365	0.616
combined	dist-graph	4	6	0.000	0.813	0.077	0.486	0.494
combined	dist-graph	5	6	0.167	0.865	0.070	0.247	0.554
combined	dist-graph	6	6	0.000	0.880	0.049	0.122	0.774
combined	dist-graph	7	6	0.000	0.839	0.083	0.514	0.395
chain	dist-graph	0	6	0.000	0.930	0.030	0.000	0.427
chain	dist-graph	1	6	0.000	0.976	0.005	0.000	0.137
chain	dist-graph	2	6	0.000	0.963	0.010	0.000	0.352
chain	dist-graph	3	6	0.000	0.955	0.013	0.000	0.461
chain	dist-graph	4	6	0.000	0.954	0.010	0.000	0.185
chain	dist-graph	5	6	0.000	0.933	0.030	0.000	0.607
chain	dist-graph	6	6	0.167	0.898	0.046	0.000	0.379
chain	dist-graph	7	6	0.167	0.938	0.023	0.000	0.503
noise	dist-graph	0	6	0.333	0.958	0.013	0.000	0.793
noise	dist-graph	1	6	0.000	0.965	0.010	0.000	0.791
noise	dist-graph	2	6	0.167	0.899	0.043	0.118	0.743
noise	dist-graph	3	6	0.167	0.949	0.016	0.000	0.645

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
noise	dist-graph	4	6	0.333	0.959	0.015	0.135	0.771
noise	dist-graph	5	6	0.167	0.960	0.012	0.000	0.784
noise	dist-graph	6	6	0.167	0.945	0.019	0.000	0.796
noise	dist-graph	7	6	0.333	0.898	0.033	0.000	0.761
delayed	dist-graph	0	6	0.000	0.970	0.010	0.812	0.123
delayed	dist-graph	1	6	0.000	0.913	0.044	0.812	0.000
delayed	dist-graph	2	6	0.000	0.877	0.067	0.816	0.142
delayed	dist-graph	3	6	0.000	0.939	0.030	0.809	0.000
delayed	dist-graph	4	6	0.000	0.920	0.046	0.812	0.109
delayed	dist-graph	5	6	0.000	0.897	0.062	0.812	0.000
delayed	dist-graph	6	6	0.000	0.884	0.062	0.812	0.000
delayed	dist-graph	7	6	0.000	0.878	0.066	0.812	0.117
distractor	dist-graph	0	6	0.000	0.878	0.041	0.000	0.159
distractor	dist-graph	1	6	0.000	0.929	0.029	0.000	0.438
distractor	dist-graph	2	6	0.167	0.830	0.071	0.000	0.542
distractor	dist-graph	3	6	0.000	0.913	0.035	0.000	0.301
distractor	dist-graph	4	6	0.167	0.919	0.032	0.000	0.554
distractor	dist-graph	5	6	0.000	0.836	0.057	0.000	0.141
distractor	dist-graph	6	6	0.000	0.844	0.055	0.000	0.035
distractor	dist-graph	7	6	0.167	0.867	0.054	0.000	0.426
wall-jam	dist-graph	0	6	0.000	0.978	0.003	0.000	0.000
wall-jam	dist-graph	1	6	0.000	0.968	0.004	0.000	0.000
wall-jam	dist-graph	2	6	0.000	0.986	0.002	0.000	0.000
wall-jam	dist-graph	3	6	0.000	0.954	0.008	0.000	0.000
wall-jam	dist-graph	4	6	0.000	0.950	0.008	0.000	0.000
wall-jam	dist-graph	5	6	0.000	0.931	0.011	0.000	0.000
wall-jam	dist-graph	6	6	0.000	0.938	0.009	0.000	0.000
wall-jam	dist-graph	7	6	0.000	0.959	0.010	0.000	0.000
fixture	dist-graph	0	6	0.000	0.696	0.145	0.354	0.094
fixture	dist-graph	1	6	0.000	0.795	0.087	0.472	0.234
fixture	dist-graph	2	6	0.000	0.752	0.087	0.295	0.098
fixture	dist-graph	3	6	0.000	0.770	0.128	0.201	0.195
fixture	dist-graph	4	6	0.000	0.717	0.115	0.354	0.067
fixture	dist-graph	5	6	0.000	0.586	0.208	0.253	0.000
fixture	dist-graph	6	6	0.000	0.647	0.165	0.156	0.090
fixture	dist-graph	7	6	0.000	0.683	0.152	0.451	0.000
friction	dist-graph	0	6	0.000	0.861	0.073	0.000	0.654
friction	dist-graph	1	6	0.167	0.882	0.045	0.000	0.779

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
friction	dist-graph	2	6	0.167	0.945	0.015	0.000	0.798
friction	dist-graph	3	6	0.167	0.958	0.009	0.000	0.531
friction	dist-graph	4	6	0.167	0.900	0.037	0.000	0.812
friction	dist-graph	5	6	0.333	0.965	0.008	0.000	0.684
friction	dist-graph	6	6	0.333	0.912	0.034	0.000	0.775
friction	dist-graph	7	6	0.333	0.958	0.011	0.000	0.654
nominal	dist-graph	0	6	0.167	0.973	0.005	0.000	0.814
nominal	dist-graph	1	6	0.333	0.862	0.025	0.000	0.737
nominal	dist-graph	2	6	0.333	0.967	0.007	0.000	0.814
nominal	dist-graph	3	6	0.333	0.973	0.007	0.000	0.816
nominal	dist-graph	4	6	0.333	0.964	0.006	0.000	0.817
nominal	dist-graph	5	6	0.333	0.972	0.005	0.000	0.818
nominal	dist-graph	6	6	0.167	0.978	0.005	0.000	0.816
nominal	dist-graph	7	6	0.000	0.978	0.005	0.000	0.811
pocket	dist-graph	0	6	0.167	0.855	0.051	0.135	0.423
pocket	dist-graph	1	6	0.167	0.965	0.010	0.000	0.816
pocket	dist-graph	2	6	0.000	0.942	0.026	0.000	0.815
pocket	dist-graph	3	6	0.167	0.978	0.005	0.000	0.817
pocket	dist-graph	4	6	0.000	0.967	0.009	0.000	0.812
pocket	dist-graph	5	6	0.333	0.921	0.019	0.000	0.678
pocket	dist-graph	6	6	0.167	0.960	0.010	0.000	0.815
pocket	dist-graph	7	6	0.167	0.975	0.007	0.000	0.818
actuator	ensemble	0	6	0.000	0.722	0.109	0.764	0.112
actuator	ensemble	1	6	0.000	0.686	0.122	0.764	0.177
actuator	ensemble	2	6	0.000	0.731	0.104	0.757	0.342
actuator	ensemble	3	6	0.000	0.687	0.121	0.795	0.398
actuator	ensemble	4	6	0.000	0.626	0.113	0.663	0.644
actuator	ensemble	5	6	0.000	0.577	0.165	0.667	0.446
actuator	ensemble	6	6	0.000	0.676	0.128	0.809	0.452
actuator	ensemble	7	6	0.000	0.683	0.107	0.795	0.149
extreme	ensemble	0	6	0.000	0.825	0.052	0.257	0.121
extreme	ensemble	1	6	0.000	0.803	0.054	0.149	0.226
extreme	ensemble	2	6	0.000	0.524	0.100	0.000	0.000
extreme	ensemble	3	6	0.000	0.773	0.051	0.073	0.333
extreme	ensemble	4	6	0.000	0.823	0.051	0.184	0.264
extreme	ensemble	5	6	0.000	0.796	0.041	0.000	0.325
extreme	ensemble	6	6	0.000	0.819	0.054	0.201	0.084
extreme	ensemble	7	6	0.000	0.927	0.028	0.090	0.046

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
combined	ensemble	0	6	0.333	0.658	0.150	0.528	0.554
combined	ensemble	1	6	0.000	0.879	0.048	0.319	0.476
combined	ensemble	2	6	0.000	0.728	0.127	0.368	0.541
combined	ensemble	3	6	0.000	0.692	0.136	0.365	0.620
combined	ensemble	4	6	0.000	0.815	0.075	0.615	0.544
combined	ensemble	5	6	0.167	0.684	0.115	0.361	0.715
combined	ensemble	6	6	0.000	0.807	0.084	0.306	0.711
combined	ensemble	7	6	0.000	0.591	0.163	0.524	0.387
chain	ensemble	0	6	0.000	0.894	0.050	0.000	0.427
chain	ensemble	1	6	0.000	0.897	0.049	0.000	0.137
chain	ensemble	2	6	0.000	0.894	0.049	0.000	0.353
chain	ensemble	3	6	0.000	0.750	0.123	0.000	0.599
chain	ensemble	4	6	0.000	0.968	0.012	0.000	0.283
chain	ensemble	5	6	0.000	0.768	0.115	0.000	0.607
chain	ensemble	6	6	0.167	0.909	0.043	0.000	0.307
chain	ensemble	7	6	0.167	0.764	0.117	0.000	0.503
noise	ensemble	0	6	0.333	0.606	0.189	0.000	0.790
noise	ensemble	1	6	0.000	0.626	0.190	0.000	0.793
noise	ensemble	2	6	0.167	0.743	0.126	0.094	0.743
noise	ensemble	3	6	0.333	0.617	0.190	0.000	0.779
noise	ensemble	4	6	0.333	0.592	0.166	0.135	0.715
noise	ensemble	5	6	0.167	0.615	0.194	0.000	0.786
noise	ensemble	6	6	0.167	0.685	0.157	0.000	0.760
noise	ensemble	7	6	0.333	0.698	0.152	0.000	0.762
delayed	ensemble	0	6	0.000	0.572	0.135	0.677	0.113
delayed	ensemble	1	6	0.000	0.660	0.132	0.812	0.120
delayed	ensemble	2	6	0.000	0.580	0.137	0.816	0.262
delayed	ensemble	3	6	0.000	0.652	0.141	0.809	0.000
delayed	ensemble	4	6	0.000	0.507	0.129	0.542	0.183
delayed	ensemble	5	6	0.000	0.622	0.133	0.812	0.041
delayed	ensemble	6	6	0.000	0.666	0.129	0.812	0.000
delayed	ensemble	7	6	0.000	0.537	0.146	0.812	0.304
distractor	ensemble	0	6	0.000	0.822	0.038	0.000	0.161
distractor	ensemble	1	6	0.167	0.722	0.109	0.000	0.423
distractor	ensemble	2	6	0.333	0.707	0.148	0.000	0.550
distractor	ensemble	3	6	0.333	0.514	0.165	0.000	0.516
distractor	ensemble	4	6	0.333	0.680	0.156	0.000	0.565
distractor	ensemble	5	6	0.000	0.850	0.041	0.000	0.138

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
distractor	ensemble	6	6	0.000	0.963	0.015	0.000	0.049
distractor	ensemble	7	6	0.167	0.688	0.085	0.000	0.301
wall-jam	ensemble	0	6	0.000	0.912	0.019	0.000	0.000
wall-jam	ensemble	1	6	0.000	0.880	0.025	0.000	0.000
wall-jam	ensemble	2	6	0.000	0.888	0.024	0.000	0.000
wall-jam	ensemble	3	6	0.000	0.737	0.048	0.000	0.000
wall-jam	ensemble	4	6	0.000	0.722	0.047	0.000	0.000
wall-jam	ensemble	5	6	0.000	0.865	0.027	0.000	0.044
wall-jam	ensemble	6	6	0.000	0.737	0.045	0.000	0.000
wall-jam	ensemble	7	6	0.000	0.917	0.018	0.000	0.000
fixture	ensemble	0	6	0.000	0.845	0.058	0.354	0.094
fixture	ensemble	1	6	0.000	0.792	0.086	0.566	0.098
fixture	ensemble	2	6	0.000	0.842	0.050	0.295	0.097
fixture	ensemble	3	6	0.167	0.845	0.067	0.222	0.332
fixture	ensemble	4	6	0.000	0.773	0.103	0.444	0.068
fixture	ensemble	5	6	0.000	0.875	0.047	0.253	0.000
fixture	ensemble	6	6	0.000	0.905	0.037	0.167	0.090
fixture	ensemble	7	6	0.000	0.803	0.086	0.438	0.000
friction	ensemble	0	6	0.167	0.681	0.152	0.000	0.657
friction	ensemble	1	6	0.167	0.765	0.074	0.000	0.644
friction	ensemble	2	6	0.167	0.738	0.119	0.000	0.796
friction	ensemble	3	6	0.167	0.677	0.129	0.000	0.527
friction	ensemble	4	6	0.000	0.741	0.122	0.000	0.674
friction	ensemble	5	6	0.333	0.664	0.140	0.000	0.676
friction	ensemble	6	6	0.333	0.706	0.109	0.000	0.817
friction	ensemble	7	6	0.667	0.691	0.150	0.000	0.672
nominal	ensemble	0	6	0.167	0.640	0.190	0.000	0.814
nominal	ensemble	1	6	0.167	0.606	0.158	0.000	0.690
nominal	ensemble	2	6	0.333	0.638	0.190	0.000	0.814
nominal	ensemble	3	6	0.333	0.502	0.146	0.000	0.499
nominal	ensemble	4	6	0.333	0.541	0.196	0.000	0.806
nominal	ensemble	5	6	0.333	0.644	0.188	0.000	0.817
nominal	ensemble	6	6	0.333	0.586	0.211	0.000	0.817
nominal	ensemble	7	6	0.000	0.611	0.201	0.000	0.811
pocket	ensemble	0	6	0.167	0.669	0.119	0.135	0.227
pocket	ensemble	1	6	0.167	0.493	0.141	0.000	0.509
pocket	ensemble	2	6	0.167	0.565	0.154	0.139	0.529
pocket	ensemble	3	6	0.000	0.734	0.082	0.000	0.275

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
pocket	ensemble	4	6	0.167	0.552	0.153	0.000	0.469
pocket	ensemble	5	6	0.000	0.523	0.171	0.000	0.496
pocket	ensemble	6	6	0.000	0.546	0.144	0.000	0.393
pocket	ensemble	7	6	0.167	0.457	0.161	0.000	0.427
actuator	hgb-topo	0	6	0.000	0.742	0.097	0.764	0.114
actuator	hgb-topo	1	6	0.000	0.751	0.094	0.764	0.212
actuator	hgb-topo	2	6	0.000	0.790	0.086	0.521	0.355
actuator	hgb-topo	3	6	0.000	0.705	0.112	0.795	0.407
actuator	hgb-topo	4	6	0.000	0.755	0.092	0.663	0.645
actuator	hgb-topo	5	6	0.000	0.586	0.137	0.670	0.443
actuator	hgb-topo	6	6	0.000	0.689	0.118	0.806	0.447
actuator	hgb-topo	7	6	0.000	0.700	0.114	0.806	0.073
extreme	hgb-topo	0	6	0.000	0.929	0.026	0.330	0.121
extreme	hgb-topo	1	6	0.000	0.936	0.019	0.205	0.000
extreme	hgb-topo	2	6	0.000	0.801	0.031	0.073	0.000
extreme	hgb-topo	3	6	0.000	0.931	0.033	0.167	0.226
extreme	hgb-topo	4	6	0.000	0.927	0.027	0.358	0.111
extreme	hgb-topo	5	6	0.000	0.942	0.022	0.264	0.112
extreme	hgb-topo	6	6	0.000	0.967	0.013	0.132	0.085
extreme	hgb-topo	7	6	0.000	0.905	0.036	0.288	0.000
combined	hgb-topo	0	6	0.333	0.811	0.092	0.260	0.675
combined	hgb-topo	1	6	0.167	0.892	0.047	0.243	0.402
combined	hgb-topo	2	6	0.000	0.752	0.119	0.233	0.621
combined	hgb-topo	3	6	0.000	0.778	0.101	0.365	0.606
combined	hgb-topo	4	6	0.000	0.852	0.056	0.569	0.385
combined	hgb-topo	5	6	0.167	0.781	0.099	0.361	0.624
combined	hgb-topo	6	6	0.000	0.780	0.109	0.122	0.773
combined	hgb-topo	7	6	0.000	0.825	0.081	0.361	0.509
chain	hgb-topo	0	6	0.000	0.918	0.040	0.000	0.427
chain	hgb-topo	1	6	0.000	0.922	0.039	0.000	0.137
chain	hgb-topo	2	6	0.000	0.918	0.040	0.000	0.352
chain	hgb-topo	3	6	0.000	0.889	0.059	0.000	0.463
chain	hgb-topo	4	6	0.000	0.990	0.002	0.000	0.186
chain	hgb-topo	5	6	0.000	0.809	0.098	0.000	0.607
chain	hgb-topo	6	6	0.167	0.850	0.073	0.000	0.380
chain	hgb-topo	7	6	0.167	0.772	0.112	0.000	0.503
noise	hgb-topo	0	6	0.333	0.707	0.150	0.000	0.795
noise	hgb-topo	1	6	0.000	0.702	0.155	0.000	0.791

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
noise	hgb-topo	2	6	0.167	0.789	0.104	0.118	0.743
noise	hgb-topo	3	6	0.167	0.706	0.145	0.000	0.645
noise	hgb-topo	4	6	0.333	0.745	0.126	0.135	0.770
noise	hgb-topo	5	6	0.167	0.657	0.171	0.000	0.784
noise	hgb-topo	6	6	0.167	0.748	0.134	0.000	0.793
noise	hgb-topo	7	6	0.333	0.756	0.130	0.000	0.778
delayed	hgb-topo	0	6	0.000	0.687	0.118	0.812	0.113
delayed	hgb-topo	1	6	0.000	0.688	0.117	0.812	0.000
delayed	hgb-topo	2	6	0.000	0.706	0.118	0.816	0.143
delayed	hgb-topo	3	6	0.000	0.683	0.120	0.812	0.000
delayed	hgb-topo	4	6	0.000	0.689	0.117	0.806	0.109
delayed	hgb-topo	5	6	0.000	0.688	0.117	0.812	0.000
delayed	hgb-topo	6	6	0.000	0.688	0.117	0.812	0.000
delayed	hgb-topo	7	6	0.000	0.687	0.118	0.812	0.119
distractor	hgb-topo	0	6	0.000	0.961	0.022	0.000	0.159
distractor	hgb-topo	1	6	0.000	0.804	0.100	0.000	0.439
distractor	hgb-topo	2	6	0.167	0.744	0.130	0.000	0.542
distractor	hgb-topo	3	6	0.000	0.764	0.117	0.000	0.437
distractor	hgb-topo	4	6	0.167	0.735	0.134	0.000	0.554
distractor	hgb-topo	5	6	0.000	0.918	0.040	0.000	0.141
distractor	hgb-topo	6	6	0.000	0.991	0.003	0.000	0.035
distractor	hgb-topo	7	6	0.167	0.769	0.113	0.000	0.426
wall-jam	hgb-topo	0	6	0.000	1.000	0.000	0.000	0.000
wall-jam	hgb-topo	1	6	0.000	0.950	0.007	0.000	0.000
wall-jam	hgb-topo	2	6	0.000	0.955	0.006	0.000	0.000
wall-jam	hgb-topo	3	6	0.000	0.875	0.016	0.000	0.000
wall-jam	hgb-topo	4	6	0.000	0.900	0.013	0.000	0.000
wall-jam	hgb-topo	5	6	0.000	0.816	0.023	0.000	0.000
wall-jam	hgb-topo	6	6	0.000	0.801	0.026	0.000	0.000
wall-jam	hgb-topo	7	6	0.000	0.816	0.023	0.000	0.000
fixture	hgb-topo	0	6	0.000	0.917	0.031	0.354	0.094
fixture	hgb-topo	1	6	0.000	0.840	0.060	0.562	0.098
fixture	hgb-topo	2	6	0.000	0.905	0.029	0.295	0.098
fixture	hgb-topo	3	6	0.000	0.940	0.023	0.201	0.178
fixture	hgb-topo	4	6	0.000	0.899	0.038	0.358	0.069
fixture	hgb-topo	5	6	0.000	0.946	0.020	0.253	0.000
fixture	hgb-topo	6	6	0.000	0.963	0.014	0.156	0.090
fixture	hgb-topo	7	6	0.000	0.884	0.043	0.424	0.000

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
friction	hgb-topo	0	6	0.000	0.810	0.097	0.000	0.654
friction	hgb-topo	1	6	0.167	0.769	0.113	0.000	0.777
friction	hgb-topo	2	6	0.167	0.768	0.114	0.000	0.799
friction	hgb-topo	3	6	0.167	0.775	0.111	0.000	0.531
friction	hgb-topo	4	6	0.167	0.773	0.112	0.000	0.675
friction	hgb-topo	5	6	0.333	0.750	0.126	0.000	0.684
friction	hgb-topo	6	6	0.333	0.819	0.092	0.000	0.821
friction	hgb-topo	7	6	0.333	0.744	0.129	0.000	0.654
nominal	hgb-topo	0	6	0.167	0.813	0.114	0.000	0.814
nominal	hgb-topo	1	6	0.333	0.792	0.112	0.000	0.736
nominal	hgb-topo	2	6	0.333	0.815	0.112	0.000	0.814
nominal	hgb-topo	3	6	0.333	0.811	0.114	0.000	0.816
nominal	hgb-topo	4	6	0.333	0.808	0.115	0.000	0.817
nominal	hgb-topo	5	6	0.333	0.818	0.111	0.000	0.818
nominal	hgb-topo	6	6	0.167	0.802	0.118	0.000	0.816
nominal	hgb-topo	7	6	0.000	0.804	0.118	0.000	0.811
pocket	hgb-topo	0	6	0.167	0.757	0.117	0.135	0.423
pocket	hgb-topo	1	6	0.167	0.723	0.151	0.000	0.816
pocket	hgb-topo	2	6	0.000	0.721	0.154	0.000	0.815
pocket	hgb-topo	3	6	0.167	0.681	0.168	0.000	0.817
pocket	hgb-topo	4	6	0.000	0.632	0.191	0.000	0.812
pocket	hgb-topo	5	6	0.333	0.606	0.166	0.000	0.815
pocket	hgb-topo	6	6	0.167	0.688	0.166	0.000	0.815
pocket	hgb-topo	7	6	0.167	0.727	0.149	0.000	0.818
actuator	persist	0	6	0.000	0.858	0.051	0.764	0.113
actuator	persist	1	6	0.000	0.830	0.062	0.764	0.181
actuator	persist	2	6	0.000	0.875	0.046	0.521	0.352
actuator	persist	3	6	0.000	0.894	0.040	0.788	0.392
actuator	persist	4	6	0.000	0.914	0.029	0.663	0.644
actuator	persist	5	6	0.000	0.908	0.033	0.670	0.445
actuator	persist	6	6	0.000	0.927	0.026	0.806	0.428
actuator	persist	7	6	0.000	0.907	0.032	0.806	0.064
extreme	persist	0	6	0.000	0.820	0.066	0.309	0.125
extreme	persist	1	6	0.000	0.876	0.042	0.250	0.000
extreme	persist	2	6	0.000	0.824	0.040	0.066	0.000
extreme	persist	3	6	0.000	0.855	0.052	0.149	0.245
extreme	persist	4	6	0.000	0.807	0.061	0.274	0.000
extreme	persist	5	6	0.000	0.779	0.074	0.264	0.109

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
extreme	persist	6	6	0.000	0.852	0.044	0.132	0.140
extreme	persist	7	6	0.000	0.861	0.040	0.233	0.000
combined	persist	0	6	0.333	0.874	0.047	0.260	0.793
combined	persist	1	6	0.167	0.874	0.046	0.250	0.497
combined	persist	2	6	0.000	0.884	0.044	0.236	0.629
combined	persist	3	6	0.000	0.836	0.065	0.365	0.601
combined	persist	4	6	0.000	0.792	0.076	0.476	0.598
combined	persist	5	6	0.167	0.807	0.069	0.361	0.632
combined	persist	6	6	0.000	0.848	0.057	0.236	0.776
combined	persist	7	6	0.000	0.905	0.034	0.497	0.395
chain	persist	0	6	0.000	0.955	0.012	0.000	0.427
chain	persist	1	6	0.000	0.963	0.008	0.000	0.137
chain	persist	2	6	0.000	0.959	0.012	0.000	0.357
chain	persist	3	6	0.000	0.957	0.011	0.000	0.464
chain	persist	4	6	0.000	0.948	0.010	0.000	0.185
chain	persist	5	6	0.000	0.956	0.011	0.000	0.607
chain	persist	6	6	0.167	0.955	0.011	0.000	0.379
chain	persist	7	6	0.167	0.955	0.012	0.000	0.510
noise	persist	0	6	0.333	0.924	0.034	0.000	0.794
noise	persist	1	6	0.000	0.951	0.021	0.000	0.816
noise	persist	2	6	0.333	0.904	0.036	0.000	0.784
noise	persist	3	6	0.167	0.912	0.036	0.000	0.643
noise	persist	4	6	0.333	0.924	0.028	0.135	0.731
noise	persist	5	6	0.167	0.909	0.039	0.000	0.785
noise	persist	6	6	0.167	0.915	0.031	0.000	0.790
noise	persist	7	6	0.333	0.895	0.038	0.000	0.778
delayed	persist	0	6	0.000	0.948	0.019	0.812	0.229
delayed	persist	1	6	0.000	0.960	0.012	0.812	0.000
delayed	persist	2	6	0.000	0.948	0.019	0.816	0.021
delayed	persist	3	6	0.000	0.941	0.023	0.812	0.000
delayed	persist	4	6	0.000	0.949	0.017	0.809	0.108
delayed	persist	5	6	0.000	0.945	0.019	0.812	0.000
delayed	persist	6	6	0.000	0.947	0.016	0.812	0.000
delayed	persist	7	6	0.000	0.943	0.018	0.812	0.118
distractor	persist	0	6	0.000	0.933	0.020	0.000	0.159
distractor	persist	1	6	0.000	0.951	0.015	0.000	0.438
distractor	persist	2	6	0.167	0.932	0.021	0.000	0.542
distractor	persist	3	6	0.000	0.941	0.021	0.000	0.438

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
distractor	persist	4	6	0.167	0.938	0.019	0.000	0.554
distractor	persist	5	6	0.000	0.923	0.019	0.000	0.141
distractor	persist	6	6	0.000	0.924	0.020	0.000	0.035
distractor	persist	7	6	0.167	0.933	0.020	0.000	0.426
wall-jam	persist	0	6	0.000	0.931	0.013	0.000	0.000
wall-jam	persist	1	6	0.000	0.935	0.011	0.000	0.000
wall-jam	persist	2	6	0.000	0.946	0.012	0.000	0.000
wall-jam	persist	3	6	0.000	0.910	0.016	0.000	0.000
wall-jam	persist	4	6	0.000	0.895	0.017	0.000	0.000
wall-jam	persist	5	6	0.000	0.922	0.013	0.000	0.000
wall-jam	persist	6	6	0.000	0.920	0.015	0.000	0.000
wall-jam	persist	7	6	0.000	0.940	0.012	0.000	0.000
fixture	persist	0	6	0.000	0.810	0.068	0.354	0.092
fixture	persist	1	6	0.000	0.847	0.053	0.569	0.099
fixture	persist	2	6	0.000	0.870	0.039	0.295	0.098
fixture	persist	3	6	0.000	0.902	0.033	0.201	0.194
fixture	persist	4	6	0.000	0.890	0.035	0.358	0.068
fixture	persist	5	6	0.000	0.877	0.038	0.253	0.000
fixture	persist	6	6	0.000	0.902	0.032	0.156	0.090
fixture	persist	7	6	0.000	0.884	0.036	0.448	0.000
friction	persist	0	6	0.000	0.948	0.017	0.000	0.654
friction	persist	1	6	0.167	0.949	0.015	0.000	0.777
friction	persist	2	6	0.167	0.926	0.022	0.000	0.798
friction	persist	3	6	0.167	0.939	0.018	0.000	0.532
friction	persist	4	6	0.167	0.918	0.020	0.000	0.852
friction	persist	5	6	0.333	0.930	0.023	0.000	0.683
friction	persist	6	6	0.333	0.926	0.024	0.000	0.819
friction	persist	7	6	0.333	0.930	0.020	0.000	0.654
nominal	persist	0	6	0.167	0.956	0.014	0.000	0.814
nominal	persist	1	6	0.333	0.933	0.017	0.000	0.724
nominal	persist	2	6	0.333	0.948	0.015	0.000	0.814
nominal	persist	3	6	0.333	0.950	0.014	0.000	0.816
nominal	persist	4	6	0.333	0.946	0.016	0.000	0.817
nominal	persist	5	6	0.333	0.947	0.015	0.000	0.818
nominal	persist	6	6	0.167	0.955	0.014	0.000	0.816
nominal	persist	7	6	0.000	0.960	0.011	0.000	0.811
pocket	persist	0	6	0.167	0.925	0.019	0.135	0.423
pocket	persist	1	6	0.167	0.954	0.014	0.000	0.816

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
pocket	persist	2	6	0.000	0.956	0.016	0.000	0.815
pocket	persist	3	6	0.167	0.950	0.016	0.000	0.817
pocket	persist	4	6	0.000	0.958	0.013	0.000	0.812
pocket	persist	5	6	0.333	0.903	0.020	0.000	0.678
pocket	persist	6	6	0.167	0.950	0.016	0.000	0.815
pocket	persist	7	6	0.167	0.951	0.014	0.000	0.818
actuator	oracle	0	6	0.333	1.000	0.000	0.000	0.758
actuator	oracle	1	6	0.167	1.000	0.000	0.000	0.654
actuator	oracle	2	6	0.167	1.000	0.000	0.000	0.681
actuator	oracle	3	6	0.167	1.000	0.000	0.000	0.768
actuator	oracle	4	6	0.167	1.000	0.000	0.000	0.720
actuator	oracle	5	6	0.167	1.000	0.000	0.000	0.651
actuator	oracle	6	6	0.333	1.000	0.000	0.000	0.689
actuator	oracle	7	6	0.000	1.000	0.000	0.000	0.640
extreme	oracle	0	6	0.167	1.000	0.000	0.000	0.849
extreme	oracle	1	6	0.500	1.000	0.000	0.000	0.871
extreme	oracle	2	6	0.000	1.000	0.000	0.000	0.395
extreme	oracle	3	6	0.000	1.000	0.000	0.000	0.567
extreme	oracle	4	6	0.000	1.000	0.000	0.000	0.815
extreme	oracle	5	6	0.000	1.000	0.000	0.000	0.617
extreme	oracle	6	6	0.000	1.000	0.000	0.000	0.528
extreme	oracle	7	6	0.000	1.000	0.000	0.000	0.712
combined	oracle	0	6	0.333	1.000	0.000	0.000	0.780
combined	oracle	1	6	0.333	1.000	0.000	0.000	0.734
combined	oracle	2	6	0.167	1.000	0.000	0.000	0.815
combined	oracle	3	6	0.167	1.000	0.000	0.000	0.624
combined	oracle	4	6	0.167	1.000	0.000	0.000	0.696
combined	oracle	5	6	0.333	1.000	0.000	0.000	0.599
combined	oracle	6	6	0.000	1.000	0.000	0.000	0.762
combined	oracle	7	6	0.167	1.000	0.000	0.000	0.664
chain	oracle	0	6	0.000	1.000	0.000	0.000	0.723
chain	oracle	1	6	0.167	1.000	0.000	0.000	0.704
chain	oracle	2	6	0.167	1.000	0.000	0.000	0.803
chain	oracle	3	6	0.000	1.000	0.000	0.000	0.567
chain	oracle	4	6	0.167	1.000	0.000	0.000	0.765
chain	oracle	5	6	0.167	1.000	0.000	0.000	0.798
chain	oracle	6	6	0.167	1.000	0.000	0.000	0.649
chain	oracle	7	6	0.167	1.000	0.000	0.000	0.670

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
noise	oracle	0	6	0.167	1.000	0.000	0.000	0.770
noise	oracle	1	6	0.167	1.000	0.000	0.000	0.540
noise	oracle	2	6	0.333	1.000	0.000	0.000	0.592
noise	oracle	3	6	0.167	1.000	0.000	0.000	0.750
noise	oracle	4	6	0.000	1.000	0.000	0.000	0.736
noise	oracle	5	6	0.167	1.000	0.000	0.000	0.809
noise	oracle	6	6	0.333	1.000	0.000	0.000	0.814
noise	oracle	7	6	0.333	1.000	0.000	0.000	0.770
delayed	oracle	0	6	0.000	1.000	0.000	0.000	0.496
delayed	oracle	1	6	0.167	1.000	0.000	0.000	0.387
delayed	oracle	2	6	0.000	1.000	0.000	0.000	0.641
delayed	oracle	3	6	0.000	1.000	0.000	0.000	0.344
delayed	oracle	4	6	0.000	1.000	0.000	0.000	0.510
delayed	oracle	5	6	0.000	1.000	0.000	0.000	0.339
delayed	oracle	6	6	0.000	1.000	0.000	0.000	0.393
delayed	oracle	7	6	0.000	1.000	0.000	0.000	0.647
distractor	oracle	0	6	0.333	1.000	0.000	0.000	0.645
distractor	oracle	1	6	0.000	1.000	0.000	0.000	0.710
distractor	oracle	2	6	0.167	1.000	0.000	0.000	0.690
distractor	oracle	3	6	0.000	1.000	0.000	0.000	0.547
distractor	oracle	4	6	0.000	1.000	0.000	0.000	0.579
distractor	oracle	5	6	0.167	1.000	0.000	0.000	0.580
distractor	oracle	6	6	0.167	1.000	0.000	0.000	0.581
distractor	oracle	7	6	0.167	1.000	0.000	0.000	0.710
wall-jam	oracle	0	6	0.000	1.000	0.000	0.000	0.444
wall-jam	oracle	1	6	0.000	1.000	0.000	0.000	0.284
wall-jam	oracle	2	6	0.000	1.000	0.000	0.000	0.326
wall-jam	oracle	3	6	0.167	1.000	0.000	0.000	0.302
wall-jam	oracle	4	6	0.000	1.000	0.000	0.000	0.452
wall-jam	oracle	5	6	0.167	1.000	0.000	0.000	0.582
wall-jam	oracle	6	6	0.167	1.000	0.000	0.000	0.632
wall-jam	oracle	7	6	0.000	1.000	0.000	0.000	0.329
fixture	oracle	0	6	0.167	1.000	0.000	0.000	0.618
fixture	oracle	1	6	0.167	1.000	0.000	0.000	0.536
fixture	oracle	2	6	0.000	1.000	0.000	0.000	0.429
fixture	oracle	3	6	0.333	1.000	0.000	0.000	0.612
fixture	oracle	4	6	0.167	1.000	0.000	0.000	0.682
fixture	oracle	5	6	0.000	1.000	0.000	0.000	0.670

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
fixture	oracle	6	6	0.000	1.000	0.000	0.000	0.640
fixture	oracle	7	6	0.000	1.000	0.000	0.000	0.368
friction	oracle	0	6	0.167	1.000	0.000	0.000	0.667
friction	oracle	1	6	0.500	1.000	0.000	0.000	0.797
friction	oracle	2	6	0.333	1.000	0.000	0.000	0.715
friction	oracle	3	6	0.500	1.000	0.000	0.000	0.815
friction	oracle	4	6	0.333	1.000	0.000	0.000	0.805
friction	oracle	5	6	0.500	1.000	0.000	0.000	0.787
friction	oracle	6	6	0.167	1.000	0.000	0.000	0.682
friction	oracle	7	6	0.333	1.000	0.000	0.000	0.802
nominal	oracle	0	6	0.000	1.000	0.000	0.000	0.698
nominal	oracle	1	6	0.167	1.000	0.000	0.000	0.756
nominal	oracle	2	6	0.333	1.000	0.000	0.000	0.811
nominal	oracle	3	6	0.333	1.000	0.000	0.000	0.816
nominal	oracle	4	6	0.167	1.000	0.000	0.000	0.742
nominal	oracle	5	6	0.333	1.000	0.000	0.000	0.696
nominal	oracle	6	6	0.167	1.000	0.000	0.000	0.672
nominal	oracle	7	6	0.000	1.000	0.000	0.000	0.656
pocket	oracle	0	6	0.167	1.000	0.000	0.000	0.562
pocket	oracle	1	6	0.000	1.000	0.000	0.000	0.459
pocket	oracle	2	6	0.000	1.000	0.000	0.000	0.628
pocket	oracle	3	6	0.000	1.000	0.000	0.000	0.721
pocket	oracle	4	6	0.167	1.000	0.000	0.000	0.779
pocket	oracle	5	6	0.167	1.000	0.000	0.000	0.695
pocket	oracle	6	6	0.000	1.000	0.000	0.000	0.645
pocket	oracle	7	6	0.000	1.000	0.000	0.000	0.580
actuator	pairwise	0	6	0.000	0.640	0.165	0.753	0.113
actuator	pairwise	1	6	0.000	0.624	0.167	0.764	0.190
actuator	pairwise	2	6	0.000	0.696	0.136	0.521	0.352
actuator	pairwise	3	6	0.000	0.708	0.124	0.660	0.458
actuator	pairwise	4	6	0.000	0.572	0.133	0.660	0.654
actuator	pairwise	5	6	0.000	0.582	0.188	0.806	0.432
actuator	pairwise	6	6	0.000	0.633	0.159	0.809	0.435
actuator	pairwise	7	6	0.000	0.614	0.177	0.806	0.118
extreme	pairwise	0	6	0.000	0.833	0.064	0.413	0.261
extreme	pairwise	1	6	0.000	0.686	0.101	0.181	0.328
extreme	pairwise	2	6	0.000	0.767	0.073	0.139	0.000
extreme	pairwise	3	6	0.000	0.662	0.127	0.264	0.269

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
extreme	pairwise	4	6	0.000	0.840	0.063	0.389	0.198
extreme	pairwise	5	6	0.000	0.779	0.088	0.410	0.110
extreme	pairwise	6	6	0.000	0.681	0.099	0.135	0.336
extreme	pairwise	7	6	0.000	0.751	0.086	0.278	0.000
combined	pairwise	0	6	0.333	0.770	0.113	0.260	0.746
combined	pairwise	1	6	0.167	0.867	0.061	0.243	0.380
combined	pairwise	2	6	0.167	0.689	0.170	0.233	0.645
combined	pairwise	3	6	0.000	0.704	0.135	0.365	0.717
combined	pairwise	4	6	0.000	0.786	0.087	0.472	0.446
combined	pairwise	5	6	0.167	0.761	0.118	0.361	0.645
combined	pairwise	6	6	0.000	0.695	0.137	0.236	0.743
combined	pairwise	7	6	0.000	0.634	0.163	0.524	0.377
chain	pairwise	0	6	0.000	0.901	0.047	0.000	0.426
chain	pairwise	1	6	0.000	0.938	0.032	0.000	0.137
chain	pairwise	2	6	0.000	0.934	0.033	0.000	0.354
chain	pairwise	3	6	0.000	0.909	0.050	0.000	0.463
chain	pairwise	4	6	0.000	0.877	0.032	0.000	0.185
chain	pairwise	5	6	0.000	0.867	0.076	0.000	0.607
chain	pairwise	6	6	0.167	0.860	0.071	0.000	0.381
chain	pairwise	7	6	0.167	0.854	0.081	0.000	0.510
noise	pairwise	0	6	0.667	0.676	0.190	0.000	0.811
noise	pairwise	1	6	0.167	0.654	0.167	0.000	0.781
noise	pairwise	2	6	0.667	0.690	0.180	0.000	0.797
noise	pairwise	3	6	0.167	0.700	0.169	0.000	0.624
noise	pairwise	4	6	0.500	0.644	0.165	0.135	0.774
noise	pairwise	5	6	0.333	0.690	0.197	0.000	0.786
noise	pairwise	6	6	0.333	0.632	0.182	0.000	0.802
noise	pairwise	7	6	0.167	0.681	0.149	0.000	0.662
delayed	pairwise	0	6	0.000	0.516	0.230	0.812	0.113
delayed	pairwise	1	6	0.000	0.513	0.242	0.812	0.000
delayed	pairwise	2	6	0.000	0.622	0.185	0.816	0.143
delayed	pairwise	3	6	0.000	0.553	0.217	0.809	0.000
delayed	pairwise	4	6	0.000	0.483	0.233	0.806	0.143
delayed	pairwise	5	6	0.000	0.514	0.241	0.809	0.000
delayed	pairwise	6	6	0.000	0.468	0.229	0.812	0.009
delayed	pairwise	7	6	0.000	0.428	0.207	0.812	0.359
distractor	pairwise	0	6	0.000	0.926	0.038	0.000	0.159
distractor	pairwise	1	6	0.167	0.680	0.143	0.000	0.444

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
distractor	pairwise	2	6	0.167	0.736	0.135	0.000	0.530
distractor	pairwise	3	6	0.000	0.866	0.070	0.000	0.389
distractor	pairwise	4	6	0.167	0.829	0.099	0.000	0.554
distractor	pairwise	5	6	0.000	0.925	0.038	0.000	0.141
distractor	pairwise	6	6	0.000	0.941	0.023	0.000	0.036
distractor	pairwise	7	6	0.167	0.801	0.107	0.000	0.426
wall-jam	pairwise	0	6	0.000	0.834	0.038	0.000	0.000
wall-jam	pairwise	1	6	0.000	0.644	0.083	0.000	0.000
wall-jam	pairwise	2	6	0.000	0.850	0.035	0.000	0.000
wall-jam	pairwise	3	6	0.000	0.592	0.097	0.000	0.000
wall-jam	pairwise	4	6	0.000	0.550	0.091	0.000	0.000
wall-jam	pairwise	5	6	0.000	0.414	0.127	0.000	0.000
wall-jam	pairwise	6	6	0.000	0.478	0.101	0.000	0.000
wall-jam	pairwise	7	6	0.000	0.653	0.069	0.000	0.000
fixture	pairwise	0	6	0.000	0.804	0.074	0.354	0.093
fixture	pairwise	1	6	0.000	0.757	0.093	0.562	0.346
fixture	pairwise	2	6	0.000	0.705	0.096	0.295	0.097
fixture	pairwise	3	6	0.000	0.660	0.096	0.219	0.265
fixture	pairwise	4	6	0.000	0.806	0.086	0.358	0.070
fixture	pairwise	5	6	0.000	0.833	0.063	0.253	0.175
fixture	pairwise	6	6	0.000	0.919	0.031	0.149	0.186
fixture	pairwise	7	6	0.000	0.773	0.100	0.413	0.000
friction	pairwise	0	6	0.000	0.689	0.168	0.000	0.654
friction	pairwise	1	6	0.333	0.752	0.139	0.000	0.780
friction	pairwise	2	6	0.000	0.724	0.143	0.000	0.791
friction	pairwise	3	6	0.167	0.750	0.138	0.000	0.526
friction	pairwise	4	6	0.167	0.794	0.130	0.000	0.676
friction	pairwise	5	6	0.333	0.782	0.143	0.000	0.683
friction	pairwise	6	6	0.333	0.785	0.130	0.000	0.823
friction	pairwise	7	6	0.333	0.702	0.168	0.000	0.526
nominal	pairwise	0	6	0.167	0.617	0.155	0.000	0.805
nominal	pairwise	1	6	0.333	0.685	0.135	0.000	0.741
nominal	pairwise	2	6	0.333	0.701	0.141	0.000	0.694
nominal	pairwise	3	6	0.333	0.617	0.160	0.000	0.690
nominal	pairwise	4	6	0.167	0.679	0.159	0.000	0.813
nominal	pairwise	5	6	0.333	0.806	0.123	0.000	0.818
nominal	pairwise	6	6	0.167	0.780	0.130	0.000	0.813
nominal	pairwise	7	6	0.167	0.584	0.191	0.000	0.800

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
pocket	pairwise	0	6	0.167	0.638	0.151	0.135	0.423
pocket	pairwise	1	6	0.000	0.646	0.209	0.000	0.809
pocket	pairwise	2	6	0.167	0.742	0.177	0.000	0.820
pocket	pairwise	3	6	0.333	0.753	0.170	0.000	0.823
pocket	pairwise	4	6	0.167	0.535	0.234	0.000	0.772
pocket	pairwise	5	6	0.167	0.486	0.221	0.000	0.654
pocket	pairwise	6	6	0.167	0.632	0.220	0.000	0.809
pocket	pairwise	7	6	0.667	0.715	0.201	0.000	0.839
actuator	rf-topo	0	6	0.000	0.429	0.162	0.764	0.114
actuator	rf-topo	1	6	0.000	0.518	0.161	0.764	0.209
actuator	rf-topo	2	6	0.000	0.541	0.142	0.635	0.350
actuator	rf-topo	3	6	0.000	0.436	0.172	0.660	0.457
actuator	rf-topo	4	6	0.000	0.283	0.207	0.663	0.649
actuator	rf-topo	5	6	0.000	0.319	0.223	0.670	0.448
actuator	rf-topo	6	6	0.000	0.419	0.202	0.809	0.439
actuator	rf-topo	7	6	0.000	0.526	0.160	0.806	0.075
extreme	rf-topo	0	6	0.000	0.560	0.084	0.347	0.121
extreme	rf-topo	1	6	0.000	0.569	0.084	0.250	0.029
extreme	rf-topo	2	6	0.000	0.490	0.074	0.111	0.000
extreme	rf-topo	3	6	0.000	0.614	0.097	0.236	0.274
extreme	rf-topo	4	6	0.000	0.557	0.076	0.292	0.000
extreme	rf-topo	5	6	0.000	0.409	0.100	0.274	0.249
extreme	rf-topo	6	6	0.000	0.594	0.062	0.132	0.085
extreme	rf-topo	7	6	0.000	0.775	0.052	0.226	0.000
combined	rf-topo	0	6	0.333	0.356	0.218	0.260	0.792
combined	rf-topo	1	6	0.167	0.453	0.137	0.236	0.518
combined	rf-topo	2	6	0.000	0.495	0.175	0.236	0.621
combined	rf-topo	3	6	0.000	0.513	0.171	0.365	0.611
combined	rf-topo	4	6	0.000	0.387	0.122	0.490	0.428
combined	rf-topo	5	6	0.167	0.335	0.194	0.361	0.627
combined	rf-topo	6	6	0.000	0.426	0.169	0.236	0.770
combined	rf-topo	7	6	0.000	0.301	0.215	0.503	0.394
chain	rf-topo	0	6	0.000	0.107	0.144	0.000	0.427
chain	rf-topo	1	6	0.000	0.409	0.104	0.000	0.137
chain	rf-topo	2	6	0.000	0.137	0.139	0.000	0.357
chain	rf-topo	3	6	0.000	0.247	0.154	0.000	0.472
chain	rf-topo	4	6	0.000	0.285	0.091	0.000	0.185
chain	rf-topo	5	6	0.000	0.262	0.183	0.000	0.607

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
chain	rf-topo	6	6	0.167	0.337	0.139	0.000	0.381
chain	rf-topo	7	6	0.167	0.263	0.178	0.000	0.502
noise	rf-topo	0	6	0.333	0.592	0.192	0.000	0.793
noise	rf-topo	1	6	0.000	0.574	0.199	0.000	0.791
noise	rf-topo	2	6	0.167	0.590	0.148	0.118	0.744
noise	rf-topo	3	6	0.167	0.608	0.165	0.000	0.647
noise	rf-topo	4	6	0.333	0.512	0.199	0.135	0.774
noise	rf-topo	5	6	0.167	0.585	0.198	0.000	0.784
noise	rf-topo	6	6	0.167	0.592	0.192	0.000	0.790
noise	rf-topo	7	6	0.333	0.566	0.195	0.000	0.777
delayed	rf-topo	0	6	0.000	0.078	0.259	0.812	0.114
delayed	rf-topo	1	6	0.000	0.100	0.227	0.812	0.010
delayed	rf-topo	2	6	0.000	0.314	0.203	0.677	0.241
delayed	rf-topo	3	6	0.000	0.113	0.230	0.809	0.000
delayed	rf-topo	4	6	0.000	0.197	0.227	0.809	0.110
delayed	rf-topo	5	6	0.000	0.135	0.246	0.812	0.000
delayed	rf-topo	6	6	0.000	0.200	0.209	0.812	0.000
delayed	rf-topo	7	6	0.000	0.197	0.249	0.812	0.238
distractor	rf-topo	0	6	0.000	0.804	0.058	0.000	0.159
distractor	rf-topo	1	6	0.000	0.655	0.135	0.000	0.438
distractor	rf-topo	2	6	0.167	0.666	0.153	0.000	0.542
distractor	rf-topo	3	6	0.000	0.707	0.118	0.000	0.468
distractor	rf-topo	4	6	0.167	0.616	0.162	0.000	0.554
distractor	rf-topo	5	6	0.000	0.870	0.048	0.000	0.141
distractor	rf-topo	6	6	0.000	0.911	0.016	0.000	0.036
distractor	rf-topo	7	6	0.333	0.710	0.124	0.000	0.427
wall-jam	rf-topo	0	6	0.000	0.847	0.020	0.000	0.000
wall-jam	rf-topo	1	6	0.000	0.816	0.024	0.000	0.000
wall-jam	rf-topo	2	6	0.000	0.976	0.003	0.000	0.000
wall-jam	rf-topo	3	6	0.000	0.912	0.013	0.000	0.000
wall-jam	rf-topo	4	6	0.000	0.800	0.026	0.000	0.000
wall-jam	rf-topo	5	6	0.000	0.821	0.023	0.000	0.000
wall-jam	rf-topo	6	6	0.000	0.853	0.020	0.000	0.000
wall-jam	rf-topo	7	6	0.000	0.847	0.020	0.000	0.000
fixture	rf-topo	0	6	0.000	0.251	0.120	0.354	0.094
fixture	rf-topo	1	6	0.000	0.375	0.138	0.569	0.273
fixture	rf-topo	2	6	0.000	0.397	0.102	0.295	0.098
fixture	rf-topo	3	6	0.000	0.255	0.117	0.194	0.175

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
fixture	rf-topo	4	6	0.000	0.560	0.111	0.448	0.085
fixture	rf-topo	5	6	0.000	0.579	0.072	0.253	0.000
fixture	rf-topo	6	6	0.000	0.605	0.060	0.160	0.089
fixture	rf-topo	7	6	0.000	0.469	0.109	0.441	0.000
friction	rf-topo	0	6	0.000	0.222	0.196	0.000	0.655
friction	rf-topo	1	6	0.167	0.354	0.166	0.000	0.776
friction	rf-topo	2	6	0.167	0.303	0.175	0.000	0.798
friction	rf-topo	3	6	0.167	0.311	0.173	0.000	0.533
friction	rf-topo	4	6	0.167	0.390	0.160	0.000	0.676
friction	rf-topo	5	6	0.333	0.380	0.189	0.000	0.683
friction	rf-topo	6	6	0.333	0.291	0.174	0.000	0.821
friction	rf-topo	7	6	0.333	0.383	0.186	0.000	0.654
nominal	rf-topo	0	6	0.167	0.514	0.232	0.000	0.814
nominal	rf-topo	1	6	0.333	0.561	0.194	0.000	0.758
nominal	rf-topo	2	6	0.333	0.512	0.229	0.000	0.814
nominal	rf-topo	3	6	0.333	0.505	0.231	0.000	0.816
nominal	rf-topo	4	6	0.333	0.502	0.231	0.000	0.817
nominal	rf-topo	5	6	0.333	0.501	0.231	0.000	0.818
nominal	rf-topo	6	6	0.167	0.505	0.234	0.000	0.816
nominal	rf-topo	7	6	0.000	0.509	0.237	0.000	0.811
pocket	rf-topo	0	6	0.167	0.652	0.142	0.135	0.423
pocket	rf-topo	1	6	0.167	0.520	0.231	0.000	0.816
pocket	rf-topo	2	6	0.000	0.510	0.237	0.000	0.815
pocket	rf-topo	3	6	0.167	0.529	0.228	0.000	0.817
pocket	rf-topo	4	6	0.000	0.527	0.233	0.000	0.812
pocket	rf-topo	5	6	0.333	0.574	0.195	0.000	0.678
pocket	rf-topo	6	6	0.167	0.514	0.233	0.000	0.815
pocket	rf-topo	7	6	0.167	0.519	0.232	0.000	0.818
actuator	risk-averse	0	6	0.000	0.483	0.148	0.503	0.161
actuator	risk-averse	1	6	0.000	0.432	0.151	0.764	0.387
actuator	risk-averse	2	6	0.167	0.449	0.159	0.392	0.357
actuator	risk-averse	3	6	0.000	0.332	0.137	0.271	0.436
actuator	risk-averse	4	6	0.000	0.211	0.149	0.135	0.796
actuator	risk-averse	5	6	0.000	0.146	0.204	0.267	0.299
actuator	risk-averse	6	6	0.000	0.409	0.132	0.535	0.336
actuator	risk-averse	7	6	0.000	0.467	0.158	0.531	0.142
extreme	risk-averse	0	6	0.000	0.159	0.188	0.000	0.343
extreme	risk-averse	1	6	0.000	0.518	0.125	0.000	0.415

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
extreme	risk-averse	2	6	0.000	0.385	0.166	0.000	0.150
extreme	risk-averse	3	6	0.000	0.294	0.154	0.000	0.501
extreme	risk-averse	4	6	0.000	0.289	0.153	0.000	0.399
extreme	risk-averse	5	6	0.000	0.554	0.129	0.000	0.563
extreme	risk-averse	6	6	0.000	0.419	0.114	0.000	0.568
extreme	risk-averse	7	6	0.000	0.631	0.082	0.000	0.280
combined	risk-averse	0	6	0.167	0.272	0.157	0.132	0.558
combined	risk-averse	1	6	0.000	0.149	0.155	0.132	0.469
combined	risk-averse	2	6	0.167	0.659	0.129	0.111	0.622
combined	risk-averse	3	6	0.000	0.352	0.155	0.215	0.428
combined	risk-averse	4	6	0.000	0.337	0.141	0.253	0.622
combined	risk-averse	5	6	0.167	0.363	0.131	0.111	0.722
combined	risk-averse	6	6	0.000	0.158	0.230	0.000	0.597
combined	risk-averse	7	6	0.000	0.651	0.142	0.372	0.326
chain	risk-averse	0	6	0.000	0.389	0.089	0.000	0.399
chain	risk-averse	1	6	0.000	0.641	0.072	0.000	0.406
chain	risk-averse	2	6	0.167	0.511	0.101	0.000	0.443
chain	risk-averse	3	6	0.000	0.556	0.098	0.000	0.451
chain	risk-averse	4	6	0.000	0.277	0.107	0.000	0.392
chain	risk-averse	5	6	0.000	0.415	0.118	0.000	0.613
chain	risk-averse	6	6	0.167	0.660	0.069	0.000	0.547
chain	risk-averse	7	6	0.000	0.399	0.151	0.000	0.577
noise	risk-averse	0	6	0.167	0.475	0.141	0.000	0.532
noise	risk-averse	1	6	0.000	0.604	0.176	0.000	0.807
noise	risk-averse	2	6	0.500	0.677	0.139	0.000	0.825
noise	risk-averse	3	6	0.333	0.643	0.165	0.000	0.780
noise	risk-averse	4	6	0.333	0.335	0.167	0.000	0.646
noise	risk-averse	5	6	0.167	0.564	0.157	0.000	0.631
noise	risk-averse	6	6	0.167	0.364	0.169	0.000	0.552
noise	risk-averse	7	6	0.500	0.580	0.161	0.000	0.776
delayed	risk-averse	0	6	0.000	0.102	0.218	0.052	0.547
delayed	risk-averse	1	6	0.000	0.096	0.179	0.000	0.492
delayed	risk-averse	2	6	0.000	0.091	0.168	0.000	0.304
delayed	risk-averse	3	6	0.000	0.098	0.190	0.000	0.457
delayed	risk-averse	4	6	0.000	0.079	0.178	0.135	0.457
delayed	risk-averse	5	6	0.000	0.075	0.179	0.000	0.507
delayed	risk-averse	6	6	0.000	0.155	0.182	0.000	0.394
delayed	risk-averse	7	6	0.000	0.069	0.159	0.000	0.607

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
distractor	risk-averse	0	6	0.000	0.444	0.137	0.000	0.333
distractor	risk-averse	1	6	0.000	0.251	0.164	0.000	0.292
distractor	risk-averse	2	6	0.000	0.256	0.197	0.000	0.366
distractor	risk-averse	3	6	0.000	0.228	0.166	0.000	0.210
distractor	risk-averse	4	6	0.000	0.270	0.180	0.000	0.419
distractor	risk-averse	5	6	0.000	0.282	0.153	0.000	0.129
distractor	risk-averse	6	6	0.000	0.109	0.205	0.000	0.030
distractor	risk-averse	7	6	0.000	0.161	0.186	0.000	0.274
wall-jam	risk-averse	0	6	0.000	0.374	0.137	0.000	0.187
wall-jam	risk-averse	1	6	0.000	0.155	0.204	0.000	0.351
wall-jam	risk-averse	2	6	0.000	0.374	0.145	0.000	0.320
wall-jam	risk-averse	3	6	0.000	0.462	0.145	0.000	0.244
wall-jam	risk-averse	4	6	0.000	0.323	0.167	0.000	0.421
wall-jam	risk-averse	5	6	0.000	0.225	0.187	0.000	0.199
wall-jam	risk-averse	6	6	0.000	0.209	0.223	0.000	0.332
wall-jam	risk-averse	7	6	0.000	0.189	0.249	0.125	0.382
fixture	risk-averse	0	6	0.000	0.589	0.144	0.267	0.262
fixture	risk-averse	1	6	0.000	0.650	0.104	0.396	0.112
fixture	risk-averse	2	6	0.000	0.645	0.109	0.174	0.128
fixture	risk-averse	3	6	0.000	0.506	0.129	0.170	0.440
fixture	risk-averse	4	6	0.000	0.574	0.115	0.170	0.165
fixture	risk-averse	5	6	0.000	0.709	0.102	0.097	0.303
fixture	risk-averse	6	6	0.000	0.579	0.146	0.000	0.486
fixture	risk-averse	7	6	0.000	0.629	0.086	0.226	0.253
friction	risk-averse	0	6	0.000	0.350	0.158	0.000	0.619
friction	risk-averse	1	6	0.000	0.107	0.163	0.000	0.629
friction	risk-averse	2	6	0.167	0.382	0.148	0.000	0.734
friction	risk-averse	3	6	0.333	0.600	0.126	0.000	0.510
friction	risk-averse	4	6	0.167	0.492	0.141	0.000	0.654
friction	risk-averse	5	6	0.333	0.598	0.118	0.000	0.700
friction	risk-averse	6	6	0.167	0.237	0.172	0.000	0.721
friction	risk-averse	7	6	0.000	0.284	0.156	0.000	0.563
nominal	risk-averse	0	6	0.167	0.717	0.126	0.000	0.708
nominal	risk-averse	1	6	0.000	0.331	0.194	0.000	0.598
nominal	risk-averse	2	6	0.333	0.511	0.167	0.000	0.691
nominal	risk-averse	3	6	0.500	0.606	0.137	0.000	0.655
nominal	risk-averse	4	6	0.333	0.403	0.174	0.000	0.644
nominal	risk-averse	5	6	0.333	0.603	0.159	0.000	0.794

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
nominal	risk-averse	6	6	0.167	0.368	0.199	0.000	0.745
nominal	risk-averse	7	6	0.333	0.531	0.173	0.000	0.815
pocket	risk-averse	0	6	0.000	0.227	0.203	0.000	0.670
pocket	risk-averse	1	6	0.500	0.684	0.144	0.000	0.856
pocket	risk-averse	2	6	0.000	0.523	0.151	0.139	0.741
pocket	risk-averse	3	6	0.500	0.540	0.169	0.000	0.812
pocket	risk-averse	4	6	0.167	0.583	0.146	0.000	0.599
pocket	risk-averse	5	6	0.167	0.353	0.189	0.000	0.614
pocket	risk-averse	6	6	0.000	0.623	0.147	0.000	0.713
pocket	risk-averse	7	6	0.167	0.570	0.143	0.000	0.696
actuator	robust-mpc	0	6	0.000	0.747	0.142	0.753	0.113
actuator	robust-mpc	1	6	0.000	0.761	0.135	0.764	0.200
actuator	robust-mpc	2	6	0.000	0.816	0.105	0.521	0.356
actuator	robust-mpc	3	6	0.000	0.934	0.030	0.663	0.470
actuator	robust-mpc	4	6	0.000	0.833	0.056	0.663	0.648
actuator	robust-mpc	5	6	0.000	0.797	0.120	0.670	0.444
actuator	robust-mpc	6	6	0.000	0.895	0.058	0.806	0.431
actuator	robust-mpc	7	6	0.000	0.886	0.060	0.806	0.064
extreme	robust-mpc	0	6	0.000	0.577	0.211	0.337	0.120
extreme	robust-mpc	1	6	0.000	0.628	0.166	0.250	0.000
extreme	robust-mpc	2	6	0.000	0.466	0.156	0.052	0.000
extreme	robust-mpc	3	6	0.000	0.703	0.116	0.219	0.210
extreme	robust-mpc	4	6	0.000	0.482	0.243	0.295	0.000
extreme	robust-mpc	5	6	0.000	0.649	0.174	0.267	0.111
extreme	robust-mpc	6	6	0.000	0.504	0.280	0.132	0.083
extreme	robust-mpc	7	6	0.000	0.737	0.111	0.295	0.000
combined	robust-mpc	0	6	0.333	0.896	0.041	0.260	0.792
combined	robust-mpc	1	6	0.167	0.822	0.079	0.247	0.372
combined	robust-mpc	2	6	0.000	0.872	0.068	0.236	0.620
combined	robust-mpc	3	6	0.000	0.810	0.096	0.365	0.624
combined	robust-mpc	4	6	0.000	0.754	0.118	0.545	0.347
combined	robust-mpc	5	6	0.167	0.843	0.082	0.361	0.643
combined	robust-mpc	6	6	0.000	0.831	0.080	0.236	0.764
combined	robust-mpc	7	6	0.000	0.834	0.086	0.514	0.394
chain	robust-mpc	0	6	0.000	0.920	0.033	0.000	0.426
chain	robust-mpc	1	6	0.000	0.979	0.004	0.000	0.137
chain	robust-mpc	2	6	0.000	0.955	0.013	0.000	0.351
chain	robust-mpc	3	6	0.000	0.966	0.008	0.000	0.464

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
chain	robust-mpc	4	6	0.000	0.969	0.009	0.000	0.185
chain	robust-mpc	5	6	0.000	0.973	0.009	0.000	0.607
chain	robust-mpc	6	6	0.167	0.924	0.030	0.000	0.379
chain	robust-mpc	7	6	0.167	0.972	0.006	0.000	0.503
noise	robust-mpc	0	6	0.333	0.956	0.015	0.000	0.793
noise	robust-mpc	1	6	0.000	0.949	0.028	0.000	0.792
noise	robust-mpc	2	6	0.167	0.883	0.052	0.118	0.743
noise	robust-mpc	3	6	0.333	0.914	0.047	0.000	0.680
noise	robust-mpc	4	6	0.333	0.942	0.024	0.135	0.772
noise	robust-mpc	5	6	0.167	0.960	0.013	0.000	0.784
noise	robust-mpc	6	6	0.167	0.905	0.056	0.000	0.792
noise	robust-mpc	7	6	0.333	0.932	0.028	0.000	0.778
delayed	robust-mpc	0	6	0.000	0.932	0.034	0.809	0.225
delayed	robust-mpc	1	6	0.000	0.819	0.118	0.812	0.000
delayed	robust-mpc	2	6	0.000	0.873	0.083	0.816	0.142
delayed	robust-mpc	3	6	0.000	0.892	0.069	0.812	0.000
delayed	robust-mpc	4	6	0.000	0.869	0.084	0.806	0.108
delayed	robust-mpc	5	6	0.000	0.834	0.102	0.812	0.000
delayed	robust-mpc	6	6	0.000	0.855	0.082	0.812	0.000
delayed	robust-mpc	7	6	0.000	0.839	0.105	0.812	0.116
distractor	robust-mpc	0	6	0.000	0.873	0.043	0.000	0.159
distractor	robust-mpc	1	6	0.000	0.959	0.011	0.000	0.438
distractor	robust-mpc	2	6	0.167	0.834	0.071	0.000	0.542
distractor	robust-mpc	3	6	0.000	0.946	0.016	0.000	0.438
distractor	robust-mpc	4	6	0.167	0.925	0.030	0.000	0.554
distractor	robust-mpc	5	6	0.000	0.773	0.080	0.000	0.141
distractor	robust-mpc	6	6	0.000	0.828	0.060	0.000	0.035
distractor	robust-mpc	7	6	0.167	0.866	0.056	0.000	0.426
wall-jam	robust-mpc	0	6	0.000	0.972	0.004	0.000	0.000
wall-jam	robust-mpc	1	6	0.000	0.968	0.004	0.000	0.000
wall-jam	robust-mpc	2	6	0.000	0.972	0.004	0.000	0.000
wall-jam	robust-mpc	3	6	0.000	0.944	0.009	0.000	0.000
wall-jam	robust-mpc	4	6	0.000	0.914	0.012	0.000	0.000
wall-jam	robust-mpc	5	6	0.000	0.931	0.010	0.000	0.000
wall-jam	robust-mpc	6	6	0.000	0.911	0.012	0.000	0.000
wall-jam	robust-mpc	7	6	0.000	0.990	0.001	0.000	0.000
fixture	robust-mpc	0	6	0.000	0.657	0.180	0.354	0.093
fixture	robust-mpc	1	6	0.000	0.736	0.122	0.566	0.100

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
fixture	robust-mpc	2	6	0.000	0.733	0.106	0.295	0.098
fixture	robust-mpc	3	6	0.000	0.760	0.132	0.201	0.195
fixture	robust-mpc	4	6	0.000	0.689	0.137	0.354	0.069
fixture	robust-mpc	5	6	0.000	0.502	0.294	0.253	0.000
fixture	robust-mpc	6	6	0.000	0.627	0.191	0.163	0.089
fixture	robust-mpc	7	6	0.000	0.631	0.197	0.438	0.000
friction	robust-mpc	0	6	0.000	0.908	0.051	0.000	0.654
friction	robust-mpc	1	6	0.167	0.880	0.060	0.000	0.777
friction	robust-mpc	2	6	0.167	0.938	0.021	0.000	0.798
friction	robust-mpc	3	6	0.167	0.958	0.009	0.000	0.532
friction	robust-mpc	4	6	0.167	0.907	0.036	0.000	0.801
friction	robust-mpc	5	6	0.333	0.972	0.007	0.000	0.683
friction	robust-mpc	6	6	0.333	0.923	0.032	0.000	0.821
friction	robust-mpc	7	6	0.333	0.955	0.012	0.000	0.662
nominal	robust-mpc	0	6	0.167	0.984	0.003	0.000	0.814
nominal	robust-mpc	1	6	0.333	0.833	0.037	0.000	0.746
nominal	robust-mpc	2	6	0.333	0.971	0.006	0.000	0.814
nominal	robust-mpc	3	6	0.333	0.977	0.005	0.000	0.816
nominal	robust-mpc	4	6	0.333	0.979	0.003	0.000	0.817
nominal	robust-mpc	5	6	0.333	0.978	0.004	0.000	0.818
nominal	robust-mpc	6	6	0.167	0.979	0.005	0.000	0.816
nominal	robust-mpc	7	6	0.000	0.986	0.003	0.000	0.811
pocket	robust-mpc	0	6	0.167	0.856	0.040	0.135	0.423
pocket	robust-mpc	1	6	0.167	0.966	0.010	0.000	0.816
pocket	robust-mpc	2	6	0.000	0.941	0.027	0.000	0.815
pocket	robust-mpc	3	6	0.167	0.979	0.005	0.000	0.817
pocket	robust-mpc	4	6	0.000	0.963	0.011	0.000	0.812
pocket	robust-mpc	5	6	0.333	0.910	0.021	0.000	0.678
pocket	robust-mpc	6	6	0.167	0.955	0.014	0.000	0.815
pocket	robust-mpc	7	6	0.167	0.977	0.006	0.000	0.818
actuator	state-dyn	0	6	0.000	0.889	0.050	0.753	0.114
actuator	state-dyn	1	6	0.000	0.891	0.047	0.764	0.210
actuator	state-dyn	2	6	0.000	0.869	0.075	0.635	0.351
actuator	state-dyn	3	6	0.000	0.943	0.035	0.660	0.463
actuator	state-dyn	4	6	0.167	0.931	0.030	0.663	0.612
actuator	state-dyn	5	6	0.000	0.895	0.068	0.670	0.449
actuator	state-dyn	6	6	0.000	0.967	0.018	0.677	0.544
actuator	state-dyn	7	6	0.000	0.948	0.027	0.806	0.117

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
extreme	state-dyn	0	6	0.000	0.667	0.150	0.330	0.121
extreme	state-dyn	1	6	0.000	0.746	0.108	0.333	0.000
extreme	state-dyn	2	6	0.000	0.726	0.089	0.080	0.000
extreme	state-dyn	3	6	0.000	0.775	0.104	0.243	0.277
extreme	state-dyn	4	6	0.000	0.624	0.165	0.264	0.000
extreme	state-dyn	5	6	0.000	0.688	0.151	0.267	0.111
extreme	state-dyn	6	6	0.000	0.642	0.192	0.267	0.140
extreme	state-dyn	7	6	0.000	0.779	0.091	0.243	0.000
combined	state-dyn	0	6	0.333	0.914	0.056	0.260	0.762
combined	state-dyn	1	6	0.167	0.862	0.061	0.240	0.520
combined	state-dyn	2	6	0.000	0.915	0.046	0.233	0.622
combined	state-dyn	3	6	0.000	0.876	0.075	0.365	0.603
combined	state-dyn	4	6	0.000	0.876	0.056	0.458	0.429
combined	state-dyn	5	6	0.167	0.875	0.076	0.358	0.642
combined	state-dyn	6	6	0.000	0.933	0.039	0.122	0.778
combined	state-dyn	7	6	0.000	0.879	0.075	0.517	0.483
chain	state-dyn	0	6	0.000	0.987	0.006	0.000	0.428
chain	state-dyn	1	6	0.000	0.996	0.002	0.000	0.137
chain	state-dyn	2	6	0.000	0.992	0.003	0.000	0.353
chain	state-dyn	3	6	0.000	0.985	0.006	0.000	0.463
chain	state-dyn	4	6	0.000	0.989	0.005	0.000	0.284
chain	state-dyn	5	6	0.000	0.965	0.024	0.000	0.607
chain	state-dyn	6	6	0.167	0.988	0.006	0.000	0.388
chain	state-dyn	7	6	0.167	0.993	0.003	0.000	0.502
noise	state-dyn	0	6	0.333	0.976	0.014	0.000	0.794
noise	state-dyn	1	6	0.000	0.908	0.078	0.000	0.816
noise	state-dyn	2	6	0.167	0.918	0.036	0.219	0.746
noise	state-dyn	3	6	0.167	0.957	0.032	0.000	0.647
noise	state-dyn	4	6	0.333	0.945	0.036	0.135	0.772
noise	state-dyn	5	6	0.167	0.961	0.029	0.000	0.784
noise	state-dyn	6	6	0.167	0.926	0.058	0.000	0.794
noise	state-dyn	7	6	0.333	0.960	0.028	0.000	0.778
delayed	state-dyn	0	6	0.000	0.996	0.002	0.812	0.113
delayed	state-dyn	1	6	0.000	0.997	0.001	0.812	0.000
delayed	state-dyn	2	6	0.000	0.992	0.004	0.816	0.144
delayed	state-dyn	3	6	0.000	0.984	0.007	0.809	0.000
delayed	state-dyn	4	6	0.000	0.991	0.004	0.809	0.108
delayed	state-dyn	5	6	0.000	0.997	0.001	0.812	0.000

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
delayed	state-dyn	6	6	0.000	0.995	0.003	0.812	0.000
delayed	state-dyn	7	6	0.000	0.994	0.003	0.812	0.118
distractor	state-dyn	0	6	0.000	0.900	0.047	0.000	0.159
distractor	state-dyn	1	6	0.000	0.981	0.007	0.000	0.438
distractor	state-dyn	2	6	0.167	0.921	0.028	0.000	0.542
distractor	state-dyn	3	6	0.000	0.969	0.011	0.000	0.437
distractor	state-dyn	4	6	0.167	0.975	0.010	0.000	0.554
distractor	state-dyn	5	6	0.000	0.858	0.051	0.000	0.141
distractor	state-dyn	6	6	0.000	0.860	0.051	0.000	0.035
distractor	state-dyn	7	6	0.167	0.935	0.026	0.000	0.426
wall-jam	state-dyn	0	6	0.000	0.999	0.000	0.000	0.000
wall-jam	state-dyn	1	6	0.000	0.955	0.006	0.000	0.000
wall-jam	state-dyn	2	6	0.000	0.993	0.001	0.000	0.000
wall-jam	state-dyn	3	6	0.000	0.946	0.008	0.000	0.000
wall-jam	state-dyn	4	6	0.000	0.935	0.010	0.000	0.000
wall-jam	state-dyn	5	6	0.000	0.918	0.013	0.000	0.000
wall-jam	state-dyn	6	6	0.000	0.944	0.007	0.000	0.000
wall-jam	state-dyn	7	6	0.000	0.949	0.008	0.000	0.000
fixture	state-dyn	0	6	0.000	0.728	0.127	0.354	0.092
fixture	state-dyn	1	6	0.000	0.790	0.091	0.573	0.099
fixture	state-dyn	2	6	0.000	0.784	0.073	0.295	0.099
fixture	state-dyn	3	6	0.000	0.791	0.129	0.201	0.195
fixture	state-dyn	4	6	0.000	0.735	0.112	0.354	0.068
fixture	state-dyn	5	6	0.000	0.658	0.161	0.253	0.000
fixture	state-dyn	6	6	0.000	0.684	0.157	0.142	0.090
fixture	state-dyn	7	6	0.000	0.738	0.116	0.434	0.000
friction	state-dyn	0	6	0.000	0.985	0.006	0.000	0.655
friction	state-dyn	1	6	0.167	0.977	0.009	0.000	0.776
friction	state-dyn	2	6	0.167	0.965	0.015	0.000	0.800
friction	state-dyn	3	6	0.167	0.986	0.008	0.000	0.532
friction	state-dyn	4	6	0.167	0.974	0.012	0.000	0.677
friction	state-dyn	5	6	0.333	0.962	0.023	0.000	0.683
friction	state-dyn	6	6	0.333	0.950	0.029	0.000	0.838
friction	state-dyn	7	6	0.333	0.983	0.009	0.000	0.655
nominal	state-dyn	0	6	0.167	0.883	0.099	0.000	0.814
nominal	state-dyn	1	6	0.333	0.839	0.071	0.000	0.727
nominal	state-dyn	2	6	0.333	0.902	0.080	0.000	0.814
nominal	state-dyn	3	6	0.333	0.877	0.095	0.000	0.816

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
nominal	state-dyn	4	6	0.333	0.883	0.099	0.000	0.817
nominal	state-dyn	5	6	0.333	0.889	0.093	0.000	0.818
nominal	state-dyn	6	6	0.167	0.922	0.064	0.000	0.816
nominal	state-dyn	7	6	0.000	0.862	0.116	0.000	0.811
pocket	state-dyn	0	6	0.167	0.974	0.007	0.135	0.423
pocket	state-dyn	1	6	0.167	0.966	0.023	0.000	0.816
pocket	state-dyn	2	6	0.000	0.935	0.045	0.000	0.815
pocket	state-dyn	3	6	0.167	0.946	0.044	0.000	0.817
pocket	state-dyn	4	6	0.000	0.980	0.012	0.000	0.812
pocket	state-dyn	5	6	0.333	0.926	0.045	0.000	0.678
pocket	state-dyn	6	6	0.167	0.966	0.026	0.000	0.815
pocket	state-dyn	7	6	0.167	0.902	0.082	0.000	0.818
actuator	no-memory	0	6	0.000	0.639	0.164	0.753	0.114
actuator	no-memory	1	6	0.000	0.626	0.166	0.764	0.186
actuator	no-memory	2	6	0.167	0.683	0.143	0.521	0.355
actuator	no-memory	3	6	0.000	0.707	0.124	0.660	0.467
actuator	no-memory	4	6	0.000	0.571	0.132	0.663	0.611
actuator	no-memory	5	6	0.000	0.473	0.243	0.670	0.468
actuator	no-memory	6	6	0.000	0.635	0.158	0.806	0.434
actuator	no-memory	7	6	0.000	0.612	0.179	0.806	0.118
extreme	no-memory	0	6	0.000	0.863	0.054	0.420	0.299
extreme	no-memory	1	6	0.000	0.706	0.089	0.205	0.306
extreme	no-memory	2	6	0.000	0.752	0.080	0.108	0.073
extreme	no-memory	3	6	0.000	0.558	0.138	0.205	0.311
extreme	no-memory	4	6	0.000	0.707	0.077	0.278	0.277
extreme	no-memory	5	6	0.000	0.819	0.069	0.271	0.111
extreme	no-memory	6	6	0.000	0.657	0.061	0.073	0.395
extreme	no-memory	7	6	0.000	0.746	0.095	0.319	0.000
combined	no-memory	0	6	0.333	0.776	0.112	0.260	0.727
combined	no-memory	1	6	0.167	0.874	0.056	0.247	0.520
combined	no-memory	2	6	0.000	0.775	0.118	0.233	0.529
combined	no-memory	3	6	0.000	0.656	0.148	0.365	0.597
combined	no-memory	4	6	0.000	0.771	0.092	0.594	0.383
combined	no-memory	5	6	0.167	0.812	0.099	0.226	0.606
combined	no-memory	6	6	0.000	0.721	0.128	0.236	0.767
combined	no-memory	7	6	0.000	0.622	0.163	0.517	0.386
chain	no-memory	0	6	0.000	0.900	0.047	0.000	0.433
chain	no-memory	1	6	0.000	0.935	0.033	0.000	0.137

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
chain	no-memory	2	6	0.000	0.931	0.034	0.000	0.352
chain	no-memory	3	6	0.000	0.910	0.050	0.000	0.463
chain	no-memory	4	6	0.000	0.865	0.033	0.000	0.185
chain	no-memory	5	6	0.000	0.865	0.076	0.000	0.607
chain	no-memory	6	6	0.167	0.922	0.038	0.000	0.245
chain	no-memory	7	6	0.167	0.856	0.080	0.000	0.502
noise	no-memory	0	6	0.333	0.600	0.191	0.000	0.745
noise	no-memory	1	6	0.000	0.554	0.205	0.000	0.669
noise	no-memory	2	6	0.500	0.720	0.161	0.118	0.766
noise	no-memory	3	6	0.167	0.738	0.152	0.000	0.637
noise	no-memory	4	6	0.333	0.651	0.156	0.135	0.770
noise	no-memory	5	6	0.333	0.696	0.185	0.000	0.787
noise	no-memory	6	6	0.333	0.636	0.179	0.000	0.778
noise	no-memory	7	6	0.167	0.680	0.146	0.000	0.769
delayed	no-memory	0	6	0.000	0.544	0.222	0.812	0.113
delayed	no-memory	1	6	0.000	0.466	0.231	0.812	0.058
delayed	no-memory	2	6	0.000	0.619	0.186	0.816	0.142
delayed	no-memory	3	6	0.000	0.556	0.216	0.809	0.000
delayed	no-memory	4	6	0.000	0.473	0.225	0.802	0.188
delayed	no-memory	5	6	0.000	0.475	0.254	0.812	0.000
delayed	no-memory	6	6	0.000	0.461	0.233	0.812	0.067
delayed	no-memory	7	6	0.000	0.471	0.213	0.812	0.231
distractor	no-memory	0	6	0.000	0.924	0.039	0.000	0.159
distractor	no-memory	1	6	0.167	0.707	0.128	0.000	0.442
distractor	no-memory	2	6	0.000	0.730	0.120	0.000	0.423
distractor	no-memory	3	6	0.167	0.793	0.099	0.000	0.372
distractor	no-memory	4	6	0.333	0.825	0.101	0.000	0.565
distractor	no-memory	5	6	0.000	0.922	0.039	0.000	0.141
distractor	no-memory	6	6	0.000	0.943	0.023	0.000	0.035
distractor	no-memory	7	6	0.167	0.811	0.105	0.000	0.426
wall-jam	no-memory	0	6	0.000	0.847	0.036	0.000	0.000
wall-jam	no-memory	1	6	0.000	0.690	0.062	0.000	0.000
wall-jam	no-memory	2	6	0.000	0.923	0.029	0.000	0.000
wall-jam	no-memory	3	6	0.000	0.392	0.121	0.000	0.000
wall-jam	no-memory	4	6	0.000	0.512	0.102	0.000	0.000
wall-jam	no-memory	5	6	0.000	0.497	0.105	0.000	0.000
wall-jam	no-memory	6	6	0.000	0.688	0.063	0.000	0.000
wall-jam	no-memory	7	6	0.000	0.788	0.046	0.000	0.000

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
fixture	no-memory	0	6	0.000	0.804	0.074	0.354	0.093
fixture	no-memory	1	6	0.000	0.725	0.123	0.576	0.098
fixture	no-memory	2	6	0.000	0.700	0.099	0.295	0.098
fixture	no-memory	3	6	0.000	0.652	0.098	0.201	0.194
fixture	no-memory	4	6	0.000	0.783	0.094	0.351	0.174
fixture	no-memory	5	6	0.000	0.818	0.069	0.253	0.000
fixture	no-memory	6	6	0.000	0.915	0.033	0.160	0.089
fixture	no-memory	7	6	0.000	0.718	0.099	0.444	0.175
friction	no-memory	0	6	0.000	0.671	0.168	0.000	0.652
friction	no-memory	1	6	0.333	0.759	0.137	0.000	0.780
friction	no-memory	2	6	0.167	0.797	0.126	0.000	0.799
friction	no-memory	3	6	0.167	0.812	0.127	0.000	0.532
friction	no-memory	4	6	0.167	0.803	0.123	0.000	0.678
friction	no-memory	5	6	0.333	0.781	0.144	0.000	0.683
friction	no-memory	6	6	0.333	0.764	0.146	0.000	0.820
friction	no-memory	7	6	0.333	0.685	0.187	0.000	0.655
nominal	no-memory	0	6	0.000	0.534	0.179	0.000	0.802
nominal	no-memory	1	6	0.333	0.688	0.121	0.000	0.722
nominal	no-memory	2	6	0.333	0.612	0.164	0.000	0.683
nominal	no-memory	3	6	0.333	0.617	0.160	0.000	0.692
nominal	no-memory	4	6	0.333	0.790	0.138	0.000	0.817
nominal	no-memory	5	6	0.500	0.804	0.123	0.000	0.828
nominal	no-memory	6	6	0.167	0.803	0.124	0.000	0.816
nominal	no-memory	7	6	0.167	0.585	0.178	0.000	0.800
pocket	no-memory	0	6	0.167	0.620	0.154	0.135	0.423
pocket	no-memory	1	6	0.000	0.645	0.209	0.000	0.809
pocket	no-memory	2	6	0.167	0.743	0.175	0.000	0.820
pocket	no-memory	3	6	0.333	0.750	0.172	0.000	0.823
pocket	no-memory	4	6	0.167	0.557	0.232	0.000	0.697
pocket	no-memory	5	6	0.167	0.461	0.225	0.000	0.660
pocket	no-memory	6	6	0.167	0.612	0.232	0.000	0.803
pocket	no-memory	7	6	0.667	0.718	0.198	0.000	0.839
actuator	topo-v4	0	6	0.000	0.644	0.158	0.753	0.114
actuator	topo-v4	1	6	0.000	0.569	0.157	0.764	0.120
actuator	topo-v4	2	6	0.000	0.716	0.126	0.521	0.353
actuator	topo-v4	3	6	0.000	0.705	0.124	0.660	0.462
actuator	topo-v4	4	6	0.000	0.492	0.134	0.667	0.515
actuator	topo-v4	5	6	0.000	0.390	0.225	0.670	0.335

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
actuator	topo-v4	6	6	0.000	0.602	0.136	0.806	0.343
actuator	topo-v4	7	6	0.000	0.570	0.168	0.806	0.108
extreme	topo-v4	0	6	0.000	0.649	0.101	0.160	0.243
extreme	topo-v4	1	6	0.000	0.338	0.181	0.066	0.000
extreme	topo-v4	2	6	0.000	0.400	0.166	0.000	0.000
extreme	topo-v4	3	6	0.000	0.647	0.095	0.108	0.169
extreme	topo-v4	4	6	0.000	0.557	0.127	0.069	0.000
extreme	topo-v4	5	6	0.000	0.649	0.101	0.069	0.111
extreme	topo-v4	6	6	0.000	0.609	0.103	0.184	0.141
extreme	topo-v4	7	6	0.000	0.727	0.078	0.000	0.000
combined	topo-v4	0	6	0.167	0.622	0.130	0.260	0.786
combined	topo-v4	1	6	0.000	0.762	0.078	0.243	0.404
combined	topo-v4	2	6	0.167	0.801	0.106	0.236	0.631
combined	topo-v4	3	6	0.000	0.636	0.137	0.372	0.427
combined	topo-v4	4	6	0.000	0.832	0.068	0.455	0.553
combined	topo-v4	5	6	0.167	0.686	0.106	0.368	0.613
combined	topo-v4	6	6	0.000	0.544	0.147	0.122	0.607
combined	topo-v4	7	6	0.000	0.707	0.142	0.524	0.407
chain	topo-v4	0	6	0.000	0.878	0.043	0.000	0.423
chain	topo-v4	1	6	0.000	0.845	0.046	0.000	0.125
chain	topo-v4	2	6	0.000	0.851	0.045	0.000	0.350
chain	topo-v4	3	6	0.000	0.913	0.049	0.000	0.463
chain	topo-v4	4	6	0.000	0.969	0.012	0.000	0.184
chain	topo-v4	5	6	0.000	0.723	0.105	0.000	0.592
chain	topo-v4	6	6	0.000	0.935	0.031	0.000	0.327
chain	topo-v4	7	6	0.000	0.727	0.089	0.000	0.402
noise	topo-v4	0	6	0.333	0.826	0.107	0.000	0.793
noise	topo-v4	1	6	0.167	0.785	0.131	0.000	0.818
noise	topo-v4	2	6	0.167	0.820	0.095	0.118	0.742
noise	topo-v4	3	6	0.167	0.735	0.122	0.000	0.669
noise	topo-v4	4	6	0.333	0.709	0.126	0.135	0.764
noise	topo-v4	5	6	0.167	0.763	0.126	0.000	0.781
noise	topo-v4	6	6	0.167	0.701	0.147	0.000	0.802
noise	topo-v4	7	6	0.333	0.807	0.091	0.000	0.680
delayed	topo-v4	0	6	0.000	0.260	0.177	0.444	0.382
delayed	topo-v4	1	6	0.000	0.142	0.182	0.542	0.301
delayed	topo-v4	2	6	0.000	0.273	0.207	0.677	0.231
delayed	topo-v4	3	6	0.000	0.330	0.183	0.674	0.240

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
delayed	topo-v4	4	6	0.000	0.153	0.181	0.677	0.527
delayed	topo-v4	5	6	0.000	0.127	0.180	0.542	0.369
delayed	topo-v4	6	6	0.000	0.313	0.207	0.812	0.141
delayed	topo-v4	7	6	0.000	0.130	0.173	0.406	0.318
distractor	topo-v4	0	6	0.000	0.714	0.067	0.000	0.244
distractor	topo-v4	1	6	0.000	0.231	0.175	0.000	0.364
distractor	topo-v4	2	6	0.167	0.461	0.196	0.000	0.536
distractor	topo-v4	3	6	0.167	0.376	0.154	0.000	0.201
distractor	topo-v4	4	6	0.167	0.328	0.194	0.000	0.478
distractor	topo-v4	5	6	0.000	0.447	0.121	0.000	0.159
distractor	topo-v4	6	6	0.000	0.455	0.135	0.000	0.088
distractor	topo-v4	7	6	0.167	0.287	0.188	0.000	0.441
wall-jam	topo-v4	0	6	0.000	0.966	0.013	0.000	0.000
wall-jam	topo-v4	1	6	0.000	0.751	0.052	0.000	0.000
wall-jam	topo-v4	2	6	0.000	0.881	0.033	0.000	0.000
wall-jam	topo-v4	3	6	0.000	0.638	0.081	0.000	0.000
wall-jam	topo-v4	4	6	0.000	0.509	0.102	0.000	0.000
wall-jam	topo-v4	5	6	0.000	0.495	0.112	0.000	0.000
wall-jam	topo-v4	6	6	0.000	0.467	0.123	0.000	0.000
wall-jam	topo-v4	7	6	0.000	0.679	0.067	0.000	0.000
fixture	topo-v4	0	6	0.000	0.695	0.099	0.240	0.093
fixture	topo-v4	1	6	0.000	0.819	0.073	0.396	0.111
fixture	topo-v4	2	6	0.000	0.685	0.105	0.295	0.176
fixture	topo-v4	3	6	0.000	0.697	0.095	0.201	0.175
fixture	topo-v4	4	6	0.000	0.855	0.053	0.236	0.068
fixture	topo-v4	5	6	0.000	0.774	0.072	0.188	0.097
fixture	topo-v4	6	6	0.000	0.855	0.043	0.059	0.201
fixture	topo-v4	7	6	0.000	0.786	0.074	0.226	0.049
friction	topo-v4	0	6	0.000	0.673	0.109	0.000	0.518
friction	topo-v4	1	6	0.000	0.554	0.114	0.000	0.604
friction	topo-v4	2	6	0.167	0.538	0.124	0.000	0.710
friction	topo-v4	3	6	0.167	0.774	0.087	0.000	0.518
friction	topo-v4	4	6	0.000	0.501	0.142	0.000	0.808
friction	topo-v4	5	6	0.333	0.656	0.113	0.000	0.599
friction	topo-v4	6	6	0.167	0.543	0.135	0.000	0.737
friction	topo-v4	7	6	0.167	0.683	0.111	0.000	0.613
nominal	topo-v4	0	6	0.167	0.810	0.119	0.000	0.814
nominal	topo-v4	1	6	0.167	0.474	0.163	0.000	0.714

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
nominal	topo-v4	2	6	0.333	0.689	0.151	0.000	0.805
nominal	topo-v4	3	6	0.333	0.579	0.155	0.000	0.739
nominal	topo-v4	4	6	0.500	0.704	0.137	0.000	0.807
nominal	topo-v4	5	6	0.500	0.809	0.119	0.000	0.819
nominal	topo-v4	6	6	0.167	0.710	0.135	0.000	0.813
nominal	topo-v4	7	6	0.000	0.708	0.152	0.000	0.807
pocket	topo-v4	0	6	0.333	0.605	0.141	0.135	0.423
pocket	topo-v4	1	6	0.167	0.601	0.138	0.000	0.623
pocket	topo-v4	2	6	0.167	0.674	0.153	0.000	0.804
pocket	topo-v4	3	6	0.333	0.525	0.226	0.000	0.812
pocket	topo-v4	4	6	0.333	0.558	0.169	0.000	0.819
pocket	topo-v4	5	6	0.333	0.564	0.208	0.000	0.693
pocket	topo-v4	6	6	0.333	0.663	0.187	0.000	0.818
pocket	topo-v4	7	6	0.500	0.489	0.202	0.000	0.811
actuator	topo-v5	0	6	0.000	0.646	0.158	0.771	0.114
actuator	topo-v5	1	6	0.000	0.530	0.164	0.764	0.230
actuator	topo-v5	2	6	0.000	0.713	0.128	0.524	0.351
actuator	topo-v5	3	6	0.000	0.704	0.125	0.660	0.351
actuator	topo-v5	4	6	0.000	0.410	0.141	0.531	0.686
actuator	topo-v5	5	6	0.000	0.275	0.255	0.670	0.328
actuator	topo-v5	6	6	0.000	0.607	0.138	0.806	0.398
actuator	topo-v5	7	6	0.000	0.567	0.165	0.806	0.000
extreme	topo-v5	0	6	0.000	0.431	0.153	0.000	0.124
extreme	topo-v5	1	6	0.000	0.478	0.145	0.073	0.000
extreme	topo-v5	2	6	0.000	0.437	0.148	0.052	0.000
extreme	topo-v5	3	6	0.000	0.461	0.144	0.111	0.176
extreme	topo-v5	4	6	0.000	0.616	0.109	0.149	0.132
extreme	topo-v5	5	6	0.000	0.690	0.086	0.094	0.258
extreme	topo-v5	6	6	0.000	0.669	0.076	0.000	0.107
extreme	topo-v5	7	6	0.000	0.787	0.057	0.000	0.000
combined	topo-v5	0	6	0.167	0.592	0.143	0.260	0.664
combined	topo-v5	1	6	0.000	0.767	0.083	0.236	0.375
combined	topo-v5	2	6	0.000	0.704	0.120	0.222	0.567
combined	topo-v5	3	6	0.000	0.506	0.145	0.354	0.454
combined	topo-v5	4	6	0.000	0.785	0.085	0.576	0.499
combined	topo-v5	5	6	0.167	0.700	0.112	0.361	0.624
combined	topo-v5	6	6	0.000	0.570	0.146	0.122	0.701
combined	topo-v5	7	6	0.000	0.678	0.131	0.517	0.549

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEEdit	Safety	Prog.
chain	topo-v5	0	6	0.000	0.961	0.016	0.000	0.320
chain	topo-v5	1	6	0.000	0.852	0.045	0.000	0.131
chain	topo-v5	2	6	0.000	0.915	0.034	0.000	0.350
chain	topo-v5	3	6	0.167	0.906	0.050	0.000	0.465
chain	topo-v5	4	6	0.000	0.859	0.037	0.000	0.190
chain	topo-v5	5	6	0.000	0.749	0.087	0.000	0.564
chain	topo-v5	6	6	0.000	0.947	0.025	0.000	0.287
chain	topo-v5	7	6	0.000	0.808	0.071	0.000	0.421
noise	topo-v5	0	6	0.333	0.707	0.129	0.000	0.658
noise	topo-v5	1	6	0.000	0.600	0.171	0.000	0.685
noise	topo-v5	2	6	0.333	0.791	0.116	0.000	0.791
noise	topo-v5	3	6	0.167	0.701	0.142	0.000	0.675
noise	topo-v5	4	6	0.333	0.663	0.136	0.135	0.759
noise	topo-v5	5	6	0.167	0.743	0.138	0.000	0.784
noise	topo-v5	6	6	0.000	0.683	0.135	0.000	0.787
noise	topo-v5	7	6	0.333	0.676	0.138	0.000	0.766
delayed	topo-v5	0	6	0.000	0.116	0.201	0.667	0.448
delayed	topo-v5	1	6	0.000	0.161	0.172	0.538	0.249
delayed	topo-v5	2	6	0.000	0.195	0.201	0.434	0.282
delayed	topo-v5	3	6	0.000	0.268	0.176	0.406	0.359
delayed	topo-v5	4	6	0.000	0.147	0.178	0.674	0.255
delayed	topo-v5	5	6	0.000	0.106	0.185	0.670	0.350
delayed	topo-v5	6	6	0.000	0.257	0.199	0.674	0.299
delayed	topo-v5	7	6	0.000	0.180	0.173	0.542	0.434
distractor	topo-v5	0	6	0.000	0.615	0.085	0.000	0.213
distractor	topo-v5	1	6	0.000	0.182	0.214	0.000	0.441
distractor	topo-v5	2	6	0.333	0.410	0.202	0.000	0.537
distractor	topo-v5	3	6	0.000	0.299	0.169	0.000	0.232
distractor	topo-v5	4	6	0.000	0.255	0.210	0.000	0.525
distractor	topo-v5	5	6	0.000	0.433	0.133	0.000	0.070
distractor	topo-v5	6	6	0.000	0.411	0.144	0.000	0.075
distractor	topo-v5	7	6	0.167	0.281	0.182	0.000	0.396
wall-jam	topo-v5	0	6	0.000	0.950	0.016	0.000	0.000
wall-jam	topo-v5	1	6	0.000	0.689	0.079	0.000	0.000
wall-jam	topo-v5	2	6	0.000	0.858	0.037	0.000	0.000
wall-jam	topo-v5	3	6	0.000	0.542	0.101	0.000	0.000
wall-jam	topo-v5	4	6	0.000	0.410	0.109	0.000	0.000
wall-jam	topo-v5	5	6	0.000	0.645	0.085	0.000	0.000

Table 12 continued

Split	Method	Seed	Eps	Succ.	Edge F1	GEdit	Safety	Prog.
wall-jam	topo-v5	6	6	0.000	0.557	0.096	0.000	0.000
wall-jam	topo-v5	7	6	0.000	0.816	0.043	0.000	0.000
fixture	topo-v5	0	6	0.000	0.689	0.093	0.132	0.094
fixture	topo-v5	1	6	0.000	0.800	0.078	0.396	0.109
fixture	topo-v5	2	6	0.000	0.682	0.105	0.302	0.098
fixture	topo-v5	3	6	0.000	0.789	0.075	0.194	0.291
fixture	topo-v5	4	6	0.000	0.797	0.061	0.250	0.069
fixture	topo-v5	5	6	0.000	0.730	0.085	0.201	0.000
fixture	topo-v5	6	6	0.000	0.786	0.062	0.000	0.089
fixture	topo-v5	7	6	0.000	0.775	0.061	0.174	0.027
friction	topo-v5	0	6	0.000	0.561	0.119	0.000	0.435
friction	topo-v5	1	6	0.000	0.525	0.150	0.000	0.611
friction	topo-v5	2	6	0.167	0.551	0.121	0.000	0.714
friction	topo-v5	3	6	0.000	0.666	0.105	0.000	0.422
friction	topo-v5	4	6	0.167	0.625	0.115	0.000	0.697
friction	topo-v5	5	6	0.333	0.738	0.101	0.000	0.618
friction	topo-v5	6	6	0.167	0.377	0.145	0.000	0.777
friction	topo-v5	7	6	0.167	0.648	0.121	0.000	0.594
nominal	topo-v5	0	6	0.167	0.810	0.118	0.000	0.814
nominal	topo-v5	1	6	0.167	0.479	0.185	0.000	0.710
nominal	topo-v5	2	6	0.333	0.700	0.142	0.000	0.798
nominal	topo-v5	3	6	0.167	0.407	0.161	0.000	0.630
nominal	topo-v5	4	6	0.333	0.644	0.142	0.000	0.804
nominal	topo-v5	5	6	0.333	0.655	0.149	0.000	0.690
nominal	topo-v5	6	6	0.000	0.592	0.158	0.000	0.797
nominal	topo-v5	7	6	0.000	0.672	0.148	0.000	0.801
pocket	topo-v5	0	6	0.000	0.604	0.141	0.135	0.423
pocket	topo-v5	1	6	0.167	0.589	0.128	0.000	0.698
pocket	topo-v5	2	6	0.000	0.668	0.157	0.000	0.801
pocket	topo-v5	3	6	0.333	0.615	0.163	0.000	0.684
pocket	topo-v5	4	6	0.167	0.382	0.198	0.000	0.780
pocket	topo-v5	5	6	0.333	0.558	0.143	0.000	0.661
pocket	topo-v5	6	6	0.167	0.603	0.177	0.000	0.809
pocket	topo-v5	7	6	0.333	0.698	0.151	0.000	0.682

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